

ENHANCING MOBILE HEALTH SENSING WITH AUTHORIZATION TECHNIQUES,
PERSISTENT REMINDERS, AND ADAPTIVE EMA TIMING

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To my wife,
Katie Slade,
for all of the support.

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I used large language models (LLMs) in the preparation of this dissertation. For the written text, I used LLMs to copyedit and revise prose: in most cases I wrote several paragraphs and had the model revise them, and in other cases I supplied an outline and had the model draft a paragraph from it. All content is derived from my original work. I also used AI assistance in developing the study applications and in generating the analysis scripts. All study design decisions were my own, and I verified the generated code and evaluated and edited all text output. I take full responsibility for the content of this dissertation.

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ABSTRACT

Mobile and ubiquitous devices, combined with machine learning (ML), have introduced new ways to collect health data in real-world settings outside traditional labs or clinics. These devices enable the collection of both active data, in which participants are directly involved, and passive data, which is collected without active participation. Active data is typically obtained through Ecological Momentary Assessments (EMA), where participants are prompted to report their symptoms and behavior. Passive data comprises sensor and usage data from mobile and ubiquitous devices collected in the background. This mobile data is used to train ML models to predict EMA-captured outcomes, supporting health applications such as digital diagnostics, monitoring systems, and adaptive interventions.

The success of health applications that leverage ML models depends on consistent, accurate collection of active and passive data. However, several challenges impede data collection, particularly in real-world settings. In active data collection, participants do not always notice and respond to EMA notifications. Passive data collection, meanwhile, requires appropriate permissions and consistent background operations, both of which are limited by constraints in widely used consumer mobile and wearable operating systems (e.g., iOS and Android). This thesis examines challenges in both active and passive data collection in health applications.

I conducted a systematic review of existing literature on active and passive data collection in mobile sensing. The review included 77 studies that collect active and passive data to inform ML-based symptom detection. Results show that active data collection is hampered primarily by low EMA compliance, whereas passive data collection faces challenges in obtaining proper authorizations and ensuring data consistency. The current literature indicates a need for advancements at the intersection of Human-Computer Interaction (HCI) and ML to improve mobile sensing capabilities for effective symptom detection that could support adaptive health applications. To address these challenges, I conducted two user studies examining four approaches designed to improve data collection in mobile sensing: (1) proper authorization techniques, (2) persistent reminders (permanent UI features such as a home screen widget), (3) interaction-aware prompting, and (4) real-time active learning.

The first user study examined the effects of different authorization methods on mobile sensing and evaluated the role of persistent reminders in enhancing both active and passive data collection. This study included 145 college students who installed a mobile app and participated for 30 days. The app collected active data through a daily EMA and gathered passive data via background events, such as location changes and periodic polling through background tasks. Additionally, updates to a home screen widget facilitated polling, which served as a persistent reminder. Contextual prompting, which triggered authorization prompts when a location event occurred, was also implemented. Findings from the study indicate that contextual prompting enhances event-based

data collection, and a persistent reminder supports passive data collection more effectively than traditional background tasks.

Building on the first study, I conducted a second user study to improve active data collection. This study integrated methods from the literature and the previous study with real-time active learning and *unlock-triggered EMA* to improve user compliance. I recruited 38 college students across three cohorts to install a mobile application that collected vitals, activity, and sleep data from a connected Apple Watch and used EMAs to collect self-reports of momentary stress. An on-device logistic regression model, retrained incrementally on participant responses, suppressed EMAs when its confidence exceeded a predefined threshold. Unlock-triggered EMAs were delivered shortly after a participant unlocked their phone, leveraging natural device interactions to increase notification salience. Cohorts 1 ($n = 5$) and 2 ($n = 13$) piloted the system. They produced the training data for the base stress-prediction model deployed in Cohort 3 ($n = 20$), which employed a within-subjects 2×2 factorial design crossing active learning (on vs. off) with EMA delivery timing (unlock-triggered vs. random-timed).

Unlock-triggered EMA delivery significantly improved compliance rates (80.0% vs. 64.2%) compared to random-timed delivery. Active learning-timed delivery collected 64.7% fewer EMAs than fixed-timed delivery. However, active learning offered no advantage over random selection in sustaining model accuracy. The generalized linear mixed model (GLMM) personalized only the participant-level intercept, and with little discriminative signal in the fixed effects ($\text{AUC} = 0.566$), uncertainty-based selection had nothing to exploit beyond that intercept, a quantity a random sample estimates as well as any actively selected one. Because of the low AUC along with other limitations, I could not determine whether active learning is broadly ineffective for personalization or simply ineffective here, given the weak signal.

To test whether this was specific to my dataset, I conducted a final analysis on CardioMate, a longitudinal dataset of 20 adults with self-reported hypertension. Under a leave-one-participant-out simulation, I compared uncertainty-based selection against random selection at matched budgets of 5 to 40 labels per participant. Active learning again showed no advantage in predicting high stress or blood pressure spikes.

These results suggest that small, simple HCI adjustments can have a large effect on both passive and active data collection. The persistent reminder, the contextual prompt for background event authorization, and unlock-triggered EMA timing each significantly improved data collection, and each did so without additional infrastructure or computation. The more complicated ML intervention, active learning, showed no significant benefit over collecting randomly selected labels in any of the studies reported here. My conclusion is not that complicated interventions like active learning-timed EMAs should be set aside, since they may well pay off in the right circumstances, but that in this work the simple HCI interventions provided clear and significant improvements in data collection.

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CHAPTER 1

INTRODUCTION

Mobile and ubiquitous devices present significant opportunities for health data collection and intervention in real-world settings. These devices facilitate both active and passive data collection. Active data collection involves direct participant engagement, often through ecological momentary assessments (EMA), where individuals report symptoms via mobile devices [1]. Conversely, passive data collection occurs when the device autonomously records data without participant involvement. Machine learning models are then trained on these passive data streams to predict health outcomes identified through EMA [2, 3, 4, 5, 6]. These mobile sensing systems serve as the foundation for developing adaptive interventions [7].

To address the challenges inherent in active and passive data collection that support ML and adaptive interventions, I explored innovations at the intersection of Human-Computer Interaction (HCI) and Machine Learning (ML). Figure 1.1 provides an overview of mobile sensing systems and their associated challenges. In the context of active data collection, participants may fail to respond to EMA requests during inopportune times, and an excessive number of requests can lead to participant burden. Conversely, passive data collection is often compromised when participants do not properly authorize collection mechanisms or when mobile devices fail to capture data consistently. In this thesis, I addressed these challenges by exploring proper authorization methods, persistent reminders (permanent UI features like home screen widgets or smartwatch complications), and adaptive EMA timing.

1.1 Mobile Devices in Health Care

Healthcare researchers initially collected data in controlled lab environments through tests, interviews, and surveys to better understand health challenges. However, these methods are limited to measuring data only within the lab setting. Interviews and surveys, in particular, are prone to recall bias, where participants may not accurately remember past events or trials. EMA was first introduced via paper journals to capture data in real-world, free-living environments. With EMA, participants recorded their experiences during or shortly after an event, providing more immediate and contextually relevant data.

The advent of smartphones and other mobile devices has significantly streamlined data collection in free-living environments. Mobile devices facilitate EMA by providing accessible reminders for participants to complete their assessments and by automatically recording data and logging entry times. Additionally, these devices are equipped with a variety of sensors capable of collecting passive health data, such as location, communication, activity, and vitals, offering expanded insights into the causes and implications of health issues. Machine learning can analyze this data to diagnose and treat diseases. Such applications have been successfully employed to address a variety of health

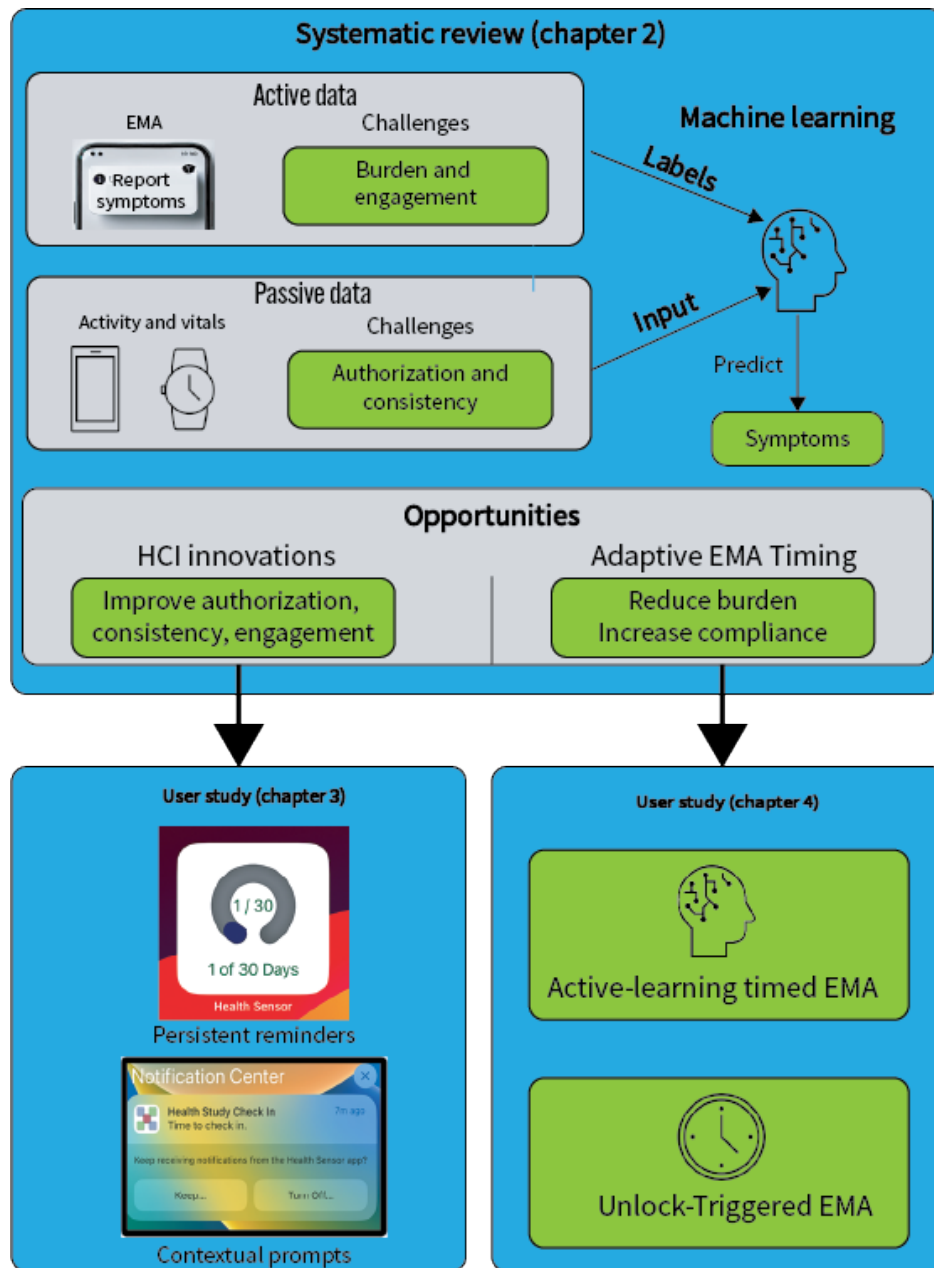


Figure 1.1: **Overview of Mobile Sensing Challenges and Solutions.** Machine learning models are trained on passively recorded data streams to predict symptoms captured by EMA. Key challenges for active data collection include participant burden and maintaining engagement, while passive data collection faces issues with obtaining authorization and ensuring data consistency. This thesis addressed these challenges through persistent reminders and contextual prompts (Chapter 3) and two adaptive EMA timing strategies: real-time active learning and unlock-triggered EMA (Chapter 4).

challenges, including mental health [8], addiction [9], and influenza [10], and to promote a more active lifestyle [11].

1.2 Active Data Collection

As stated above, active data refers to instances where the participant is actively involved in data collection. Active data is generally collected through EMA, also known as experience sampling [12], but can also be collected through other means, such as audio, video, paper surveys, and other exercises. EMA has a long history that predates the invention of the smartphone, with information traditionally collected through paper journals [1]. Despite its long history, challenges in conducting EMA on consumer mobile devices persist. Advancements in notification systems, such as notification deferral [13], notification summaries [14], and intelligent notification systems [15, 16], could cause participants to miss EMA reminders. I explored these challenges and potential solutions, which include more reliable notifications, persistent reminders, and adaptive EMA timing.

1.3 Passive Data Collection

Smartphones, smartwatches, wearables, and environmental sensors collect passive health data without the participant’s active involvement. These smart devices, equipped with multiple sensors, gather various types of passive data, including location, communication, activity, vitals, sleeping patterns, and more [17, 18, 19]. As consumer smartphones and smartwatches continue to integrate additional sensors, researchers increasingly rely on these devices for passive data collection. However, passive data collection on consumer mobile devices brings several challenges, including proper authorization to access private data and obtaining background runtime to perform scans [20, 21]. Besides exploring better authorization methods, I also explored how persistent reminders can increase the reliability of passive data scans while helping with active data collection.

1.4 Authorization Methods

Since mobile operating systems (iOS and Android) prioritize user privacy and battery conservation, active and passive data collection require appropriate authorization. Explicit authorization is necessary to collect personal data, such as location and health information, and send notifications. This authorization is typically granted via permission prompts, which can be presented during app setup or contextually when private data is accessed. Additionally, implicit authorization is required to secure background runtime for passive sensor data collection, which consumes battery life. Implicit authorization can be inferred from app usage patterns or established by installing a persistent reminder (see below). Chapter 3 presented a user study that explored how these authorization methods impact mobile sensing effectiveness.

1.5 Persistent Reminders

Persistent reminders are permanent user interface (UI) elements on mobile and ubiquitous devices, such as home screen widgets, unlocking mechanisms, persistent notifications, and smartwatch complications. These reminders serve two primary functions: first, they prompt participants to complete their EMAs, and second, as a fixed part of the UI, they enable mobile operating systems (OS) to allocate background runtime to keep them updated. This runtime facilitates passive data collection by allowing persistent reminders to access sensors. Additionally, persistent reminders can display feedback from ML models [22], including predicted outcomes and confidence levels, which may encourage participants to interact with their EMAs and adopt adaptive interventions. In addition to exploring various authorization methods, the user study presented in Chapter 3 also investigated the effectiveness of persistent reminders in supporting passive data collection.

1.6 Unlock-Triggered EMA

In addition to ML-driven timing, interaction-aware strategies can improve EMA compliance by aligning prompts with moments of natural device engagement. Unlock-triggered EMA delivers prompts shortly after a participant unlocks their phone, based on prior work on *unlock journaling* [23] but adopting a lighter-weight variant that does not block device access. By leveraging the increased *notification salience* at the moment of active device use, unlock-triggered EMA aims to increase response rates without increasing the overall number of prompts. Chapter 4 evaluated this HCI-driven strategy alongside real-time active learning to examine how these approaches can be combined to improve active data collection.

1.7 Real-Time Active Learning

One major challenge in training machine learning models is obtaining sufficient labels to develop an accurate model, particularly when those labels are provided through EMAs. Real-time active learning can mitigate this burden by reducing the number of labels needed for model training. Active learning begins by training an initial model on a small set of labeled data. It then uses a selection algorithm to identify the most informative samples—those that, if labeled, would most improve the model’s performance. In the context of mobile health sensing, this selection process can be applied in real time to govern EMA timing: when the model is confident in its prediction, the EMA is suppressed, and when confidence is low, the participant is prompted for a label. I refer to this strategy as *active learning-timed EMA*: by selectively requesting labels only when they are most informative, it aims to reduce participant burden while maintaining or improving model performance. I use that term throughout for EMAs whose delivery is governed by model uncertainty, in contrast to *fixed-timed EMA*, in which every scheduled prompt is delivered. Chapter

4 evaluated this approach in a live, closed-loop system deployed on consumer mobile devices.

1.8 Overview

In Chapter 2, I conducted a systematic review of studies that utilize mobile devices to collect health data through active and passive methods. Active data collection requires participant engagement, such as completing EMAs to report symptoms, whereas passive data collection leverages sensors to gather data unobtrusively. The review identified key challenges, including maintaining subject engagement, minimizing the burden in active data collection, assuring consistent data flow, and securing necessary authorizations for passive data collection. The results highlighted gaps and opportunities for innovations in HCI, such as persistent reminders, and ML, such as EMA timing algorithms, to enhance mobile health sensing.

In Chapter 3, I conducted a user study to evaluate methods for improving passive and active data collection on mobile devices, addressing the gaps and opportunities identified in Chapter 2. Specifically, I explored HCI methods, such as persistent reminders and authorization techniques, to enhance mobile sensing. The study investigated three key areas: (1) persistent reminders versus background tasks for passive data collection, (2) contextual prompting versus setup prompting for event-driven passive data collection, and (3) whether persistent reminders could enhance user engagement for active data collection.

In Chapter 4, I conducted a user study to address the gaps and opportunities identified in Chapter 2 by exploring ML and HCI methods for adaptive EMA timing. The study evaluated the effectiveness of active learning–timed EMAs compared to fixed-timed EMAs, as well as unlock-triggered EMA delivery compared to random-timed delivery. EMAs were used to collect self-reports of momentary stress as labels, while vitals, activity, and sleep data were passively collected from a connected Apple Watch to serve as inputs to an on-device logistic regression model. The study was conducted across multiple cohorts. The first two cohorts employed a within-subjects design comparing active learning–timed EMAs against fixed-timed EMAs, with unlock-triggered EMA introduced as a secondary condition. Based on initial findings, a third cohort employed a within-subjects 2×2 factorial design crossing active learning (on vs. off) with EMA delivery timing (unlock-triggered vs. random-timed). This study examined whether combining ML-driven and HCI-driven timing strategies could reduce participant burden and improve EMA compliance, supporting more adaptive and participant-friendly mobile health sensing systems.

Chapter 5 draws the interventions together. Some changed how EMAs were delivered; others changed how the model chose when to ask. The delivery changes (persistent reminders, contextual authorization, and unlock-triggered delivery) improved data collection wherever I applied them, and none of them asked anything of the model. However, the results of active learning–timed EMA were inconclusive.

CHAPTER 2

SYSTEMATIC REVIEW

2.1 Introduction

In the ever-evolving landscape of healthcare technology, mobile and ubiquitous health sensing systems have emerged as useful tools for real-time health monitoring and intervention. Mobile devices provide opportunities to collect health data “in the free-living environment” rather than in the lab. Traditionally, participants in research studies have been required to actively engage in collecting subjective measures related to psychiatry and lifestyle through ecological momentary assessments (EMAs). EMAs involve “repeated sampling of subjects’ current behaviors and experiences in real-time in the subjects’ natural environment” [1]. Consumer devices integrated with various sensors can collect data passively without active participation by the end user [24]. The combination of active and passive data streams leads to rich multimodal datasets that can aid researchers in learning nuanced patterns of everyday life [25]. These rich datasets provide fertile ground for machine learning (ML) to enhance healthcare outcomes [26, 27], where machine learning models are trained using passively recorded data streams as input to predict health outcomes captured by EMA [2, 3, 4, 5, 6] (see Figure 2.1).

Machine learning models are trained on passively recorded data streams to predict health outcomes captured by EMA. Challenges to active and passive data collection persist, such as reducing burden and ensuring data consistency. Our systematic review explores these challenges and provides insights into potential solutions. The development of machine learning-based adaptive interventions necessitates robust data collection methodologies. Despite recent advancements in mobile sensing and machine learning, significant challenges persist in active and passive data collection on mobile and ubiquitous devices. Stone et al. [28], through a review of EMA, identified privacy, compliance, and measurement reliability issues. Similarly, Doherty et al. [29] reported challenges concerning compliance, user burden, user engagement, and data validity. Passive data collection also faces obstacles; Boonstra et al. [30] demonstrated that Android and iOS devices completed only 55% and 45% of scheduled scans, respectively, and mobile sensing has been shown to decrease smartphone battery life [31]. Moreover, creating cross-platform sensing applications for Android and iOS introduces additional complexities [32, 33]. Other significant challenges include privacy concerns [34] and usability issues [35]. These challenges may hinder the efficacy of machine learning-based adaptive interventions.

In this study, we examined existing research on active and passive sensing to understand how challenges affect the overall efficacy of these sensing modalities. We aimed to expand upon existing reviews that either focused on specific health dimensions or did not include studies involving smartwatches, wearables, and environmental sensors. Through our investigation, we identified strategies

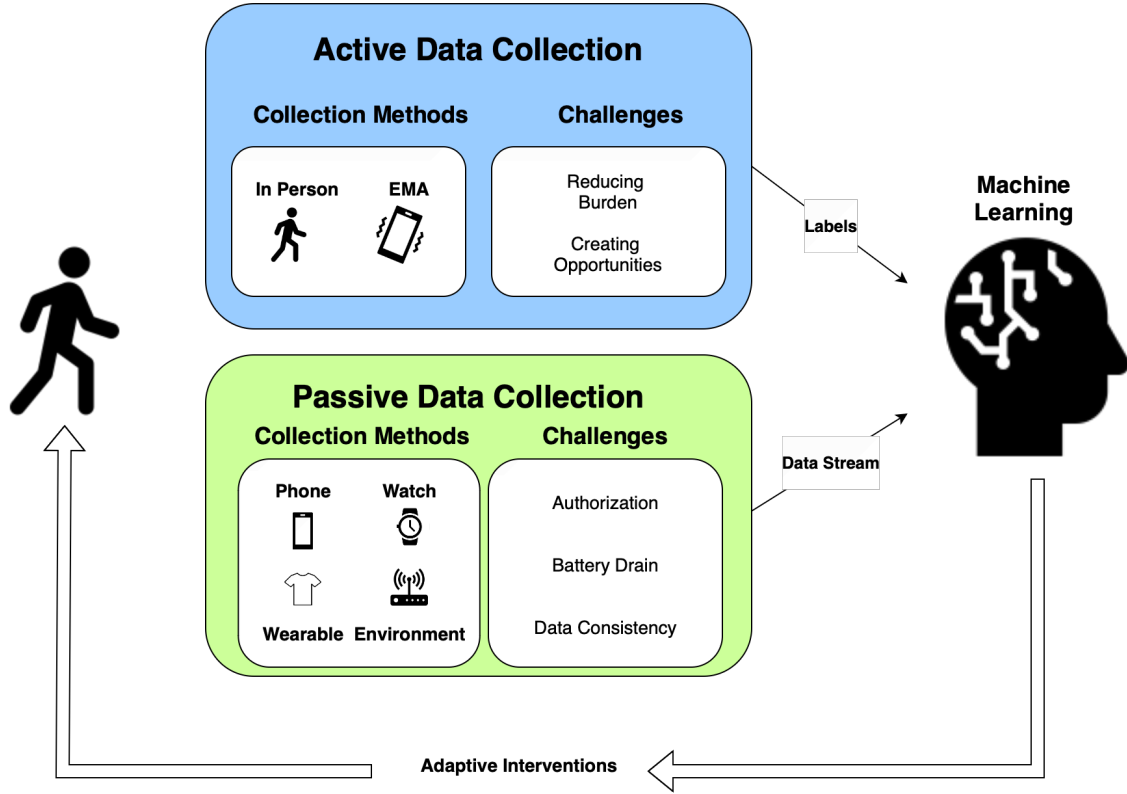


Figure 2.1: **Overview of active and passive data collection methods.** Machine learning models are trained on passively recorded data streams to predict health outcomes captured by EMA. Challenges to active and passive data collection persist, such as reducing burden and ensuring data consistency. Our systematic review explores these challenges and provides insights into potential solutions.

to mitigate or overcome obstacles in mobile sensing to enhance machine learning-based adaptive interventions. Specifically, we addressed the following research questions:

RQ1: What methodologies are employed to collect active data from participants? What strategies can be implemented to enhance the success of active data collection?

RQ2: What types of passive data are collected, and which sensors are utilized for this purpose? What strategies are adopted to improve collection rates and usability?

2.2 Related Work

Most previous systematic reviews on EMA and mobile sensing have primarily concentrated on specific use cases or diseases rather than on mobile sensing as a general field. Several reviews, for instance, have focused on mental health and digital phenotyping. Seppälä et al. [17] reviewed

studies involving mobile phones and wearable sensors for psychiatric disorders and symptoms, Elwirehardja et al. [18] examined sensing studies related to depressive disorders, Benoit et al. [19] focused on schizophrenia and bipolar disorders, and Teepe et al. [7] reviewed depression studies employing just-in-time adaptive mechanisms. Thieme et al. [36] explored the intersection of machine learning and human-computer interaction (HCI) for mental health. Reviews in the domain of affective computing, where intelligent systems aim to understand human emotions through mobile sensing, have also been conducted [37, 38, 39]. Beyond mental health, Hussain et al. [40] reviewed articles on sleep, Saboor et al. [41] focused on gait, and Stuijt et al. [42] studied mobile sensing in cancer patients. These reviews, however, are limited to specific diseases or disease families rather than providing a broader perspective on mobile sensing methodologies applicable across various healthcare domains.

Reviews on general mobile sensing, such as those by Kumar et al. [43] and Trifan et al. [44], provide valuable insights. Kumar et al. [43] conducted a comprehensive literature review to trace the historical evolution of passive sensing and mobile sensing frameworks, addressing aspects such as the adoption and maintenance of sensing software. While reviewing general mobile sensing literature, Trifan et al. [44] concentrated on smartphone applications. Our contribution extends this work by reviewing articles encompassing smartphones, watches, wearables, and environmental sensors integrated with machine learning or adaptive interventions. Additionally, we incorporated recent studies to identify current trends and gaps in the literature.

2.3 Methods

We adhered closely to the PRISMA guidelines [45, 46] to structure the identification and selection of relevant articles. The primary author reviewed all search results, while secondary authors independently reviewed a subset, ensuring each paper was evaluated by both the primary and a secondary author. In cases of disagreement regarding inclusion, the default decision was to include the paper for final analysis and data extraction. Similarly, data extraction was verified by at least two authors to minimize biases.

2.3.1 Record Identification

We identified relevant papers by searching ACM, IEEE, and Web of Science on May 5, 2024, using search terms related to mobile sensing and machine learning. We selected ACM and IEEE for their extensive computing and HCI literature coverage, and we chose Web of Science for its broad interdisciplinary scope. To ensure our focus was on current work, we limited the results to publications from 2016 to the present. The search terms used were: “mobile or ubiquitous,” “EMA,” “health,” “passive,” and “deep learning or machine learning” (see full query syntax for

ACM¹ and IEEE²). We also searched Google Scholar (see full query syntax³), restricting results to ACM and IEEE publications to ensure comprehensiveness within these venues. Additionally, we searched Web of Science for papers outside of IEEE and ACM venues using the same search terms. However, this search yielded an unmanageable number of results (307,694). To refine the search, we added “adaptive interventions” to the search terms⁴, which reduced the results to 303 papers.

The search yielded 774 results. ACM and IEEE returned 167 and 92 results, respectively. Google Scholar added 212 results, and Web of Science returned 303 papers.

2.3.2 Record Selection

We screened the titles and abstracts of the 774 resulting articles to determine their relevance to using mobile sensing to improve health conditions (see Table 2.1 for inclusion and exclusion criteria). Our analysis included articles if they utilized mobile or ubiquitous devices to collect health data (IC1). We excluded papers that did not collect both active and passive data to concentrate on studies offering a comprehensive approach to mobile health sensing, specifically those that aim to train machine learning models to predict actively sensed data from passive sensor streams. We excluded all review papers and those that did not employ mobile devices for data collection or were unrelated to health issues. This screening process resulted in 280 papers (ACM 89, IEEE 33, Google Scholar 115, Web of Science 43).

Next, we conducted a full-text review of the remaining 280 papers, excluding those that used previously collected datasets (EC1), such as Jiang et al. [47] or did not collect both active and passive data [48, 49, 50] (EC2). After applying these exclusion criteria, we identified 77 articles for inclusion in the systematic review. The Google Scholar search contributed 46 articles (12 IEEE and 36 ACM). The ACM search yielded 34 articles, including 18 that overlapped with Google Scholar results. The IEEE search provided 8 articles, which also overlapped with Google Scholar results. Web of Science contributed 13 additional articles (see Figure 2.2).

2.3.3 Data Extraction

We developed an extraction sheet to systematically gather data from the selected papers (see Table A.1 in the appendix). The sheet includes columns to identify each paper by author and year and to record the following details: disease or health issue addressed, study details (number

¹AllField:((mobile OR ubiquitous) AND ema AND health AND passive AND ("machine learning" OR "deep learning")) with Publication dates limited to 2016 to Present.

²("Full Text Only": "Mobile" OR "Full Text Only": "ubiquitous") AND ("Full Text Only": ema) AND ("Full Text Only": health) AND ("Full Text Only": passive) AND ("Full Text Only": "machine learning" OR "Full Text Only": "deep learning")

³[[mobile or ubiquitous] AND ema AND health AND passive AND [deep learning or machine learning] [source:ACM OR source:IEEE], with a date range from 2016 to Present

⁴ALL=((mobile OR ubiquitous) AND EMA AND health AND passive AND [deep learning OR machine learning] AND (Adaptive Interventions)) Limited to 2016 to Present

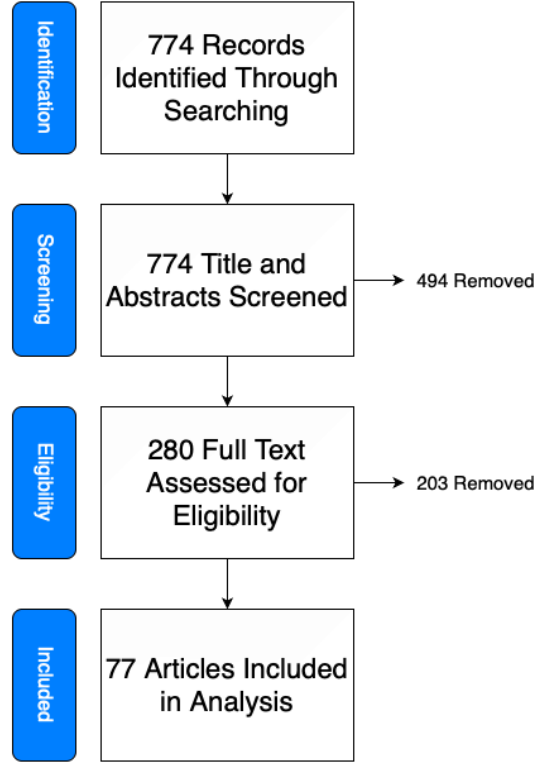


Figure 2.2: **Flowchart of article identification for inclusion and exclusion.** Following this process, we identified 77 articles to include in our analysis.

Table 2.1: **List of inclusion and exclusion criteria.**

IC1	The article needed to include data collection related to health or diseases with mobile, ubiquitous, or environmental devices.
EC1	The article used a data set that they did not collect or did not provide details on data collection.
EC2	Did not collect both active and passive data.

of participants and study duration), types of active data collected, types of passive data collected, and the analysis and results (including ML and statistical analysis). Additionally, we included a column for details on strategies for improving active data collection or adapting to missing data and a similar column for passive data collection.

The data extraction sheet was organized hierarchically by device type and sorted chronologically within each group. The hierarchy begins with the phone group, which includes all studies utilizing a phone exclusively for data collection. The next level is the watch group, comprising studies that use a watch with or without an accompanying phone. For this study, any device worn on the wrist is considered a watch. The wearables group is the third level, encompassing studies using other

types of wearable devices, which may also involve a watch or phone. The final and most inclusive group consists of studies employing sensors placed in the environment, such as home or workplace sensors, which may also involve any combination of wearable devices, watches, and phones.

2.4 Results

2.4.1 Quantitative Results

Our search results indicate that the number of published papers on the use of mobile and ubiquitous sensing continues to rise (see Figure 2.3), consistent with findings from previous reviews [44]. This trend underscores the relevance of our review by demonstrating the increasing interest in utilizing mobile sensing to study and improve health outcomes.

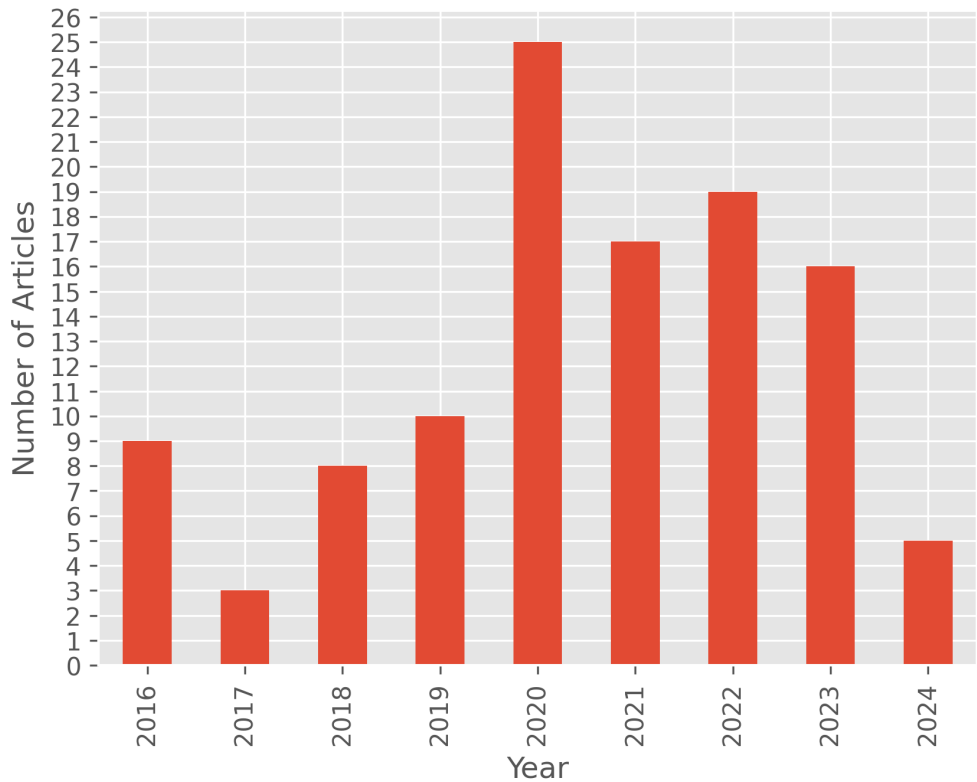


Figure 2.3: **Number of articles per year.** This chart illustrates the general trend of increasing studies involving mobile sensing over the years. Note that the data for 2024 includes only papers published before May 2024.

Although various health issues were included in the review, mental health emerged as the most frequently studied area (see Figure 2.4), accounting for over 36% of the studies. Other categories examined include stress, drug and alcohol abuse, eating habits, physical activity, sleep, and brain

function. Following mental health, drug and alcohol abuse and stress were the next most studied issues in our review. Additionally, several studies applied machine learning and passive sensing technology to understand and improve other health conditions.

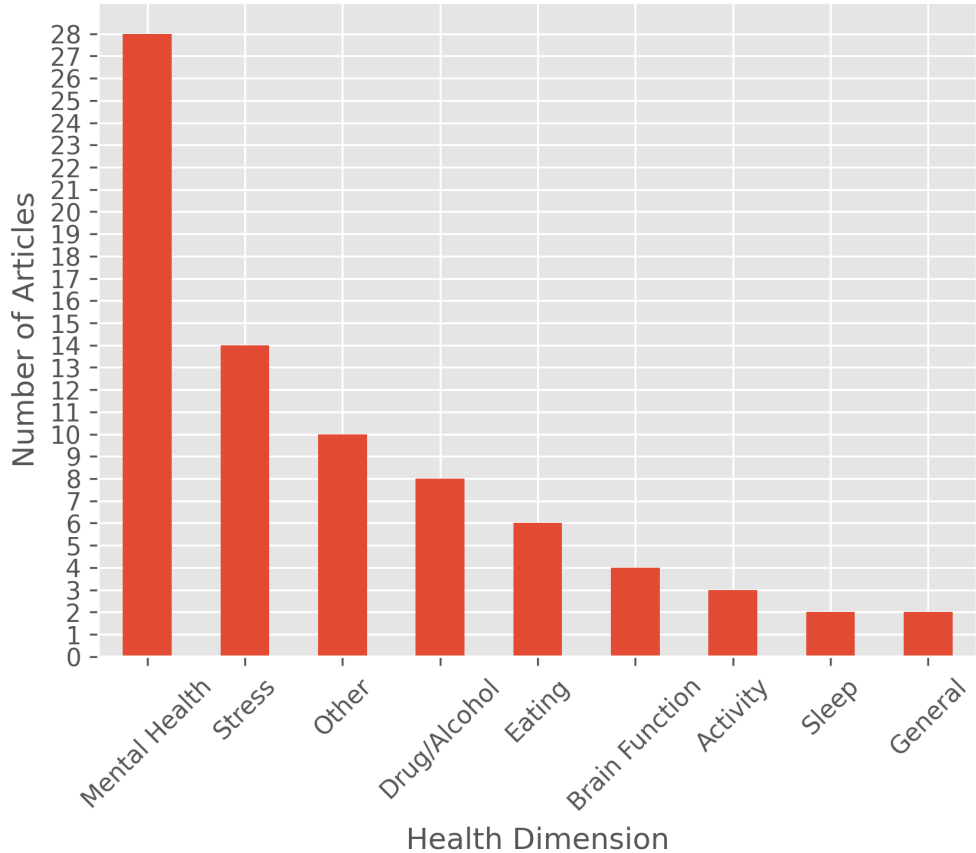


Figure 2.4: **Number of articles by health dimension.** Mental health was the most studied health dimension. Health dimensions included in the “other” dimension include chronic pain, hyperactivity, academic performance, couples conflict, heart health, medicine adherence, and multiple sclerosis. (n=77)

Our search revealed several trends in the types of most frequently used sensing technologies (see Figure 2.5). Smartphones have consistently been the primary sensing technology, but wrist-worn devices, such as smartwatches, have steadily increased. Other wearables (excluding watches) have also gained popularity, although recent trends indicate a decline. This decline could be attributed to commercial and clinical-grade smartwatches’ enhanced capabilities and sensors, which may reduce the need for additional wearables. Environmental sensors, placed in the environment rather than connected to the participant, are also gaining popularity. However, the increased capabilities of smartwatches, combined with advancements in machine learning, have potentially diminished the need for additional wearable and environmental sensors to collect passive data.

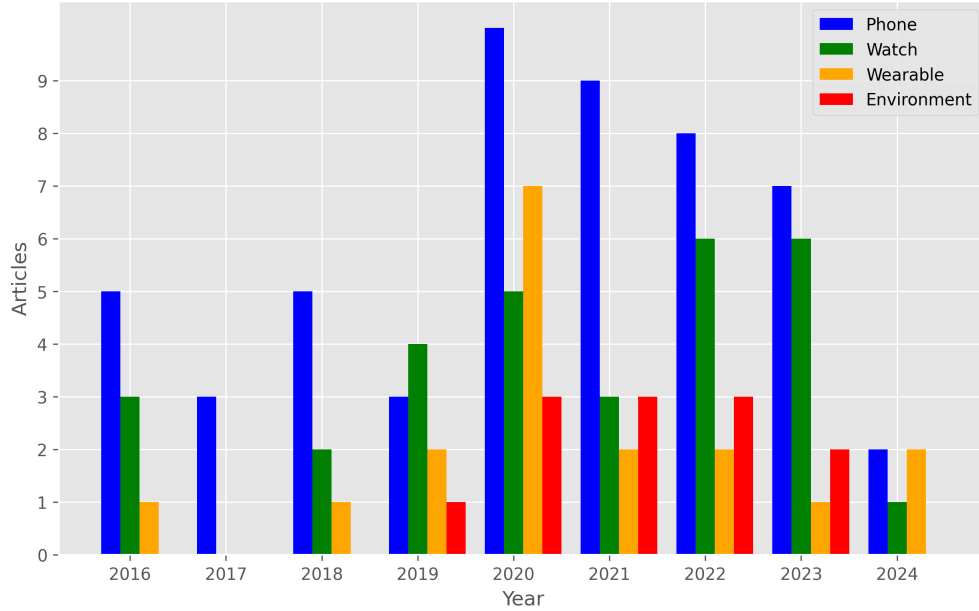


Figure 2.5: **Number of articles by year and device.** The number of articles involving wearables shows a decreasing trend in more recent years, whereas watches tend to be increasing. (n=77)

We also analyzed the data types collected through passive sensing, as shown in Figure 2.6. Location and activity data were the most commonly collected, followed closely by vitals, accelerometer data, sleep patterns, and phone usage. Vitals were typically collected using watches or other wearables, while location and phone usage data were primarily collected via smartphones. Sleep data had the most varied collection methods, sensed through phone usage, smartwatches, Wi-Fi, and environmental sensors.

In addition to passive data collection, we also reviewed active data collection methods. Most papers (56 out of 77) employed EMA to gather data. Figure 2.7 presents a box plot of the average number of EMAs requested per day. Excluding papers that did not perform EMAs or did not report EMA frequency, the average number of EMAs requested per day was 4.89, with a standard deviation of 5.78. One study [51] requested an EMA every 30 minutes, while another [52] only requested an EMA every two weeks. Other forms of active data collection included in-person surveys, tests, and videos.

2.4.2 RQ1: Methods to Improve Active Data Collection

While reviewing the full text of each paper, we documented any attempts to improve active data collection or how missing data were managed during analysis. Our notes are included in the data extraction table. One method to enhance EMA compliance mentioned by two of the 77 papers [53, 54] was linking monetary incentives to the number of completed assessments. Although

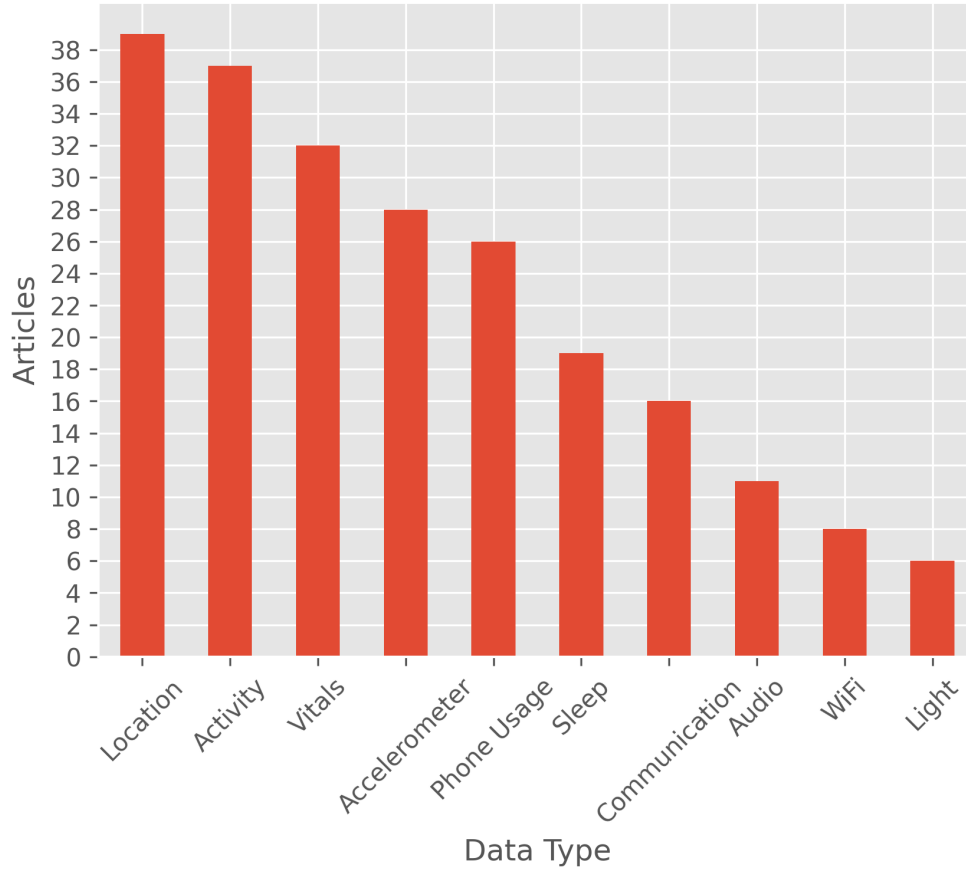


Figure 2.6: **Bar chart of the types of passive data collected.** Location, activity, and vitals are among the most commonly collected data types, followed by accelerometer and phone usage. (n=77)

only explicitly mentioned by two of the 77 papers [55, 56], a common method to improve compliance was conducting in-person check-ins or having the research team contact participants when missing EMAs were detected. The most prevalent approach to handling missing data, mentioned by at least six of the 77 papers, was to ignore time periods with missing data or exclude participants who did not provide sufficient active data [57, 56, 58, 59, 60, 61]. These methods are common in the literature, and their advantages and disadvantages have been discussed in other review papers [28, 29].

In addition to the previously mentioned approaches, we identified several novel methods to improve active data collection. Four of the 77 papers [62, 58, 63, 64] used text messages to initiate an EMA instead of application notifications. Text messages may improve EMA compliance rates as participants might respond more reliably to text notifications, which are generally more prominent than application notifications. Another strategy to enhance EMA compliance is allowing participants to help select EMA timings. Kim et al. [22] observed that late-night EMA requests were

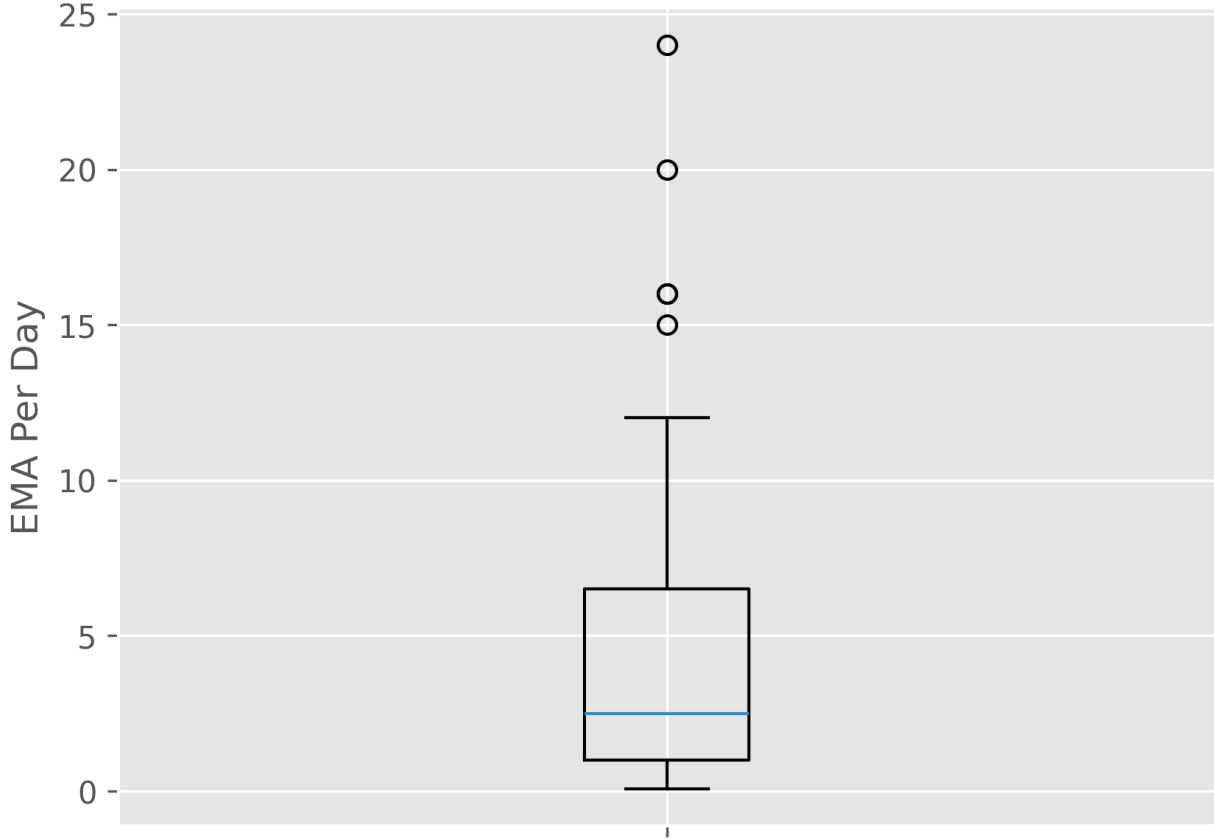


Figure 2.7: **Boxplot of the average number of EMAs taken per day.** Data excludes all articles that did not report EMA per day or did not collect data through EMA. The average was 4.89, with a standard deviation of 5.78. 56 of the 77 papers collected active data via EMA. (n=77)

often missed. Jacobson et al. [65] allowed participants to input their waking hours before scheduling EMAs. These methods could improve compliance by adapting to the participants' needs and preferences.

Machine learning was also employed to adapt EMAs to the participant's needs and preferences. Theilig et al. [66] used machine learning to adjust EMA timings to be less disruptive. Lustrek et al. [67] developed a machine-learning framework to autofill EMA questions, reducing participants' burden. King et al. [68] utilized machine learning to pre-select EMA questions, minimizing the number of questions required for each EMA. Finally, Arakawa et al. [69] demonstrated that unsupervised learning can reduce or eliminate the need for active data collection. These studies indicate that machine learning offers a promising avenue for enhancing active data collection.

Unique HCI innovations can also improve active data collection. Three of 77 papers [70, 71, 72] provided participants with a user interface that displayed data collection rates, which helped motivate users to provide additional data. Kunchay et al. [73] explored using microEMAs, where a

one-question EMA could be answered on a smartwatch. Saha et al. [74] also examined microEMAs combined with unlock journaling, where participants answer an EMA question to unlock their phones. Additionally, Kim et al. [22] tested the use of explainable AI to facilitate participant self-reflection. They developed a system that explains how the ML algorithm computed participants' stress levels based on passively collected data, displaying prominent features to help participants understand the causes of their increased stress levels. This unique combination of machine learning and HCI presents a promising direction for future research.

2.4.3 RQ2: Strategies to Improve Passive Data Collection

During our search, we found several challenges with passive sensing on smartphones. Four of the 77 studies reported issues with participants turning off features or not authorizing data collection [75, 76, 77, 78], and four of the 77 studies had challenges exporting or syncing data to servers [62, 60, 79, 75], particularly when executing background tasks on iOS devices. Two studies noted that participants sometimes did not wear the device [80, 81]. Other challenges included ensuring timestamps were synchronized [82] and managing changes resulting from mobile operating system updates [83].

We also identified several methods that improved the reliability of data collection. Several studies contacted participants to resolve issues and encouraged them to wear devices [79]. This approach can be enhanced by a dashboard for the research team that displays data collection rates [84]. Hafiz et al. [54] used monetary incentives to encourage participants to wear the devices. Adler et al. [60] utilized EMA to direct participants to start or resume syncing FitBit data, while Biel et al. [85] used EMA to allow participants to fill in missing data. These methods focused on assisting or motivating participants to enhance passive data collection.

In addition to improving passive data collection, several papers discussed machine learning strategies to handle missing data [80, 55]. Booth et al. [58] included a literature review of different methods for managing missing data. Rashid et al. [86] explored four methods of imputing missing data (regression, matrix completion, KNN, last observation carried forward), which can reduce prediction error by up to 22%. Chen et al. [55] used missing compliance/data as a feature to train the machine learning model. Baglione et al. [21] used machine learning to activate and deactivate sensors to preserve battery life. Machine learning methods can help mitigate the issue of unreliable passive data collection.

HCI methods can also enhance passive data collection. Kao et al. [70] developed a user interface (UI) that displays data quality to participants, potentially encouraging them to authorize data collection and provide higher-quality data. Four of the 77 papers mentioned using Apple's Health app to improve data collection [87, 66, 88, 11]. The Apple Health app is a central storage location for health data, allowing other mobile applications to read and write data into the app. Since the Health app is integrated into iOS, it offers a more reliable method for data collection. Similarly,

Android devices have a centralized health store called Health Connect. These centralized stores could provide a means to enhance passive data collection.

2.5 Discussion

2.5.1 Current Challenges

As mobile devices continue to evolve, EMA frameworks must adapt to accommodate emerging functionalities. Recent smartphone changes have integrated features to minimize distractions, including notification deferrals and summaries [89, 90]. These features [13, 15] and intelligent notification systems may impede EMA prompts. Active data collection must adjust to these notification system mechanisms to ensure participants receive timely reminders to complete their EMA tasks.

Through this review, we identified several innovations that improve active data collection. Unlock journaling, microEMAs, and machine learning have improved EMA compliance by reducing the burden of completion or enhancing participant opportunities and motivation through behavioral nudges. Future research should concentrate on the intersection of machine learning and HCI to further mitigate the burden and enhance opportunities for data collection and user motivation. Kim et al.’s work [22] examined how providing participants with feedback through explainable AI can increase motivation and warrants further exploration. This is particularly relevant if such feedback facilitates the implementation of adaptive interventions. Integrating machine learning with HCI presents a promising avenue for addressing the challenges associated with active data collection.

Passive data collection with consumer mobile devices also suffers several challenges, especially in ensuring continuous data collection and syncing. Adapting machine learning algorithms to handle unreliable data is common. However, machine learning can also help improve passive data collection, which can be used to help preserve battery life [21]. In addition, developing user interface features that can help create opportunities and foster user motivation to cooperate and authorize passive data collection should also be explored.

2.5.2 Design Guidelines

Our analysis revealed numerous challenges inherent in both active and passive data collection techniques. We identified strategies researchers have adopted to address these issues. We present our findings as a series of design guidelines to assist researchers in effectively collecting both passive and active data for training machine learning models.

The design guidelines for active sensing are detailed in Figure 2.8, which encompasses a variety of methods to consider when developing an active data collection system. We developed these guidelines from the results of answering RQ1. Our review identified several methods to reduce participant burden, such as microEMA and unlock journaling, which can minimize the time required to complete assessments [73, 74]. Furthermore, leveraging ML to aid participants in completing

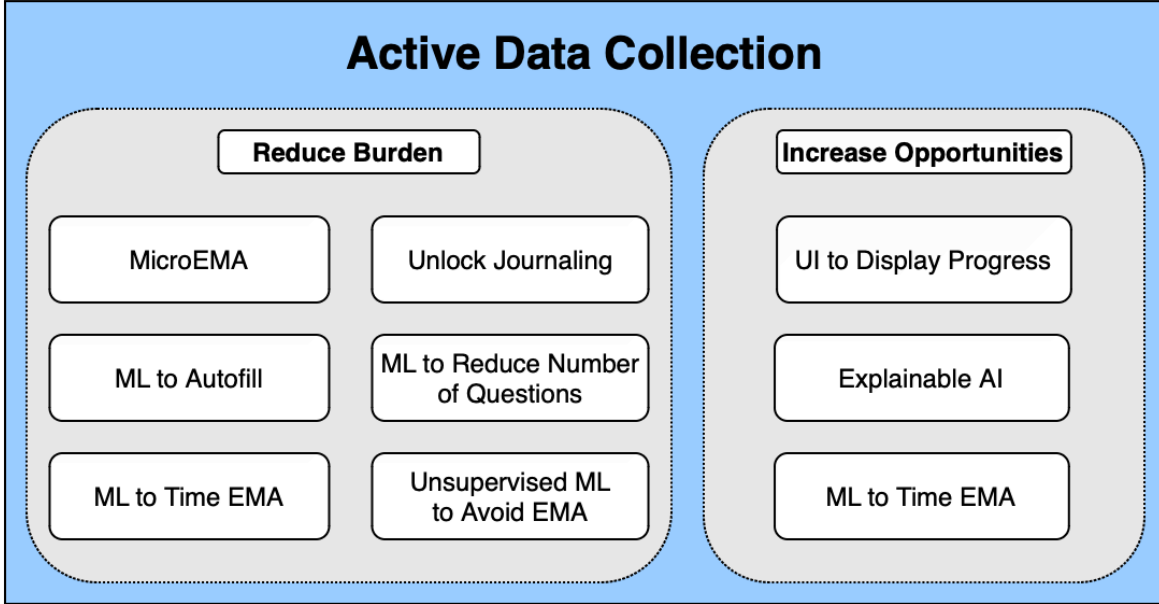


Figure 2.8: **Design Guidelines for Active Sensing.** This list highlights various strategies to reduce the burden of active data collection and increase opportunities for data provision.

their assessments presents another promising avenue. ML has been employed to optimize the timing of EMA prompts for convenience [66], select appropriate EMA questions [68], autofill responses [67], or even determine when additional active data would be unnecessary [69].

In addition to reducing participant burden, mobile sensing frameworks can enhance data collection opportunities. For example, EMA progress displays in the user interface were incorporated in several studies. Using explainable AI [22] can motivate participants to contribute more data. Timing EMAs to coincide with more convenient moments [50] or when the AI requires additional labels to boost confidence can also enhance the efficacy of adaptive intervention systems.

By answering RQ2, we also developed guidelines for developing effective passive data collection systems. During the collection phase, sensing applications can employ EMA to alert participants to assist in resuming data collection [60]. Several studies utilized the OS-provided health stores (Health App for iPhones and Health Connect for Android) to improve data availability and reliability. Feedback from machine learning algorithms can be used to deactivate sensors when data is unnecessary, thereby conserving battery life [21]. Post-collection, many studies typically exclude time frames with missing data from analysis. However, interpolation or machine learning techniques can be employed to estimate missing values [86], or alternatively, missing data can be utilized as an additional feature within machine learning models [55].

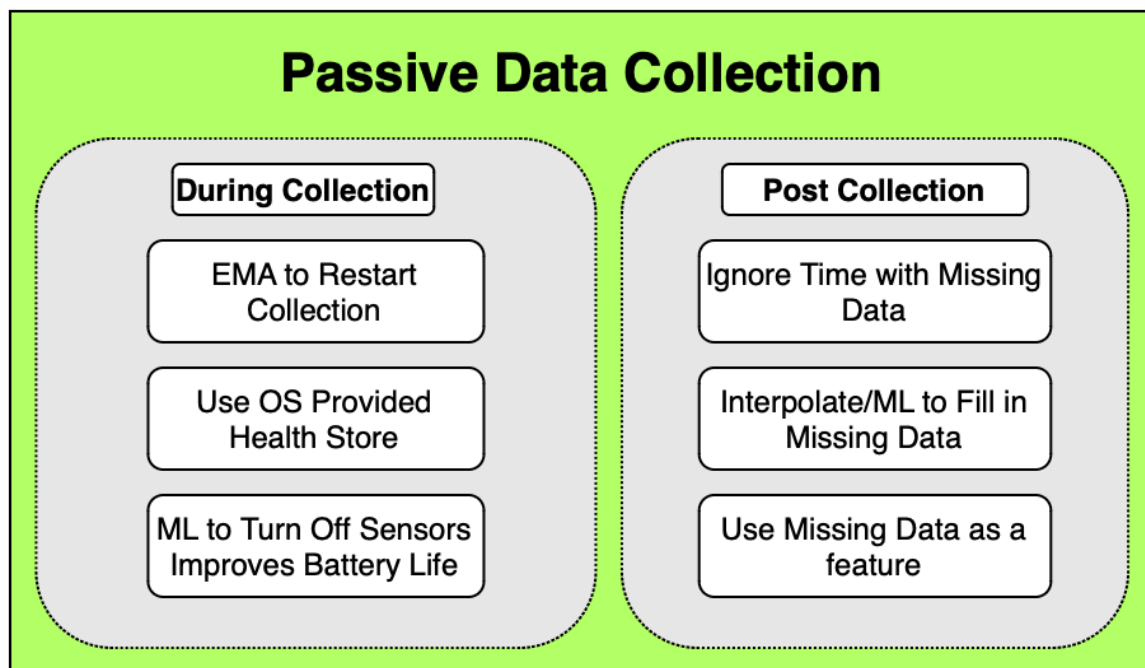


Figure 2.9: **Design guidelines for passive sensing.** This list outlines various strategies to enhance data collection rates during the acquisition phase and includes post-processing techniques to manage missing data during machine learning analysis.

2.5.3 Limitations

Although our search covered several databases and search engines, we focused on results that included active and passive sensing. Consequently, papers that only performed either active or passive sensing were excluded, even though they could have provided additional details and methods for mobile sensing.

We gleaned valuable insights from a few papers excluded from our search. Chang et al. [49], for example, did not include active data collection but used another device to provide truth labels for machine learning. Mishra et al. [91] and Morrison et al. [92] developed machine-learning models to determine the best time to interrupt users for active data collection, supporting just-in-time adaptive interventions (JITAI). Smets et al. [25] showed that accurately predicting stress and drug craving could be done with only GPS data. Torkamaan [50] also developed adaptive EMA timings to reduce participant burden and increase usability. Morshed et al. [93] developed an ML model to predict mood instead of performing EMAs.

2.6 Conclusions

This systematic review examined 774 publications integrating active and passive data collection to support machine learning for adaptive health interventions. After applying exclusion criteria, 77 studies were included in the final review. We analyzed trends across health dimensions, data collection devices, and active and passive data collection innovations. Our findings indicate that consumer smartphones and watches, in combination with machine learning, are beginning to reduce the reliance on specialized wearables and environmental sensors. However, significant challenges remain with active and passive data collection using consumer devices.

In active data collection, key challenges include reducing the EMA burden and increasing opportunities for participants to complete EMAs. These issues have been partially addressed through innovations in HCI and ML. For example, ML techniques have been applied to reduce the number of questions in EMAs, auto-fill responses, and time EMAs when participants are most likely to respond. Despite these advances, gaps remain in creating participant-friendly solutions that balance the need for frequent data collection with minimizing participant burden. Opportunities exist for further HCI innovations, such as persistent reminders, to enhance engagement, and for ML advancements, such as uncertainty-based active learning, to optimize the timing of EMAs for both model performance and user compliance.

Passive data collection also faces significant obstacles, particularly regarding consistent data capture on smartphones. Common issues include difficulties obtaining proper authorization to access private data, leading to data loss and challenges ensuring continuous background runtime for sensor data collection. These challenges reveal a gap in system-level solutions to maintain reliable data streams on consumer devices. Opportunities for improvement include developing HCI tools, such as contextual or event-driven prompts, to simplify authorization processes and leveraging persistent reminders to support passive data collection by maintaining background processes and signaling data continuity.

Overall, this review highlights key areas for innovation at the intersection of HCI and ML to address the identified challenges. Persistent reminders, improved EMA timing algorithms, and contextual prompting represent promising directions for advancing mobile health sensing systems to enable more effective adaptive health interventions.

CHAPTER 3

DESIGN GUIDELINES FOR IMPROVING MOBILE SENSING DATA COLLECTION: A PROSPECTIVE MIXED-METHODS STUDY

3.1 Introduction

Building on the findings of Chapter 2, which identified significant challenges and opportunities in active and passive data collection for mobile health sensing, Chapter 3 focuses on addressing these gaps through targeted innovations. Specifically, the systematic review highlighted the need for improved participant engagement in active data collection and consistent, reliable data streams in passive data collection. To address these issues, this chapter explores HCI methods, such as persistent reminders and contextual prompting, as well as strategies for improving authorization processes to support passive data collection. By implementing and evaluating these approaches, I aim to advance the practical application of mobile and ubiquitous devices in gathering high-quality health data for machine learning models.

Mobile and ubiquitous devices are valuable tools for gathering patient-generated health data in real-world settings [94, 95, 96, 97, 98, 99, 100, 101, 102, 3, 2]. Prior to the advent of the smartphone, mobile devices were predominantly used to collect data actively, usually through ecological momentary assessments (EMA). EMA involves “repeated sampling of subjects’ current behaviors and experiences in real-time in the subjects’ natural environment” [1, 103, 29]. EMA can include completing journals, diaries, and survey questions [104], providing audio/video samples [105, 106], or participating in interactive tests [107]. As mobile devices evolved to include more sensors and access to health data, mobile applications started to use passive data collection methods [24, 108, 109, 110], where mobile devices collect data without involving the end user. Today, machine learning models often use passively recorded sensor data streams as inputs to machine learning models that predict health outcomes or events captured through EMA [111, 112, 4, 5, 113, 114, 6] (see Figure 3.1).

Despite the growth of mobile data collection in health research, several challenges for mobile sensing on consumer mobile devices persist, as discussed in Chapter 2. These challenges are due to implementation decisions made by the developers of major mobile devices (i.e., iOS/iPhone and Android) to protect users’ privacy, preserve battery life, and minimize distractions. Passive data collection requires access to private user data, whereas active data collection needs to interrupt users to initiate an EMA. In this work, we studied how various user interface decisions for obtaining authorization at the application level affect the success of both active and passive mobile sensing. Specifically, we tested the following authorization scenarios: (1) users explicitly granting authorization to receive notifications to initiate EMA, (2) users explicitly granting authorization

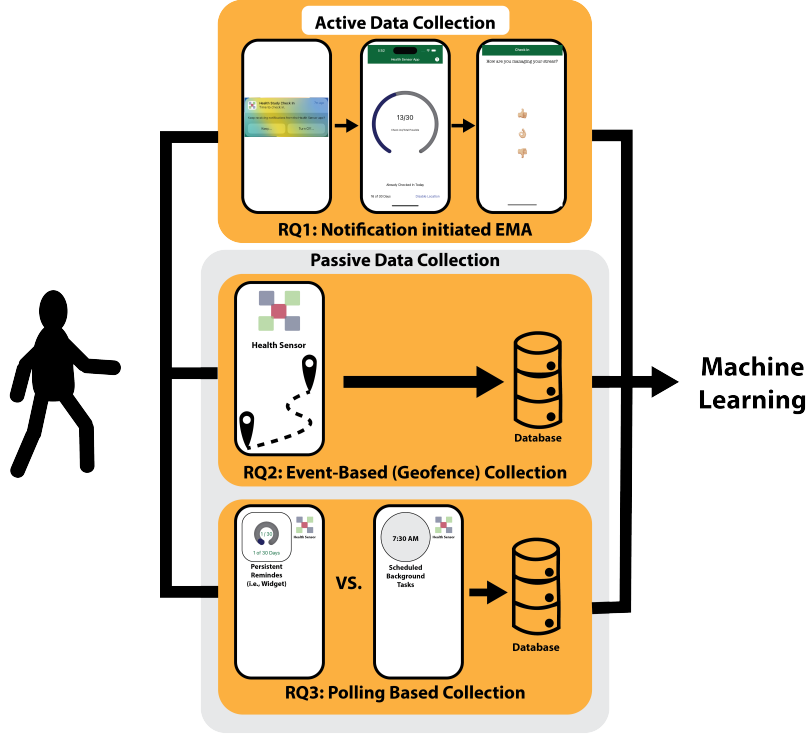


Figure 3.1: **Passive and Active Data Collection to Support Machine Learning.** Machine learning models often use passively recorded sensor streams as inputs to predict outcomes captured through EMAs. We explore the feasibility of collecting both passive and active data on consumer mobile devices to answer the following research questions: (RQ1) How do contextual prompting and setup prompting affect scheduled notification delivery and the response rate of notification-initiated EMAs? (RQ2) Which authorization paradigm, setup or contextual prompting, is more successful in leading users to grant authorization to receive background events? (RQ3) Which polling-based method, persistent reminders or scheduled background tasks, completes more background sessions?

to access background events, and (3) the system implicitly granting background runtime to collect data through polling.

The specific authorization procedure for the first two scenarios varies depending on the device. An Android device obtains authorization through setup prompting, where the user is prompted during the initial application launch. On the other hand, iOS devices use contextual prompting, where the user is prompted when the first event or notification occurs.

Background runtime for passive sensing is often obtained implicitly. Android and iOS systems determine when to run background tasks based on user actions and battery status. Another way to obtain background runtime is through a persistent reminder. A persistent reminder is a permanent UI (user interface) feature, like a home screen widget or persistent notification, that receives background runtime to update its UI.

To explore these active and passive sensing implementation scenarios, we developed mobile sensing applications for Android and iOS devices that logged both passive and active data. We tested our applications with a user study to answer the following research questions (see Figure 3.1):

RQ1: How do contextual prompting and setup prompting affect scheduled notification delivery and the response rate of notification-initiated EMA? We hypothesize that contextual prompting will lead to better EMA compliance because participants will not have to respond to a setup prompt and will have more context when authorizing notifications.

RQ2: Which authorization paradigm, setup or contextual prompting, is more successful in leading users to grant authorization to receive background events? We hypothesize that the contextual prompts will improve background event authorization because the added context will help participants understand and feel safe approving the permission prompt.

RQ3: Which polling-based method, persistent reminders or scheduled background tasks, completes more background sessions? We hypothesize that persistent reminders will be allowed to poll for data more often than background tasks because the mobile operating system (OS) is willing to expend resources to keep the UI up-to-date.

3.2 Methods

This section is organized as follows. First, we introduce the authorization procedures that can affect the success of mobile sensing studies. Next, we describe how we used these procedures in our mobile sensing application. Finally, we describe a user study we performed to answer our research questions. A summary of the authorization methods used for each research question is found in Table 3.1.

Table 3.1: **Summary of authorization methods used in each research question.** RQ1 compares authorization methods for notification-initiated EMA. RQ2 compares authorization methods for event-driven data collection. RQ3 compares polling-based collection methods.

	Condition	Authorization Method
RQ1	Android	Setup Prompting for Notifications
	iOS	Contextual Prompting for Notifications
RQ2	Android	Setup Prompting for Background Events
	iOS	Contextual Prompting for Background Events
RQ3	Background Tasks	User Actions, Device Status Implies Consent
	Persistent Reminders	Installation Implies Consent

3.2.1 Authorization Procedures

For both Android and iOS, mobile applications must gain explicit authorization before sending notifications or receiving events. Explicit authorization is when a user grants authorization through a permission prompt. At the time of writing, Android devices use setup prompting for explicit authorization, where the user is prompted to grant authorization during setup. The setup prompt can be displayed during the initial launch or when the feature requiring authorization is enabled. iOS devices, on the other hand, use contextual prompting for explicit authorization, where the user is prompted when the first event or notification occurs.

Contextual prompting provides the user with more context to enable a more informed decision. For example, with setup prompting, the user is asked for background location access after consenting, not knowing when or where the location will be accessed. With contextual prompting, the user is prompted when the location is first accessed, helping them understand when their location is accessed. Providing this additional context could help the users feel more comfortable with granting permissions, knowing that the location will be accessed only at a specific location. With setup prompting, on the other hand, we hypothesize that the user might be overwhelmed with several prompts and deny the request for location or notifications.

While explicit authorization procedures are required for notifications, implicit authorization is needed to grant the background runtime for sensing applications to automatically collect sensor and usage data. Implicit authorization is when the mobile OS implies authorization based on user actions and device status. Gaining background runtime for scheduled tasks depends on user actions, such as application usage, swiping an app out of the recent app switcher, and enabling power-saving mode. Device status indicators such as the battery level, charging state, and network connectivity are additional factors determining when background runtime is granted. iOS devices always use implicit authorization to determine when to grant runtime. Android devices vary by model and version. However, modern devices include the “adaptive battery” [115] feature, which implicitly grants background runtime.

Installing a permanent UI feature, or persistent reminder, is another way to gain background runtime through implicit authorization. A persistent reminder is part of an application that is always displayed on the mobile device’s home screen, lock screen, or notification center. A persistent reminder gains background runtime to keep its UI updated. In this case, the installation of the persistent reminder implicitly grants authorization to run in the background. The persistent reminder can also remind the participant to complete EMA tasks.

3.2.2 Mobile Sensing Applications

We developed a mobile sensing application for iOS and Android devices designed to collect active data through EMA and passive data through polling-based and event-based data collection. The native languages of each OS, Swift for iOS and Kotlin for Android, were used for development.

Besides native UI differences and authorization procedures outlined above, iOS and Android applications were designed to appear and function identically. The applications featured a home screen widget displaying the study's progress and serving as a persistent reminder. The main screen, EMA screen, and home screen widget are shown in Figure 3.2.

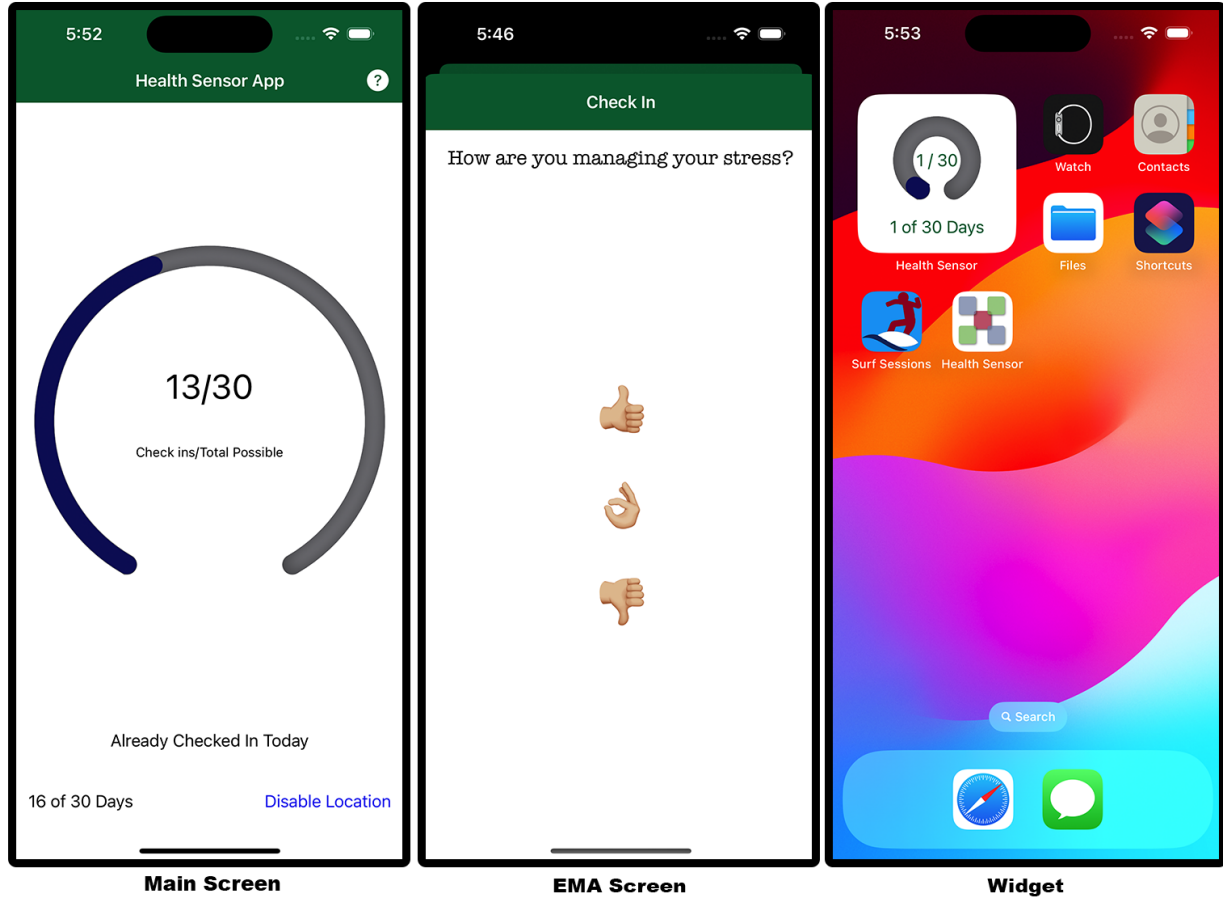


Figure 3.2: **Mobile sensing application.** Left: The main screen displays study progress. Middle: The EMA screen, which asks participants about their stress levels. Participants were asked to complete one assessment per day. Right: The home screen widget is used as a persistent reminder.

3.2.3 Development Process

We first developed the iOS application and then developed the Android application to match the look and functionality of the iOS application. We tested the applications simultaneously to ensure that Android and iOS devices reported the same data and functioned similarly. We ensured they reported the same number of location events, sent notifications at the same time, and otherwise behaved similarly.

The authorization methods we used in this study reflect each mobile OS’ preferred method for gaining authorization. At the time of writing, Android devices do not support contextual prompts for notifications or background events. iOS does not directly support setup prompting for background events and discourages setup prompting for notifications. We choose the OS’ preferred authorization method because users should be familiar with the procedure. Figure B.1, located in the appendix, highlights the authorization differences.

We originally planned to upload all data to secure storage through the background task. However, during testing, we noticed an inconsistency in uploading the data using only the background task. Thus, we also synced data whenever the application was launched in the foreground. To ensure all data was collected, the collection was confirmed before presenting a “Study Finished” screen along with instructions to take a screenshot.

3.2.4 RQ1: Notification-Initiated EMA

RQ1 compares setup and contextual prompting for notification authorization to measure their effect on EMA compliance. We compared EMA compliance using the iOS notification system, which uses contextual prompting, against Android’s notification system, which uses setup prompting. To answer RQ1, the applications used notification-initiated EMA to collect active data. We performed a simple single-question EMA once per day, for which both Android and iOS applications reminded users to complete their EMA through a notification. The Android setup prompt and iOS contextual prompt are displayed in Figure 3.3.

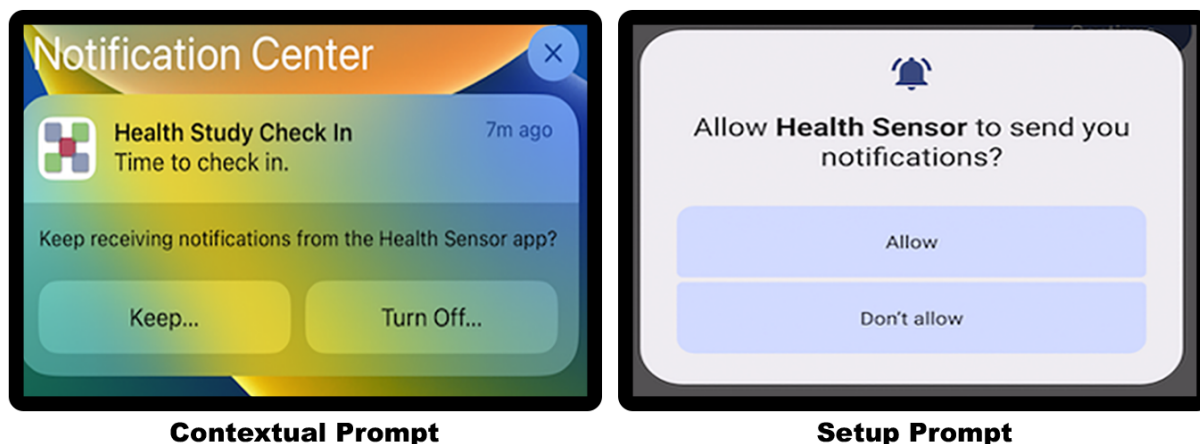


Figure 3.3: **Notification permission prompts.** Left: Contextual prompt on an iOS device. Right: Setup Prompt on an Android Device. With setup prompting, users are asked to approve notifications during the initial run of the application. Using contextual prompting, users are asked to approve notifications when the first notification arrives.

Notification systems on Android and iOS have other features that could affect compliance. iOS

15 introduced “Focus” times [89], which implements notification deferral, and notification summaries [13]. Android devices also implement notification deferral through a focus mode, pausing notifications from selected applications when activated. These features might reduce EMA compliance [15, 116].

3.2.5 RQ2: Event-Based Passive Data Collection

RQ2 explores how setup and contextual prompting affect the success of event-based passive data collection. In event-based collection, an event, such as a location change, phone call, message, or health alert, initiates the data collection. Sensing applications need explicit authorization to receive background events. Our applications used location changes as the event of interest and monitored a geofence, or circular region, where participants frequently entered and exited, triggering a location event. iOS devices used contextual prompting to authorize receipt of background location events, and Android devices used setup prompting.

3.2.6 RQ3: Polling-Based Passive Data Collection

RQ3 explores the ability of traditional background tasks and persistent reminders to collect data passively. Both Android and iOS applications implemented and scheduled a daily background task. The background task logged the time and uploaded all logs to cloud storage. We also uploaded the data upon application launch to ensure that all data was collected. For persistent reminders, we implemented a home screen widget, shown in Figure 3.2, and requested a daily refresh, which was also logged.

We explored users’ willingness to install the persistent reminder. Each participant had an equal chance of being assigned to either a control group that did not receive prompts or notifications to install the widget or an experimental group that did receive prompts and notifications. One group of participants was provided verbal instructions for installing the widget.

3.2.7 Alternative Study Designs

We used the mobile OS’ default or preferred authorization procedures to test their effectiveness in the wild, similar to other studies that test the feasibility of cross-platform mobile sensing [117, 30, 118]. The strength of this approach is that it does not limit the study to users of one particular device (iOS or Android), increasing the diversity of the participants. It also works to understand mobile sensing in the wild, revealing insights into potential pitfalls while developing cross-platform mobile sensing frameworks. However, this limits our ability to isolate study variables because other differences between the mobile OS implementations or device hardware could impact our results.

Studies focusing on a single mobile OS sometimes sacrifice user diversity in favor of isolating variables. These studies are favored when testing new features [119, 120] or focusing on specific

variables [121]. Android tends to be used in these studies due to its more open API.

3.2.8 User Study

We tested our sensing applications through a user study that aimed to identify the impact of screen time and study time on college students' stress levels. Participants installed our applications on their personal mobile devices and participated in our study for 30 days. To test event-based collection, a geofence that contained the school library and classrooms was used to calculate study time. Background tasks purported to collect screen time statistics to test polling-based collection.

Active data was collected through a simple EMA that asked participants how they managed their stress levels, as shown in Figure 3.2. A notification was sent each morning at 7:30 AM to remind users to complete their EMA. Students could complete their daily EMA until the end of the day (midnight). Participants could respond with a thumbs up, thumbs down, or neutral.

3.2.9 Ethical Considerations

The University of Hawaii Institutional Review Board (IRB) approved the study under protocol #2022-0722, and the Brigham Young University - Hawaii Institutional Review Board (IRB) approved it under protocol #22-72. All participants consented to participate in the study through the sensing application and received extra credit in their courses for participating.

The following measures were implemented to ensure user privacy and participant safety: 1) After installation, the mobile application required the participants to electronically consent before collecting any data. 2) Students who did not want to participate were given an alternative extra credit assignment, representing the same time commitment as completing the study. 3) The application anonymized all data collected before being uploaded to cloud storage, so the instructors could not identify the participants' data, including their study habits. 4) Follow-up paper surveys were collected by a volunteer, not the class instructor. 5) At the end of the study, students were instructed by the application to submit a screenshot to their courses' LMS, and all students were provided with the same amount of extra credit regardless of EMA compliance or the amount of data collected.

3.2.10 Recruitment

We recruited 145 college students in Computer Science, Information Technology, Business, and Science classes at BYU-Hawaii. The students were offered extra credit incentives to install our application on their mobile devices and actively engage in the study for 30 days.

To ensure students that their professors could not identify their study habits, specific demographic information was not collected, and the application anonymized all data before uploading logs to cloud storage. However, the general demographics of the recruitment base included college

students aged 18-26, with around 60% female and 40% male. Students represented a variety of races and cultures, with a distribution of about 40% Caucasian/non-Hispanic, 15% Native Hawaiian and Other Pacific Islanders, 25% Asian, 15% two or more races, and 5% other.

3.2.11 Exit Survey

Upon completion of the study, a follow-up survey was provided to participants, which 48 participants completed. The survey was provided on paper during the course’s final exams. To ensure privacy, a volunteer, not the instructor, distributed and collected the survey from participants. To preserve privacy, we did not correlate the survey with the data collected by the mobile application. The survey allowed us to gather qualitative insights regarding the differing success rates across the authorization paradigms. The participants were asked: 1. what reminded them to complete their assessment, 2. their general thoughts on granting background location access, 3. whether they knew about and installed the home screen widget and their thoughts on installing a widget, and 4. suggestions to improve the application.

3.3 Results

3.3.1 Data Cleaning and Analysis

We initially logged 178 (126 iOS, 52 Android) users who installed the app and consented to participate in the study. We removed users from the study if the logs showed they participated for less than five days, which meant that they deleted the application less than five days after starting. We did not detect any users who deleted the application after participating for five days. This yielded 145 participants, representing 105 iOS and 40 Android users. Each participant had an equal chance of being assigned to the widget or control group by the application. 52 individuals (34 iOS and 18 Android) were randomly assigned to the widget group, where the application prompted them to install the widget. 82 participants were randomly assigned to the control group, which did not receive the prompts. 11 participants (8 iOS, 3 Android) received verbal instructions to install the widget on their home screens.

To conduct our qualitative analysis, we coded responses to the survey based on general patterns that emerged in at least five responses. The questions and the number of themed responses are listed in Table 3.2.

3.3.2 RQ1: Notification-Initiated EMA

To answer RQ1, we compared notification authorization procedures: contextual prompting implemented in iOS versus setup prompting implemented in Android by measuring the number of completed EMAs. On average, participants across all conditions completed 23.4 EMAs out of 30,

Table 3.2: **Summary of exit survey responses for each question, separated by device.** At the end of the study, participants were administered a paper survey to complete during their final exam, of which 48 participants (33 iOS, 15 Android) completed the survey. Responses were coded based on general themes that emerged in at least five responses. Themes with less than five responses are not listed.

	iOS (33 Total)	Android (15 Total)
Question 1: What reminded you to check in?		
Notifications	12	9
Set alarm	9	0
Widget	5	0
Saw application	5	3
Question 2: Did you allow the app to use your location? Why or why not?		
Yes	25	15
No	2 (1 worried about tracking)	0
It would help the study	7	6
It was required	4	3
Safe	4	1
Question 3: Did you install the widget? Why or why not?		
Yes	15	2
No	15	13
Didn't know about it	7	4
Helped me remember to check-in	6	1
Didn't want to change homescreen/don't use widgets	4	2
Question 4: Do you have any comments or suggestions on improving the app?		
Better notifications	15	3
Select notification time	4	2

with a standard deviation of 7.38. We observed no statistically significant difference in the number of completed assessments when examining device-specific differences (see Figure 3.4) as shown by an ANOVA test ($F(1,144) = 0.0227$, $P = .88$) or when comparing the widget group to the control group ($F(1,144) = 1.33$, $P = .268$).

Although our quantitative results demonstrated no difference between devices, our qualitative results indicated that contextual prompting, notification deferral, and notification summaries on iOS devices affected notification delivery. Nine Android users reported that notifications were the primary method to remind them to complete their EMA. Despite 9 iOS users stating that the notifications helped them remember to complete the task, 15 iOS users surveyed said the notifications did not appear consistently. Nine participants even set their own alarms or reminders: as stated by P32, “Notifications weren’t working, so I had an event in my calendar to remind me.”

3.3.3 RQ2: Event-Driven Passive Data Collection Results

In RQ2, we compared the effectiveness of contextual prompting (iOS) and setup prompting (Android) in gaining authorization to access background location events. 27.5% of Android and 48.6%

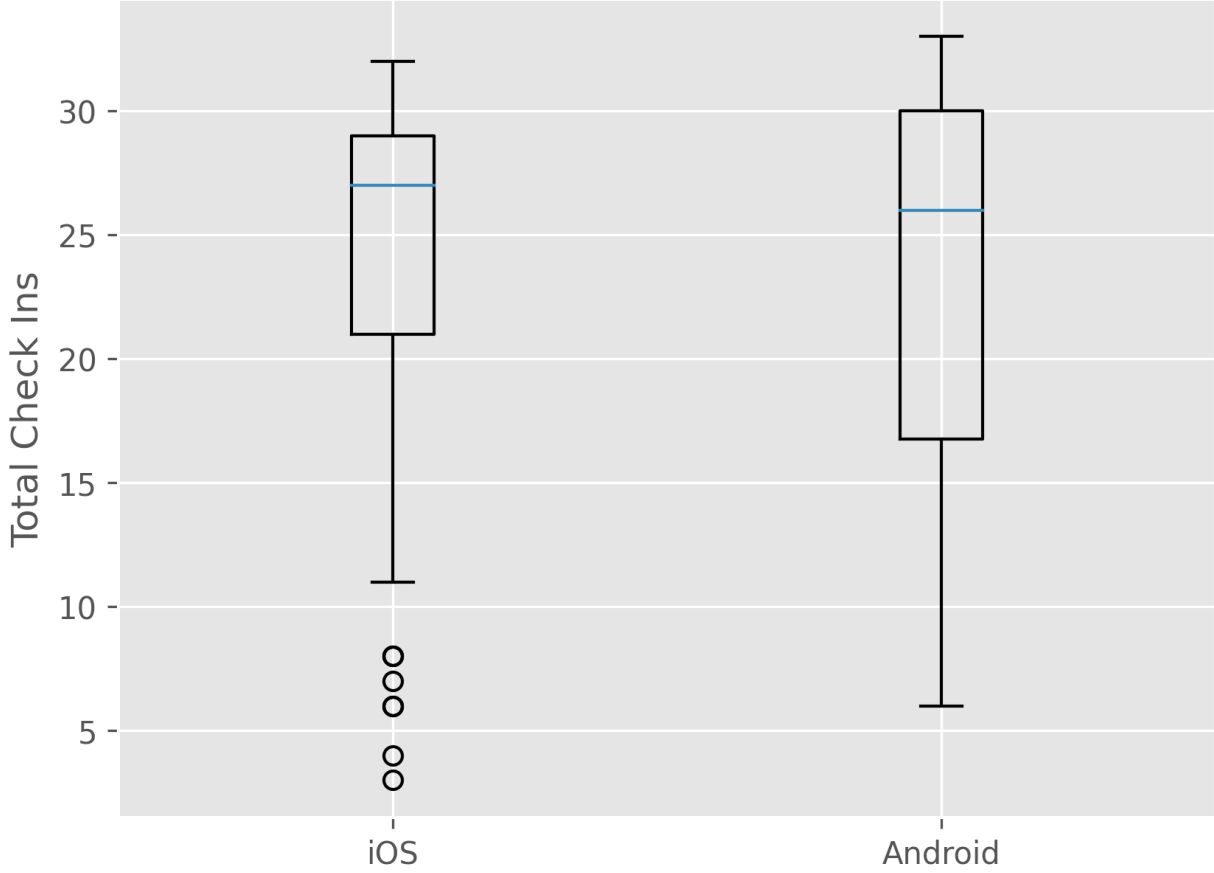


Figure 3.4: **Number of EMAs completed by device (126 iOS, 52 Android).** There was no statistical difference between devices in the total number of assessments completed. Contextual prompting and other notification system differences in iOS and Android did not affect the compliance rate. However, qualitative results revealed difficulties with contextual prompting on iOS devices.

of iOS users enabled background location permission. Contextual and setup prompting differed significantly ($\chi^2=4.42$, $df=1$, $P \leq .04$). Thus, contextual prompting was 55.5% more effective than setup prompting in gaining authorization for background location events.

Our post-study survey asked participants about their willingness to share their location with the sensing application. 40 surveyed participants said they authorized background location access vs. two participants who reported not sharing their location. Only one participant, P10 (iOS), expressed apprehension about sharing their location, stating, “I was afraid it could track me.” However, 13 participants surveyed expressed a willingness to share their location to contribute to the study’s objectives and thought they granted the needed authorization to access background location events. P46 (Android) exemplified this sentiment: “Yes, if it helps the study then I don’t really mind if it has my location.” Users’ willingness to share location data for non-research reasons

will most likely vary, so we emphasize that these results only apply to the context of research studies.

3.3.4 RQ3: Background Tasks and Persistent Reminders

RQ3 compared traditional background tasks to persistent reminders for polling-based data collection. First, we present our results on how willing users were to install the persistent reminder, and then we present results comparing persistent reminder refreshes to background tasks completed. Finally, we compare background tasks completed by device type and present our qualitative results.

We studied how willing participants were to install the persistent reminder by randomly assigning users to an experimental group where they were notified by the application to install the widget and a control group that did not receive such prompts. In addition, a subgroup of 11 users was given verbal instructions to install the widget. Of 52 participants in the widget group, only 18 participants installed the home screen widget, 16 being iOS users and 2 being Android users, as shown in Table 3.3. One iOS participant in the control group installed the widget. Of the 11 participants who received verbal instructions to install the widget, five complied accordingly (45.5%), of whom 4 were iOS users and 1 was an Android user.

Table 3.3: **The number of users who installed the persistent reminder by group.** The control group did not receive any prompts to install the widget. In the widget group, the application requested that the users install the widget. Participants were verbally instructed to install the widget in the final group.

Group	iOS (Installed/Total)	Android (Installed/Total)
Control	1/63 (1.6%)	0/19 (0%)
Widget	12/34 (35.3%)	1/18 (5.6%)
Verbal Instructions	4/8 (50%)	1/3 (33.3%)

Limiting our data to only users who installed the persistent reminder, on average, the persistent reminder was refreshed 61.2 times throughout the study with a standard deviation of 16.7, and the devices performed an average of 7.2 background tasks with a standard deviation of 12.5. The two Android devices performed 23 and 42 widget refreshes, with 8 and 6 background tasks, respectively. When users installed the widget, persistent reminders refreshed 266.5% more than background tasks were executed (see Figure 3.5).

Focusing on the difference between Android and iOS, we also observed a significant difference in background tasks completed when comparing devices. Using data from all participants, an ANOVA test revealed a significant difference in the total number of background sessions between iOS devices and Android devices ($F(1,144) = 37.8$, $P < .001$), showing that Android devices were more permissive in granting background runtime. Most iOS devices completed fewer than ten background sessions, and many did not execute a single background task. In addition, almost half of the Android devices did not complete a daily background task. This demonstrates our difficulty

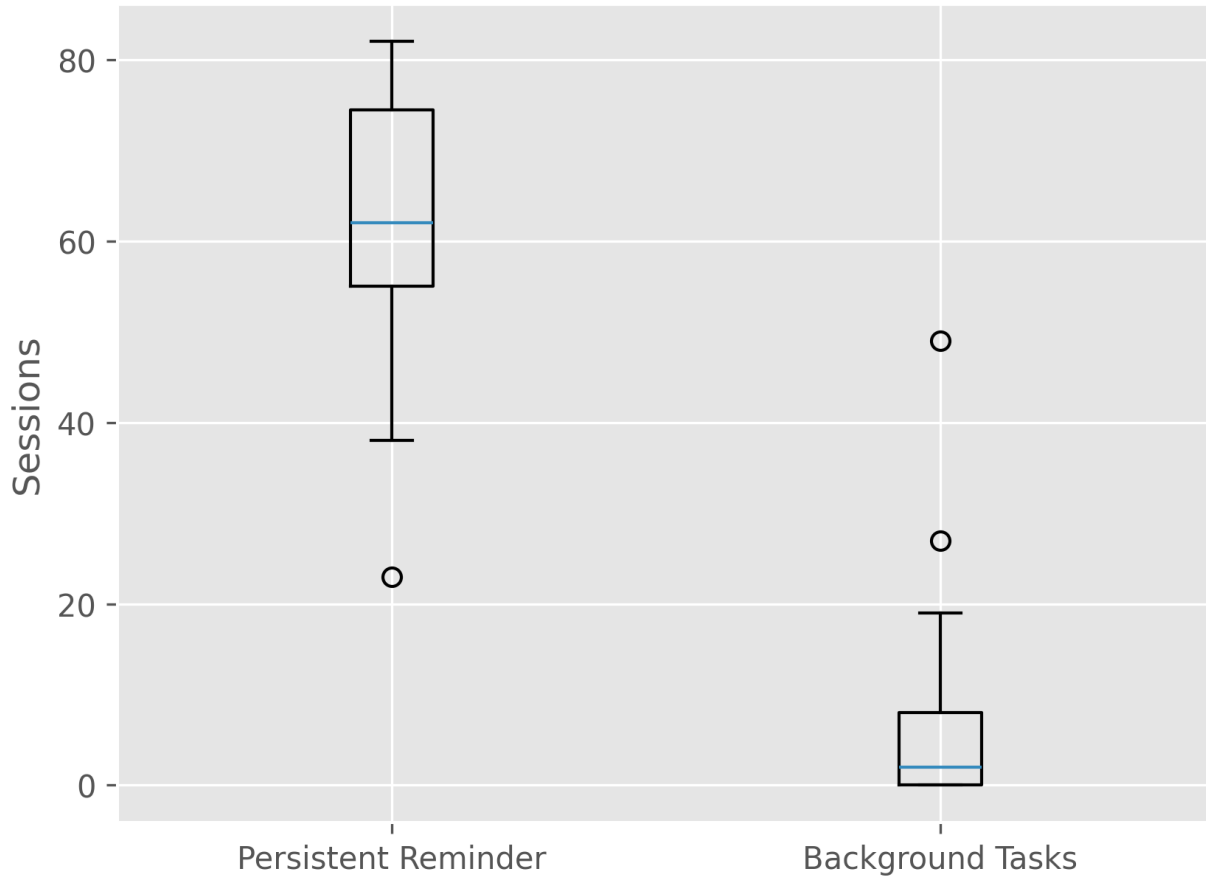


Figure 3.5: **Total polling-based data collection sessions, grouped by persistent reminders and background tasks.** Only data from participants who installed the widget are included (N=17 for both groups). This graph compares how often the application completed a passive data collection session if the participant installed the widget. Persistent reminders are given more opportunities to run in the background.

in consistently gaining background runtime, especially on iOS devices. Figure 3.6 shows the total number of background sessions conducted by device type.

Our qualitative analysis showed mixed results regarding the installation of the home screen widget. Several participants indicated that the widget helped them remember to complete their daily assessment, as expressed by seven participants. For example, P37 (iOS) stated, “The widget helped me to remember to check in every day and what the study was about.” However, six users opted not to utilize the widget due to concerns about modifying their home screen or preferring not to use widgets. P22 (iOS) mentioned, “My home screen was already full and organized,” while P35 (Android) stated, “I just don’t want to change my home screen.” iOS users may be more willing to add the widget to their home screen because iOS allows for widget stacking, a feature not found in most Android devices. This functionality enables the rotation of different widgets within a stack,

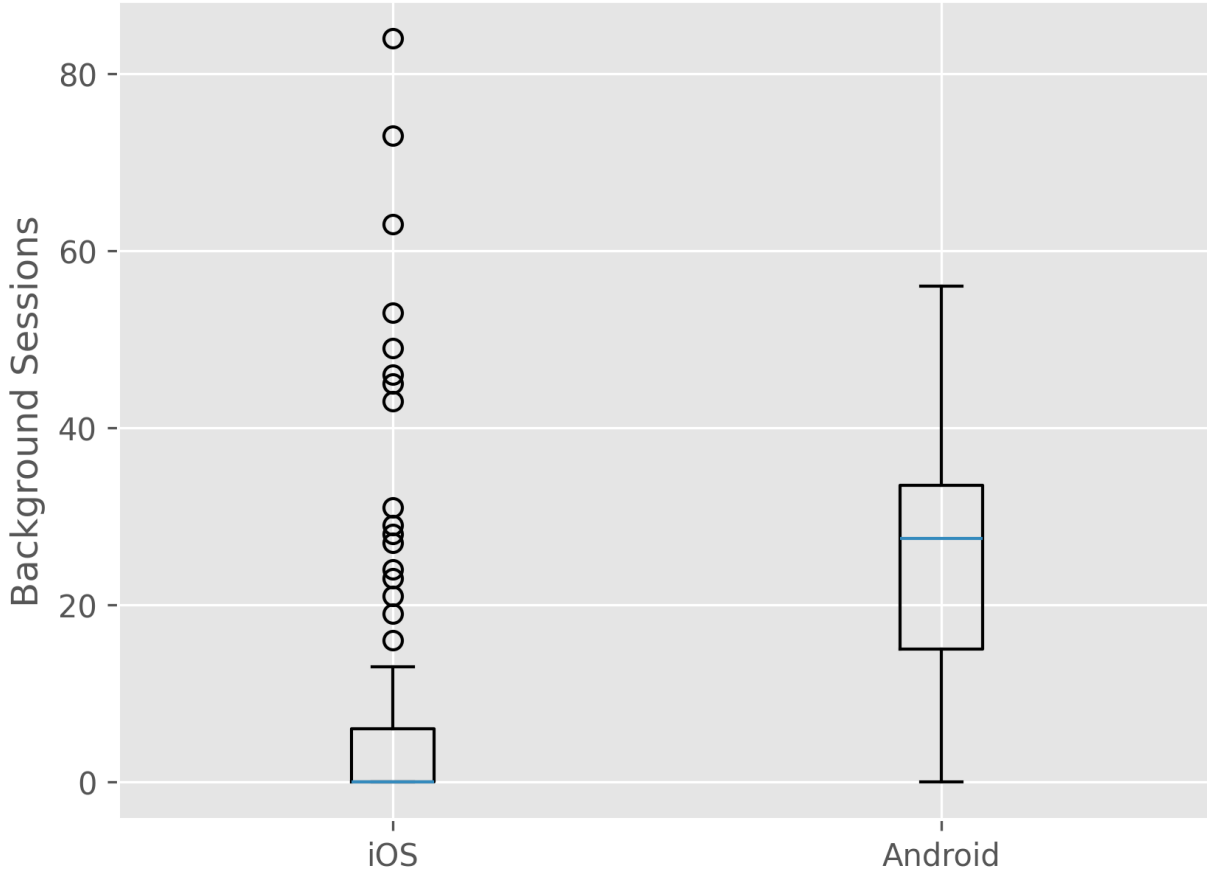


Figure 3.6: **Total number of background tasks completed, grouped by device (126 iOS, 52 Android).** Both Android and iOS devices requested background runtime once per day. About half of the iOS devices did not complete a single background task, and very few completed a task once per day. Half of Android devices completed less than one background task a day.

eliminating the need to reorganize the home screen to install the widget.

3.3.5 Effects of Stress Level

Because students experiencing a high-stress level might be more motivated to authorize data collection and complete assessments, we divided the participants into two groups: a low-stress group (N=117) whose EMA responses averaged better than neutral and a high-stress group (N=28) whose EMA responses averaged lower than neutral. We then compared the two groups' authorization and data collection rates. There was no statistical difference between the groups in the number of EMA responses ($F(1,144) = 0.928$, $P = .34$), the number of background tasks executed ($F(1,144) = 1.54$, $P = 0.217$), the number of participants who enabled location ($X^2=0.42$, $df=1$, $P = .52$), or whether they installed the widget ($X^2=0.53$, $df=1$, $P = .57$). The stress level of the participants did not

significantly impact their willingness to grant authorization, provide EMA responses, or install the widget.

3.4 Discussion

We present design principles to enhance the success of mobile passive-sensing research studies.

3.4.1 Notification-Initiated EMA

Setup prompts should be used for notifications. Although our quantitative results showed no difference between the iOS and Android notification systems, our qualitative analysis revealed that many iOS users did not consistently receive notifications to remind them to complete their EMA. Nine iOS users reported setting their own alarms or calendar reminders to remember to complete their EMA. We conjecture that if our study had required participants to complete multiple tasks per day or to follow a strict time frame for EMA tasks, then the EMA compliance on iOS devices would have suffered.

We hypothesized that contextual prompts would work more effectively. However, they lose their context when notification deferrals and summaries are enabled. Only users who searched through their notification summaries would have found the contextual prompt and authorized future notifications. With setup prompting, by contrast, users authorize the notifications immediately after consenting.

Our qualitative results show that notification deferral and notification summaries can cause delayed and missed responses to EMA notifications on iOS devices. New methods should be developed to help participants remember to complete their assessments in a timely manner. iOS allows developers to mark notifications as time-sensitive, which increases their visibility to the user and should be used for time-sensitive EMAs. Our study did not use time-sensitive notifications because our assessment did not require a time-sensitive response. Future work could analyze the effectiveness of time-sensitive notifications vs. basic notifications. However, even time-sensitive notifications could be delayed or ignored based on the user’s settings. We recommend that mobile sensing researchers review notification settings during onboarding. Persistent reminders offer an additional way to remind users to complete tasks. Seven participants (6 iOS and 1 Android) reported that the persistent reminder helped them to remember to complete their assessment and could be an area to explore further.

Besides notification deferral, notification summaries, and contextual prompting, other notification system advances can potentially disrupt notification-initiated EMA. Because notifications disrupt users, sometimes causing stress [122, 123, 124], several modifications to notification systems could be introduced to consumer mobile devices. Lin et al. [14] demonstrated how notification summaries could be improved by letting users determine the order of the notifications. Pejovic et

al. [125] explored user contexts to understand user interruptability, leading to the development of intelligent notification systems [15, 16]. Kandappu et al. [126] also explored intelligently interrupting users. Mobile sensing applications must adjust to these advanced notification systems or find a different way to initiate EMA. For example, Zhang et al. [23] explored using a persistent reminder on the lock screen to initiate EMA.

3.4.2 Event-Driven Passive Data Collection

For event-driven data collection, use carefully designed contextual prompts. Our qualitative analysis revealed that most participants were willing to authorize background location access for research studies and were under the impression that they enabled background location access. However, our quantitative results revealed that many participants did not authorize background location access. Many failed to grant authorization for background events even though they intended to authorize it. As we hypothesized, contextual prompting on iOS was 55.5% more effective in gaining authorization for background location events. This coincides with previous work that shows additional context helps users make better privacy decisions [127, 128, 119].

Even with contextual prompts, less than half of the participants authorized access to background location events. Challenges arise when users misinterpret authorization prompts, limiting the success of event-based collection [121, 128, 129, 119, 130]. To improve contextual prompt accuracy, Wijesekera et al. [120] developed a machine learning-based, context-aware permission model that improves the permission accuracy rate above that of contextual prompts, which should be considered if it becomes available on consumer mobile devices. However, the current model suggests providing “generic but well-formed data” when an app is denied access. In the context of health sensing apps, this can skew results and create adverse health interventions. We recommend that such systems be designed to communicate to the app about denials and that apps be constructed to handle denials to ensure data fidelity. Further research must be performed to balance the collection abilities of mobile sensing applications and protect the privacy and security of their users [34].

3.4.3 Polling-Based Passive Data Collection

For polling, implement a persistent reminder that can be used both as a means of collecting data and as a reminder to complete EMA tasks. Collecting data through background tasks presented significant challenges due to the use of an implied consent model for granting background runtime that is inherent to mobile OS, as described in previous work [30, 32]. Although Android devices are generally more permissive, too many factors are under consideration to consistently guarantee background runtime. However, as we hypothesized, using a persistent reminder as a secondary means to poll for data yielded more successful data collection. Installing a persistent reminder on the user’s home screen signifies to the OS the intent to allocate the necessary resources to maintain the reminder’s regular updates, which can also be used for passive data collection.

Many users did not comply with installing the widget when prompted by the app, especially among Android users. Verbal prompts to install the widget did help to improve compliance on Android and iOS. Our qualitative analysis indicated some resistance to installing the widget because users did not want to change their home screen layout, and some were unfamiliar with how to install widgets or did not use widgets.

Several methods are available to overcome the resistance to using widgets. First, participants who more directly benefit from the data collection might be more willing to install a widget. For instance, participants affected by a disease might be more willing to install a widget, especially if the application offers just-in-time, adaptive interventions. In addition, study incentives can also be increased for participants who install the widget. To assist users unfamiliar with widgets, the application can provide a video tutorial on installing the widget, or instructions can be provided during study onboarding.

3.4.4 Limitations

There are other differences between iOS and Android devices and users of those devices than what we tested for, which could be confounding factors. In addition, we studied passive sensing for research studies, and the results do not necessarily apply to passive sensing in other contexts, such as crowdsourcing or commercial purposes.

This study involved college students, and while they represented diverse academic disciplines, races, and countries of origin, they were all aged 18-26 and tended to be more familiar with mobile devices. Further work should be conducted to observe how these results would translate to a larger, more heterogeneous population. Individuals less familiar with mobile devices, such as geriatric patients, may exhibit more difficulty granting and declining authorization due to different privacy preferences or familiarity with smartphone technology.

For persistent reminders, only two Android users installed the widget, with one device refreshing more than once per day and the other refreshing 23 out of the 30 days. These preliminary results are promising. However, further work is required to ensure the results are consistent with a larger sample size.

3.4.5 Future Work

Additional modifications to the mobile applications could be implemented and tested in follow-up studies. Although Android does not allow contextual prompts, contextual prompts could be mimicked by randomly sending a notification asking for background location access or notification access. Such a study would eliminate potential confounding factors that could have influenced the results.

Our sensing applications could be used in additional health studies incorporating other populations with different motivations to comply with study procedures. Comparing our results with

studies involving more diverse health issues would be an interesting avenue for future work.

3.4.6 Comparison with Prior Work

Numerous studies have contributed to a comprehensive understanding of EMA compliance across research fields [28]. In a review of EMA studies, Wrzus et al. [131] found that compliance cannot be predicted by the number of assessments or length of the study but that financial incentives did improve compliance rates. Murray et al. [132] explored the role of participants' emotional states in affecting compliance. In the related field of crowdsourcing [133], gamification can improve response rates [134]. Other efforts to improve EMA compliance include Schneider et al.'s [135] work on just-in-time, adaptive EMA to reduce the burden on participants. Zhang et al. [23] explored unlock journaling, where users unlock their devices by completing an EMA. Our work builds on this by examining how notification permissions affect EMA compliance.

Mobile sensing depends upon gaining proper authorization to collect data and interrupting participants to initiate an EMA. Research into permissions on mobile devices has shown that users often misinterpret permission prompts [129, 130]. Wijesekera et al. [120, 119] showed that contextual prompts help users correctly interpret permission prompts. Alsoubai et al. [121] profiled users to help understand differing privacy strategies, which helps improve intelligent permission systems [120]. Our mobile sensing applications explored contextual prompts and permissions and their role in passive data collection.

Prior work has demonstrated that collecting consistent data across iOS and Android devices remains challenging [20]. Most mobile sensing studies use Android devices due to their more open API, but some work has been conducted to improve mobile sensing on iOS devices. The AWARE-iOS research team [32] explored background data collection methods on iOS and developed guidelines for sustainable data collection on iOS. AWARE-iOS has successfully collected passive data on iOS devices [33]. RADAR-base [136, 137] is an open-source mobile health platform for collecting and analyzing large-scale data from various devices, including passive and active mobile applications. We expand upon this inspirational prior work by adding design guidelines for consistent active and passive data collection across Android and iOS devices. Consistent data collection is needed to support just-in-time adaptive interventions on consumer mobile devices, necessitating further research [7].

3.5 Conclusions

We developed and tested mobile sensing applications for iOS and Android to answer our research questions: 1. How do contextual prompting and setup prompting affect scheduled notification delivery and the response rate of notification-initiated EMA? 2. Which authorization paradigm, setup or contextual prompting, is more successful in leading users to grant authorization to receive

background events? and 3. Which polling-based method, persistent reminders or scheduled background tasks, completes more background sessions? Although contextual prompts for notification authorization on iOS devices did not impact EMA compliance rates compared to setup prompts on Android devices, many iOS users reported not receiving notifications. For background event authorization, contextual prompts on iOS devices were 55.5% more effective in gaining authorization than setup prompts on Android devices. Finally, persistent reminders resulted in a completion of background sessions 266.5% more often than when using traditional background tasks. However, we observed some user resistance to installing persistent reminders. Although mobile sensing on consumer mobile devices continues to exhibit challenges, our results suggest that persistent reminders and proper authorization procedures can improve user compliance.

These findings build on the opportunities and challenges identified in Chapter 2, where gaps in consistent passive data collection and participant engagement were highlighted. By addressing these gaps, this study demonstrates the practical utility of HCI methods, such as persistent reminders and contextual prompting, in overcoming barriers to mobile sensing. The results provide a foundation for further exploration of adaptive approaches to mobile health sensing, particularly in developing strategies to balance user burden with the need for high-quality data collection, a theme further explored in Chapter 4.

3.5.1 Abbreviations

EMA: Ecological Momentary Assessment

OS: Operating System

CHAPTER 4

ADAPTIVE EMA TIMING TO REDUCE USER BURDEN AND INCREASE COMPLIANCE

4.1 Introduction

Chapter 2 identified several open challenges in mobile sensing on consumer devices. For active data collection, these challenges include minimizing participant burden while obtaining sufficient high-quality labels to support accurate ML models. Passive data collection presents complementary difficulties, particularly in maintaining consistent data streams and obtaining appropriate user authorization. Together, these challenges highlight opportunities at the intersection of HCI and ML [22].

Building on these findings, Chapter 3 examined persistent reminders and contextual prompting to improve both active and passive data collection. That study showed that persistent reminders increased the consistency of passive data collection, while contextual prompting was more effective than setup prompting for obtaining authorization for background tasks. These approaches also demonstrated potential benefits for participant engagement in active data collection. Despite these improvements, a key open challenge remains: how to reduce the burden imposed by frequent EMAs without compromising the quality of data needed for ML-based inference.

This chapter addresses that challenge by evaluating both HCI- and ML-driven methods that pursue two complementary goals: improving participant compliance and reducing the burden of answering EMAs. I conducted a study examining two approaches to EMA timing. The first is an HCI-driven, interaction-aware strategy, termed *unlock-triggered EMA*, in which EMA prompts are delivered shortly after participants unlock their phones. This builds on *unlock journaling*, which improves compliance by aligning brief self-reports with moments of natural device engagement, increasing *notification salience* [23]. Unlock-triggered EMA adopts a lighter-weight variant, presenting prompts shortly after an unlock without blocking device access, retaining the salience benefit while avoiding forced interaction. The second is an ML-driven approach that uses real-time active learning to time EMAs. Under active learning-timed EMA, prompts are selectively suppressed when model confidence is high, reducing unnecessary interruptions and lowering the number of labels the system must collect.

Active learning reduces labeling effort by choosing which samples to label rather than labeling at random. An initial model is trained on a small labeled set, and a selection algorithm then queries the samples expected to improve the model most, so the same labeling budget yields better performance than random labeling. Prior work has shown its potential for reducing EMA burden in stress monitoring, though usually through offline or simulation-based evaluation on previously collected datasets [138]. This study instead applies active learning prospectively, using model

confidence during deployment to decide when participant input is most informative, in a live closed-loop system that adapts as labels arrive (see Figure 4.1).

In the post-hoc analysis of model performance, I also examined whether personalizing the model to individual participants improved stress prediction. For this analysis, I used a generalized linear mixed model (GLMM), which keeps a shared population model but adds per-participant adjustments (*random effects*) that shift predictions toward each person’s own data, such as a *random intercept* capturing that person’s baseline stress level. Section 4.2 describes these models in more detail.

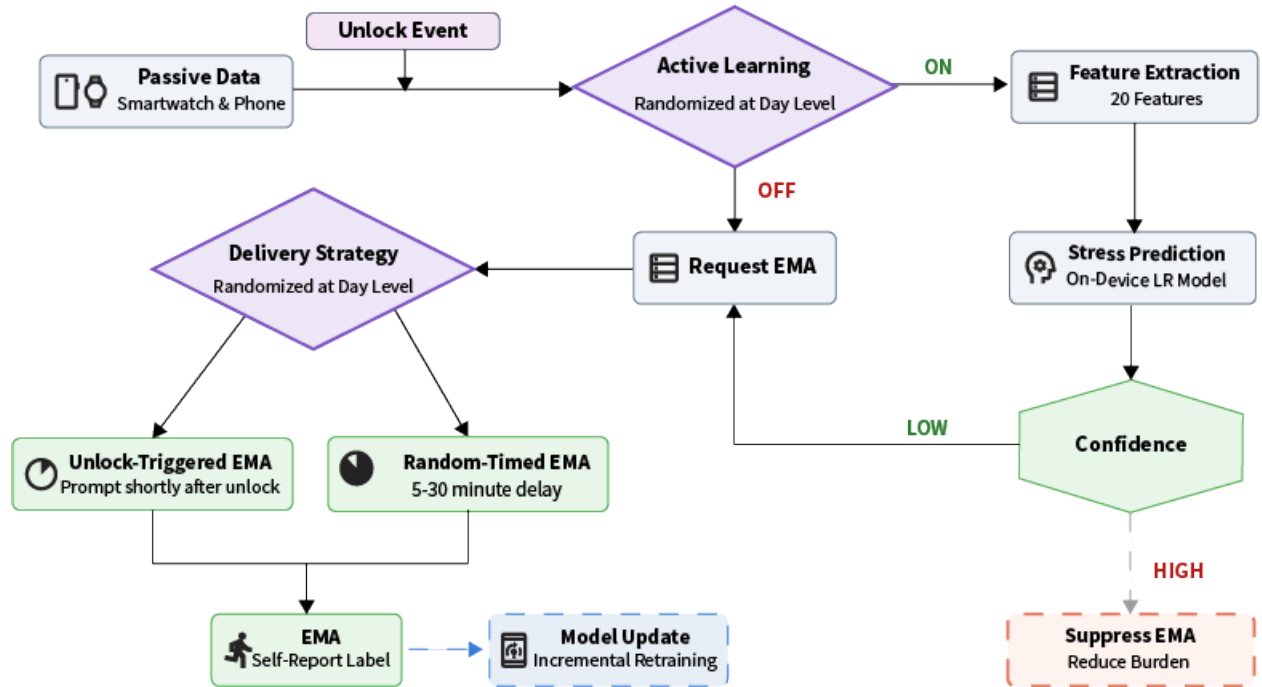


Figure 4.1: **Real-Time Active Learning and EMA Delivery Strategies.** The two purple diamonds mark the study’s independent variables. A phone unlock starts the pipeline. The first independent variable is *active learning*. With active learning off, I request an EMA without consulting the model. With it on, I extract a 20-feature vector and pass it to an on-device logistic regression model that predicts momentary stress. I then request an EMA only when model confidence is low and suppress it when confidence is high to reduce participant burden. The second independent variable, *delivery strategy*, determines when a requested EMA arrives, either unlock-triggered, where the prompt appears shortly after the unlock, or random-timed, where a random five- to thirty-minute delay approximates random-timed EMAs. Self-reported labels from answered EMAs feed incremental retraining.

I evaluated the effectiveness of active learning–timed EMAs and *unlock-triggered EMA* through a controlled user study across three cohorts. Cohorts 1 and 2 served as pilot deployments that validated the mobile sensing system and produced the labeled training data used to fit the base stress-prediction model deployed in Cohort 3. Cohort 3 then employed a within-subjects 2×2

factorial design, crossing delivery timing (unlock-triggered vs. random-timed) with active learning (on vs. off), and was used to address the three research questions defined below:

RQ1: Does *unlock-triggered EMA* improve EMA compliance compared to random-timed EMAs, and can a model that predicts participant engagement from sensor context improve on it?

RQ2: Do active learning-timed EMAs reduce the number of labels collected without substantially degrading model performance compared to fixed-timed EMAs?

RQ3: How do *unlock-triggered* and active learning-timed EMA affect the participant, in terms of prompt burden, responsiveness, perceived notification frequency, and the distribution of reported stress?

The study focused on college students and examined momentary stress as the primary outcome of interest.

4.2 Background and Motivation

Prior work has explored several strategies to reduce participant burden or increase compliance opportunities in EMA-based mobile sensing studies. These include microEMAs that reduce response effort, interaction-centric techniques such as unlock journaling [23] that align self-report with natural device use, and machine learning approaches that identify less disruptive interruption moments [66] or autofill responses [67]. These methods have shown promising improvements in compliance and user experience, and active learning-timed EMA in particular has been shown to improve label efficiency and model performance [138]. However, they are typically evaluated retrospectively or in simulation, on previously collected datasets rather than in live deployments. Prior work therefore has not examined how active learning-timed EMA is experienced by the participant in a live deployment. It does not tell us whether the approach changes compliance or responsiveness, or whether participants notice and value the reduced prompting.

A second body of work addresses participant burden not by changing *when* labels are requested but by reducing *how many* are needed to personalize a model. Individuals differ substantially in their baseline stress and in how their physiological signals map to perceived stress, so population models trained across users tend to generalize poorly to any single person [139]. Training a separate model for each person, though, usually requires more labeled data than a participant can realistically provide during a deployment. Mixed-effects models, also called hierarchical models, offer a middle ground. By adding participant-level random effects to a shared population model, they estimate each person’s parameters jointly with the population through *partial pooling*, so that participants who contribute few labels borrow strength from the population rather than overfitting to their own sparse data [140, 141]. Random-intercept formulations can recalibrate a model to an individual’s baseline from only a handful of labels, and related hierarchical and sub-population approaches have improved stress and affect modeling over both fully general and fully individual models [142].

I chose momentary stress as the prediction target because it is a practical testbed for the timing

methods rather than the contribution itself. College students experience stress regularly, can report it in a single tap, and can be studied with commodity wearables at low cost and minimal risk. This made a live deployment feasible. The timing method does not depend on which label is collected, so stress is simply the outcome I used to exercise it. Even so, I want to set realistic expectations for how well stress can be predicted in this setting. Wearable-based stress detection performs well in controlled laboratory conditions, but accuracy drops substantially in free-living environments that rely on self-reported stress as ground truth [143]. Such self-reports are subjective, sparse, and prone to recall bias, and individuals vary in how they perceive and label their own stress, making a consistent ground truth difficult to establish [144]. Large-scale studies reflect these limits: a five-day study of 1,002 participants reported an F1 score of 0.43 for stress detection [25], and even with up to 600 subject-specific labels, recall on the stressed class reached only 0.44 [138]. This modest performance is a well-documented constraint of free-living stress detection, not a shortcoming of any single model. I therefore present the stress-prediction results as a check that the pipeline behaves sensibly on a noisy target. At the same time, the method’s success rests on the compliance, burden, and label-efficiency outcomes.

4.3 Methods

4.3.1 Study Application Development

To evaluate the feasibility and effectiveness of real-time active learning and unlock-triggered EMA, I developed *Wellsense*, a mobile sensing application for iPhones that administered EMA surveys and collected physiological, activity, and sleep data from a connected Apple Watch. The application incorporated an on-device logistic regression (LR) model that predicted momentary stress in near real time and was incrementally retrained using participant-specific data to support personalization [3, 2]. Passively collected data, extracted features, EMA responses, and application logs were uploaded to a backend server, enabling the research team to monitor participant progress and data collection quality through a researcher-facing dashboard.

The application design drew on prior work identified in the literature review [145] (Chapter 2) and findings from the study described in Chapter 3, with an emphasis on improving EMA compliance while minimizing participant burden. Design decisions focused on reducing interaction cost, increasing response opportunities, and improving notification salience. Specifically, the application used setup prompting for notification authorization and time-sensitive notifications to increase visibility in the notification center [146]. The main application screen displayed daily and cumulative EMA completion alongside remaining participation days.

The application included a home screen widget that mirrored the main display and served as a persistent reminder, reflecting evidence that persistent reminders can support sustained engagement and background execution [146]. Figure 4.2 illustrates the primary user interface components of

the Wellsense application, including the main dashboard, the microEMA response interface, and the home screen widget.

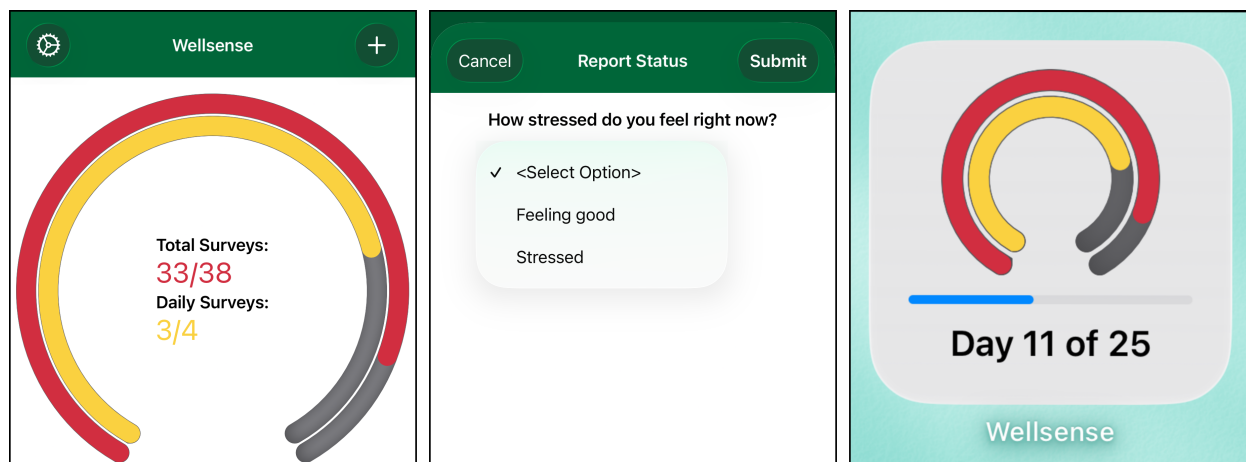


Figure 4.2: **Screenshots of the Wellsense study application.** Left: Main application dashboard displaying cumulative and daily EMA completion using circular progress indicators. Middle: MicroEMA response screen presented when participants open the application directly rather than responding via the notification. Right: Home screen widget displaying a compact version of the main dashboard, used as a persistent reminder.

Active data collection relied on a single-question microEMA. Participants were asked, “How stressed do you feel right now?” and responded using a binary choice (“Stressed” or “Feeling good”). MicroEMAs have been shown to improve compliance by reducing completion time and interaction effort, particularly on mobile and wearable devices [73, 23]. This item has also been validated for in-the-moment stress assessment and has demonstrated moderate correlation with established stress questionnaires in prior work [147, 87].

Participants could respond to EMAs directly from the notification, within the mobile application, or via the Apple Watch, further reducing response friction. If participants missed or dismissed a notification, they could manually initiate an EMA by selecting the “+” button on the main screen. Together, these interaction pathways were designed to maximize response opportunities while keeping the cost of participation low.

4.3.2 Data Collection and Feature Engineering

Passive physiological, activity, and sleep data were collected using Apple’s Health app, a centralized, system-managed repository for health data on iOS devices that periodically supplied newly available samples to the application in the background without opening it. Table 4.1 summarizes the specific health signals collected in this study, their typical sampling behavior, and the features derived from them. The operating system controls sampling frequency, which varies by signal type, user activity, and device state. Because the Health app encrypts stored health data on-device, newly recorded

samples are accessible only after the user unlocks the device, naturally aligning data availability with user interaction patterns. The application logged these unlock events and incorporated them as an additional contextual signal in the machine learning pipeline. Prior work has shown that using the Health app as an intermediary can improve the reliability and completeness of mobile health data collection compared to direct sensor polling alone [87, 66, 88, 11].

To support real-time inference, I transformed raw HealthKit samples and system events into a fixed-length feature vector, summarized in Table 4.1. I computed physiological and activity features over a rolling 1-hour window to capture short-term dynamics. In contrast, sleep, exercise history, and time-since-event features drew on longer history to preserve behavioral context. Features were normalized to $[0, 1]$, and when a source signal was unavailable the corresponding feature was assigned a neutral value of 0.5, or a capped value for time-since-event features, ensuring that the on-device logistic regression model could operate continuously and that missing data did not bias predictions toward either class.

Table 4.1: **Feature set used for logistic regression stress prediction.** The 20 scalar features input to the on-device CoreML model are computed from HealthKit signals and system events. Sampling rates are OS-controlled and vary with activity and device state. Features marked with † were computed from a rolling 1-hour window, whereas features marked with ★ were derived from longer-term history. Features were normalized to $[0, 1]$ before inference. When a source signal was unavailable, the feature was set to the neutral midpoint of 0.5.

Feature	Source (Sampling)	Computation	Missing Data
† hrAverage, hrMax, hrMin	Heart rate (5–10/hr)	Mean, maximum, and minimum heart rate	Neutral 0.5
† hrStdDev, hrPeaks	Heart rate	Standard deviation of heart rate; count of peaks >1 SD above mean.	Neutral 0.5 if <2 samples
† stepTotal, activeEnergyTotal	Steps, active energy (5–10/hr)	Total steps; total active energy (kcal)	Neutral 0.5
† exerciseMinutes	Exercise time (sessions)	Total exercise minutes	Neutral 0.5
★ timeSinceLastExercise	Workouts (events)	Minutes since last exercise event	Capped max
★ sleepDuration, sleepQuality	Sleep stages (nightly)	Duration of most recent sleep period; proportion of REM/core/deep sleep	Neutral 0.5
★ timeSinceLastSleep	Sleep stages	Minutes since last sleep ended	Capped max
★ lastHrv, minutesSinceLastHrv	HRV SDNN (every 1–1.5 hr)	Most recent SDNN value; minutes since that sample	Neutral 0.5 / capped max
†★ lastHourUnlocks, lastDayUnlocks	Phone unlock events	Unlock counts over the past hour and past 24 h	Neutral 0.5
★ hourSin, hourCos, weekdaySin, weekdayCos	System clock	Sine/cosine encodings of time-of-day and day-of-week	Never missing

4.3.3 On-Device Machine Learning System

The machine learning system deployed on device produced the real-time stress predictions used to time EMAs. HealthKit updates were typically delivered several times per hour, shortly after participants unlocked their phones. To limit battery usage, the application generated a new prediction only if at least 30 minutes had elapsed since the previous inference. When a participant submitted an EMA response, I constructed a new feature vector using data aligned to the EMA response timestamp, rather than the notification delivery time. This ensured that model inputs reflected the participant’s state at the moment of self-report. The feature vector consisted of the 20 scalar features described in Table 4.1 and served as input to a logistic regression (LR) model implemented in Apple’s CoreML framework. The model produced a binary prediction of whether the participant was experiencing elevated stress (“Stressed” vs. “Not Stressed”), together with a confidence value bounded between 0.5 and 1.0 that the active learning component used to guide EMA timing decisions.

The application evaluated recent notification history after generating each feature vector to determine whether to schedule an EMA. Participants specified their waking hours and the periods during which they wore the smartwatch, with a minimum availability of 12 hours per day, consistent with prior work showing that aligning EMA delivery with participant-defined availability can improve compliance [65]. In Cohorts 1 and 2, this window was divided into eight equal intervals, yielding a maximum of eight EMAs per day; in Cohort 3, the window was divided into twelve intervals, yielding a maximum of twelve EMAs per day. If the elapsed time since the previous EMA exceeded the minimum interval, a new notification was eligible for delivery. When active learning was enabled, I added one further constraint: an EMA was delivered only when model confidence fell below a threshold of 0.75, so prompts went out when the model was least sure. In Cohort 3, this threshold increased by 0.05 each time an EMA was suppressed, making delivery progressively more likely, and reset to 0.75 after each delivery. This escalation prevented a confident model from suppressing prompts indefinitely.

On-device model training for *personalization* was intended to run roughly once every 24 hours, but iOS limits on background execution meant it could not be triggered on a fixed schedule or after a set number of new labels. Training instead began whenever the operating system granted background runtime. If more than 24 hours had passed since the last update, the application sent a notification asking the participant to open the app to start it.

Cohorts 1 and 2 deployed an LR model initialized with random weights to serve as a low-information baseline. For Cohort 3, I trained a base LR classifier offline on the same feature vector, using data pooled from Cohorts 1 and 2. The training set comprised $N = 1,887$ labeled data points from 18 participants, with per-participant counts ranging from 20 to 181 (median 119). To address class imbalance (approximately 78% “Not Stressed”), SMOTE was applied to the training set, and features with missing values—including sleep duration, sleep quality, and others unavailable at

inference time—were assigned a neutral normalized value of 0.5, matching the imputation strategy used by the mobile application. I selected model hyperparameters and the decision threshold via 5-fold stratified cross-validation optimizing macro-F1 with L1 regularization, and calibrated predicted probabilities using temperature scaling. I exported the resulting model and calibrated decision threshold for on-device deployment in Cohort 3, where they served as the starting point for the per-participant personalization procedure described above.

4.3.4 Participants and Recruitment

I recruited participants from Business, Science, Computer Science, and Information Technology courses at Brigham Young University–Hawaii through in-class announcements directing interested students to a study information website. Eligibility required ownership of a compatible iPhone, willingness to participate in a study lasting about 28 days, and completion of the 10-item Perceived Stress Scale (PSS-10), which screened for participants experiencing routine day-to-day stress. Prospective participants completed a short screening survey covering device ownership and availability, followed by the PSS-10, and those who qualified were contacted to schedule an onboarding session, during which the mobile application was installed and configured on their personal iPhone. They were instructed on how to respond to EMA notifications and indicate their availability to answer questions. Participants without an Apple Watch received a loaner device for the duration of the study. After completing the study, participants received a \$20 meal certificate for a local restaurant as compensation for their time. This study was approved by the University of Hawaii Institutional Review Board under protocol 2022-00823 and by the Brigham Young University–Hawaii Institutional Review Board under protocol 25-54.

The Perceived Stress Scale (PSS) used for screening is a self-report instrument measuring the degree to which individuals appraise their lives as unpredictable, uncontrollable, and overloaded during the past month. I used the 10-item version (PSS-10), which has demonstrated strong reliability and validity in young adult and student samples [148] and has been used as a screening measure in prior mobile and wearable stress-sensing work (e.g., [87]). The full set of items, as presented to participants, is shown in Appendix Table E.1.

A total of 38 participants completed the study across the three cohorts (Cohort 1: $n = 5$; Cohort 2: $n = 13$; Cohort 3: $n = 20$). I excluded one Cohort 3 participant from analysis because their iPhone did not consistently allow HealthKit background delivery, preventing reliable passive data collection. The analytic sample for Cohort 3 was therefore $n = 19$, yielding 37 analyzed participants overall. Of the 38 participants, 73.7% were male ($n = 28$) and 26.3% female ($n = 10$), with ages ranging from 18 to 34 years ($M = 24.97$, $SD = 3.09$). The sample was predominantly Asian ($n = 23$, 60.5%) and Pacific Islander ($n = 9$, 23.7%), with the remainder ($n = 6$, 15.8%) identifying as Other, including White and African American.

Table 4.2: **Final post-study survey questions.** The post-study survey was administered after participants completed all study phases and collected qualitative feedback on EMA timing strategies, notification delivery, and persistent reminders. Responses were analyzed using thematic analysis.

#	Survey Question
1	What factors influence your decision to respond to or ignore a notification asking you about your stress or anxiety?
2	Can you describe any challenges or difficulties you encountered when responding to the notifications?
3	How do you feel about the frequency and timing of the notifications? What changes would make it easier or more convenient for you to respond? Did you notice a difference between the different phases of the study?
4	In what ways did the home screen widget impact your willingness to respond to notifications?
5	Do you have any suggestions for improving the mobile application?

4.3.5 Final Survey

At the end of the study, all participants completed a post-study survey with five open-ended questions (Table 4.2) covering response decisions and difficulties, perceived notification frequency and timing, the home screen widget, and suggestions for improvement. I analyzed responses using inductive thematic analysis to contextualize the quantitative findings and identify usability issues and design opportunities for improving adaptive EMA systems. Recurring ideas were grouped into themes within each question and then examined for cross-cutting patterns across the full dataset, with participant identifiers retained throughout to preserve traceability between each theme and the individual responses supporting it.

4.4 Cohorts 1 and 2: Within-Subjects Evaluation

4.4.1 Study Design

After developing the mobile application, I conducted a within-subjects user study to evaluate the feasibility of active learning–timed EMA and unlock-triggered EMA strategies. I deployed the study sequentially across cohorts, with Cohorts 1 and 2 both running a randomly initialized base model.

The first two cohorts followed a four-phase within-subjects procedure. The first phase lasted four days and used fixed-timed EMAs to collect initial labeled data for on-device personalization of the base model. The second and third phases each lasted one week and delivered fixed-timed and active learning–timed EMAs in counterbalanced order: participants were randomly assigned to one of two sequences, experiencing a week of fixed-timed EMAs followed by a week of active

learning-timed EMAs or the reverse. Before each phase, the model was reset to the phase-one personalized baseline to ensure comparability across conditions. The final phase lasted one week and used fixed-timed EMAs to collect a held-out test set for evaluating model performance, giving 25 days of data collection per participant.

I incorporated unlock-triggered EMA as a secondary within-subjects condition beginning with the second cohort. During unlock-triggered periods, EMA notifications were delivered shortly after participants unlocked their phones. As a control, I compared unlock-triggered timing with a random-timed strategy that introduced a random 5- to 30-minute delay between a phone unlock event and EMA delivery. Delaying the prompt decouples it from the unlock that triggered it, simulating the arrival of a standard random-timed, notification-based EMA. Exposure to unlock-triggered and delayed EMAs was evenly distributed across the study duration, with participants experiencing each condition for half of each study phase.

4.4.2 Results

Baseline Data Collection

Cohorts 1 ($n = 5$) and 2 ($n = 13$) were conducted primarily to collect labeled EMA data for training the logistic regression base model deployed in Cohort 3. The training set and the procedures used to fit that model, including the resulting calibrated decision threshold, are reported in the On-Device Machine Learning System subsection above.

Preliminary Findings Motivating the Cohort 3 Factorial Design

Two preliminary observations from Cohorts 1 and 2 shaped the Cohort 3 factorial design.

First, the per-participant classifiers in Cohort 2 performed at chance. Mean balanced accuracy was 0.47 (95% CI [0.37, 0.57]) for the active-learning model and 0.52 ([0.46, 0.59]) for the fixed model, with Cohen’s κ of -0.04 and $+0.02$. I then compared each model against an always-predict-majority baseline, measured as raw accuracy because that is the metric the baseline is defined on. Of the 13 Cohort 2 participants, 11 had stressed labels in their evaluation window, and two were excluded because no stressed label was available to score against. None of those 11 beat the majority baseline in either arm, falling short of the 78% “Not Stressed” share by 0.26 on average (Wilcoxon $p = .008$). No participant showed a significant prediction-label association either (Fisher’s exact, all $p \geq .17$). Per-participant modeling would not yield usable results at this sample size, which motivated the shift to a GLMM for the post-hoc analysis of Cohort 3 data.

Second, a paired comparison conducted across 12 of the 13 Cohort 2 participants, excluding one who contributed too few labels to analyze, showed that unlock-triggered EMA delivery produced a substantially higher response rate than random-timed delivery: participants answered $M = 91.0\%$ ($SD = 13.0$) of unlock-triggered EMAs versus $M = 72.0\%$ ($SD = 18.0$) of random-timed EMAs,

a difference of 19.0 percentage points, $t(11) = -3.64$, $p = .004$. Given how large this effect was, I made unlock timing a factor in the Cohort 3 factorial design.

4.5 Cohort 3: Factorial Evaluation of Unlock-Triggered EMA and Active Learning

4.5.1 Study Design

The two observations reported above shaped both the Cohort 3 study design and the planned post-hoc analysis. First, the modest predictive performance of a logistic regression model trained on data from Cohorts 1 and 2 is consistent with findings from the broader mHealth sensing literature suggesting that stress detection from wrist-worn consumer devices faces inherent performance ceilings [138]. Second, on-device personalization of an LR model required more labeled data than a single participant could realistically provide during a multi-week deployment. These observations had distinct implications for the study and its analysis. During the study, the on-device model remained a logistic regression classifier, since CoreML’s deployment constraints precluded fitting more complex hierarchical models on the device. This LR model produced the confidence values that drove real-time active learning decisions. For the post-study analysis of model accuracy, however, I adopted a GLMM with a random intercept per participant. Between-person differences in baseline stress tendency accounted for roughly one-third of the label variance in this dataset ($ICC = 0.323$), and a random intercept captures that stable variation from the few labels available per participant, while avoiding the larger samples needed to fit participant-specific slopes.

Cohort 3 employed a *within-subjects 2×2 factorial design with a randomized block sequence*. The two factors were *EMA delivery timing* (unlock-triggered vs. random-timed) and *active learning* (on vs. off), yielding four conditions. Because active learning suppresses some EMAs within a given day, total daily EMA count is a key dependent variable. I therefore assigned conditions at the day level rather than at the individual EMA level, using permuted block randomization with a four-day block. The four conditions were randomly shuffled at the start of each block, so each occurred exactly once per block in randomized order, balancing exposure across conditions while limiting carryover and day-of-week biases.

The resulting dataset supported a post-hoc, paired comparison of model accuracy under active learning–timed versus fixed-timed personalization labels; the evaluation procedure is described alongside those results below.

4.5.2 Results

Cohort 3 included 19 participants in the analytical sample. I determined EMA response status from notification log timestamps. I did not count EMAs suppressed by active learning as delivered. Ac-

tive learning–on days resulted in substantially fewer delivered EMAs than active learning–off days, reflecting the model’s suppression of prompts when prediction confidence exceeded the threshold.

RQ1: Effect of Unlock-Triggered Delivery on Response Rates

The central compliance finding is that unlock-triggered delivery substantially increased EMA engagement: participants answered 80.0% of unlock-triggered prompts versus 64.2% of random-timed prompts (a 15.8-percentage-point gain). Delivering EMAs at natural phone-unlock moments thus captured participant attention markedly better than random timing. The remainder of this subsection details the supporting evidence and the analysis that produced it.

I analyzed response rates and response times with a 2×2 within-subjects repeated-measures ANOVA, with delivery timing (unlock-triggered vs. random-timed) and active learning (on vs. off) as within-subjects factors and per-participant response rate and per-participant median log-transformed response time, $\log(1 + \text{RT})$, as the dependent variables. Because every participant experienced all four condition cells, each person serves as their own control, isolating the timing and active-learning effects from stable between-person differences in baseline responsiveness. Effect sizes for pairwise comparisons are reported as Hedges’ g .

This difference in response rates was large and statistically significant, $F(1, 18) = 23.37$, $p < .001$, $g = -1.03$, with a Bayes factor of $BF_{10} = 217$ favoring the timing effect, well past the conventional threshold of 100 for decisive evidence. The advantage was also robust across the active-learning manipulation: the active learning $\times$ timing interaction on response rates was non-significant, $F(1, 18) = 0.32$, $p = .579$, so the compliance benefit of unlock-triggered delivery held whether or not active learning was enabled. Per-condition response rates and individual participant profiles are shown in Figure 4.3.

Delivery timing also improved response times, but not by enough to matter in practice. Participants answered unlock-triggered EMAs a median of 4.8s faster than random-timed EMAs (7.2s vs. 12.0s), $F(1, 18) = 17.43$, $p < .001$, $g = 0.85$. A few seconds is not much in mobile sensing, and the gap may say more about where the response was entered (phone versus watch) than about how quickly the participant noticed the prompt, since a response entered on the watch is not timestamped until it reaches the paired phone.

RQ1: Predicting Engagement

After establishing that unlock-triggered delivery increases compliance, I next tested whether the phone could go further by predicting engagement directly. Using only the sensor context available at the moment of delivery, I modeled whether a participant would answer an EMA within 60 seconds, then asked whether prompting only at model-selected moments would outperform the simpler unlock trigger. Three findings emerged; the statistics behind each are reported in the paragraphs that follow. First, the model worked: prompting only at model-selected moments pro-

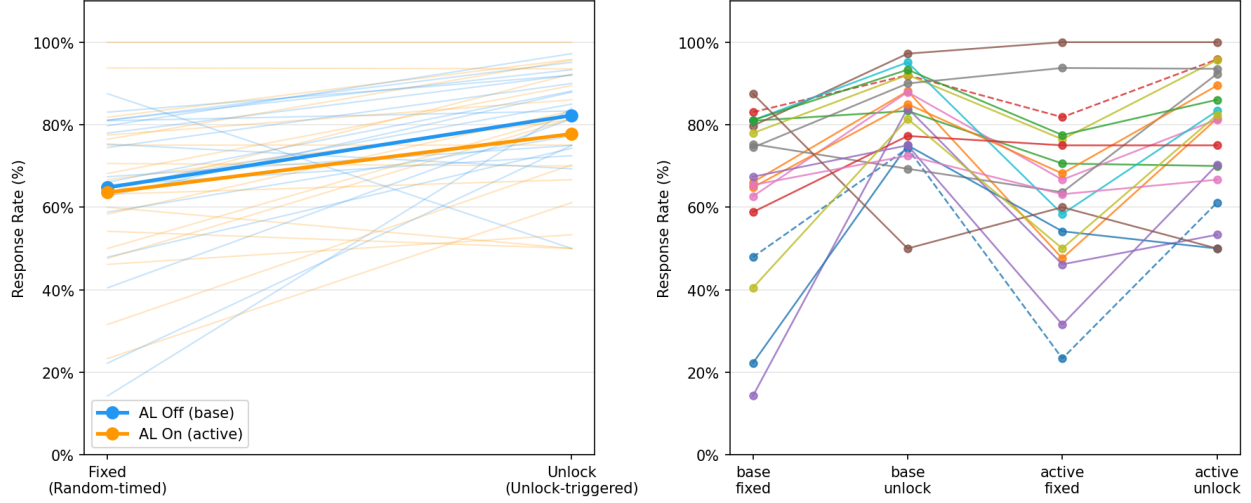


Figure 4.3: **Cohort 3 EMA response rates by condition.** Left: per-participant mean response rates ($\pm$ SE) for active learning-off (blue) and active learning-on (orange) under random-timed (left) and unlock-triggered (right) delivery. Thin lines show individual participants. Right: individual participant profiles across the four factorial cells. Dashed lines indicate participants who stopped early. Unlock-triggered delivery substantially increased response rates regardless of active-learning condition, while active learning produced only a small, non-significant change.

duced a significantly higher response rate than prompting at every opportunity. Second, it did not outperform the unlock trigger: the response rate at model-selected moments was not significantly different from the rate at unlocks, and because the model prompted on only 16.7% of opportunities, it returned roughly a fifth as many answered EMAs as prompting at every opportunity. Third, the two strategies were complementary: EMAs that were both unlock-triggered and model-selected were answered within 60 seconds 75.6% of the time, significantly higher than either strategy alone. However, requiring both conditions narrowed delivery to 9.3% of opportunities and returned the fewest labels of any strategy I compared.

Engagement was modeled post hoc as a binary outcome, whether a delivered EMA was answered within 60 seconds, using a pooled GLMM with a random intercept for each participant. I evaluated it with leave-one-participant-out (LOPO) cross-validation over 18 participants and 1,849 delivered EMAs—the held-out 70% of each participant’s timeline, after reserving the first 30% chronologically as a threshold-setting window (one participant contributed too few EMAs to split and was excluded). This partition is distinct from the personalization split used for RQ2 below, and the two sets of counts are not comparable: Cohort 3 delivered 2,647 EMAs in total. For each held-out participant, the model was fit on the pooled delivered EMAs of all remaining participants and then scored on the held-out participant’s EMAs, so that no participant’s own data contributed to the model that judged them. Discrimination was modest, with a mean AUC of 0.595, 95% CI [0.565, 0.624]. Because I wanted the model to suppress prompts rather than classify

them, I report it at an operating point that favors specificity over balanced accuracy, at which the model prompted on 16.7% of available opportunities. The threshold was fixed on the chronological threshold-setting window and then applied unchanged to the held-out 70%, so the rates reported below are prospective rather than tuned on the data that scores them. Figure 4.4 shows alternative operating points.

Selective prompting produced a higher 60-second response rate than indiscriminate prompting. Across the test window, 54.1% of all delivered EMAs were answered within 60 seconds; restricting delivery to moments the model selected raised this to 65.4%. Unlock-triggered EMAs, however, were already answered 63.9% of the time. I did not detect a significant difference between unlock triggering and the model (+0.015, 95% CI $[-0.042, +0.070]$, $p = .574$). The two strategies combined: EMAs that were both unlock-triggered and model-selected were answered within 60 seconds 75.6% of the time, a significantly higher rate than either strategy achieved alone. These rates sit below the 80.0% and 64.2% reported above because the outcome here is a response within 60 seconds rather than within the one-hour window used to score compliance. A rule that narrows delivery this far is useful only if it does not skew what gets labeled, so I also compared the prompts the combined rule retained with those it discarded. The two sets did not differ in stress composition (20.8% vs. 23.0% stressed; $\chi^2(1) = 0.22$, $p = .64$; cluster-robust GEE OR = 0.94, 95% CI $[0.54, 1.64]$). The confidence interval on this difference is wide (-12.1 to $+8.0$ percentage points), so this is an absence of detected bias rather than a demonstration of equivalence. All five comparisons are reported in Table 4.3.

Table 4.3: **Sixty-second EMA response rates by prompting strategy.** Pairwise differences in the rate at which delivered EMAs were answered within 60 seconds, over 18 participants and 1,849 delivered EMAs. Each row compares strategy A against strategy B: the rate columns give the 60-second response rate under each strategy, with the number of EMAs answered over the number prompted in parentheses, and Diff. is the rate under A minus the rate under B. An opportunity here is an EMA the application delivered, so prompting at every opportunity is the full set of delivered EMAs. Bootstrap p -values come from a cluster bootstrap over participants ($B = 20,000$), Wilcoxon p -values from a participant-level paired signed-rank test, and q -values are the Wilcoxon p -values after Benjamini–Hochberg adjustment across the five comparisons. Favoring gives the number of participants for whom the difference was positive.

Strategy		Response rate (answered/prompted)		p -value				
A	B	A	B	Diff.	boot.	Wilc.	q	Favoring
Unlock	Random-timed	0.639 (610/955)	0.436 (390/894)	+0.203	< .001	< .001	.0002	17/18
Model	Every opportunity	0.654 (202/309)	0.541 (1,000/1,849)	+0.113	< .001	.039	.048	13/18
Model	Unlock	0.654 (202/309)	0.639 (610/955)	+0.015	.574	.899	.899	10/18
Unlock + model	Unlock	0.756 (130/172)	0.639 (610/955)	+0.117	.002	.001	.002	15/18
Unlock + model	Model	0.756 (130/172)	0.654 (202/309)	+0.102	< .001	.001	.002	15/18

These rates show how good each prompt is, not how many labels each strategy returns, and the two have to be read together. Of every 100 delivery opportunities, prompting every time issues

100 prompts and yields about 54 answered EMAs, while the model’s suppression rule issues 16.7 prompts and yields about 11. A higher response rate per prompt therefore buys fewer interruptions per label collected, which is a real gain in participant burden, but it does not by itself produce more labels.

Taken together, these results indicate that a model can identify moments of likely engagement well enough to be useful, and that much of what it identifies is already captured by the unlock event itself. That is a useful result for system design: unlock-triggered delivery reached nearly the same response rate as model-selected delivery, so it improves per-prompt engagement without thinning the label set the model is meant to improve. The model earns its place by layering on top of that trigger, raising the 60-second response rate to 75.6%, a design worth considering when interruptions cost more than labels. Figure 4.4 makes the trade-off explicit across operating points: relaxing the suppression threshold from a specificity of 0.87 to 0.76 doubles what the median participant returns, from 9.5 to 19 answered EMAs, at a cost of 0.04 in positive predictive value, while prompting at every opportunity returns 62 at 0.541.

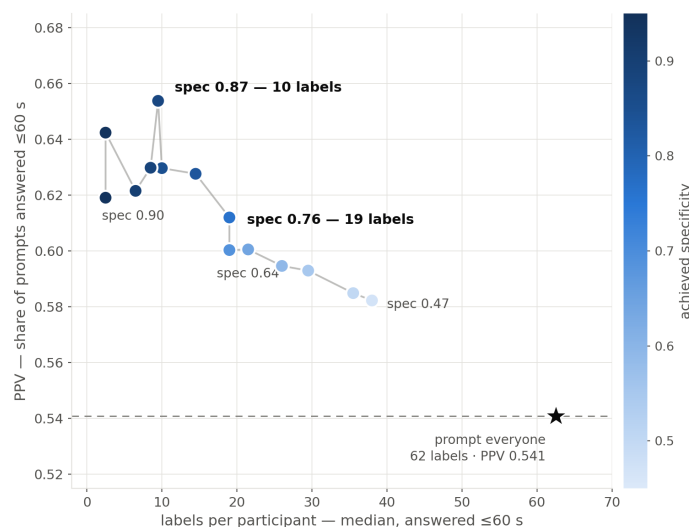


Figure 4.4: Engagement model: labels collected versus precision across operating points. Operating points of the 60-second engagement model: over 18 participants and 1,849 delivered EMAs in the chronological test window. The horizontal axis gives the number of labels the median participant yields, the vertical axis the positive predictive value (the share of prompts answered within 60 seconds), and the color the achieved specificity; the star marks prompting at every opportunity (62 labels, PPV 0.541). A specificity of 0.87 lifts PPV to 0.654 but leaves 9.5 labels per participant, 1.5 of them stress-positive (202 and 36 cohort-wide, against 1,000 and 217 when prompting everyone). A specificity of 0.76 costs 0.04 of PPV and returns 19 labels, 2.5 of them stress-positive (317 and 71 cohort-wide). Suppression does not measurably change the stressed share of the labels collected: 21.7% prompting everyone, 22.4% at specificity 0.76, and 17.8% at specificity 0.87.

RQ2: Effect of Active Learning on Model Performance

The central finding for model performance is that active learning–timed EMA cut the number of labels collected by 64.7% (518 active learning–timed vs. 1,468 fixed-timed labels, of 1,986 collected across the 19 Cohort 3 participants from 2,647 delivered EMAs), and that this deployment cannot say whether the labels it selected were any better than labels selected at random. The population model discriminated only slightly above chance (mean AUC = 0.566), and what personalization recovered was each participant’s base rate rather than anything physiological. The confidence values that drove suppression came from a model that barely separated stressed from unstressed moments, so the effectiveness of active learning–timed EMA is inconclusive from this data set.

I evaluated model accuracy post hoc under the two labeling strategies using a leave-one-participant-out protocol, with an added personalization step. I held out one Cohort 3 participant at a time. I fit a population random-intercept GLMM on the pooled labeled EMAs of every other Cohort 2 and Cohort 3 participant, using the 20-feature vector from Table 4.1. I repeated this for each Cohort 3 participant, so every participant was scored by a model that had never seen their data. I ordered the held-out participant’s labels chronologically and split them into a 70% personalization set and a 30% test set. Within the personalization set, I re-estimated the participant random intercept twice: once from active learning–timed labels and once from fixed-timed labels. I report results as the Brier score rather than AUC because re-estimating only the intercept shifts every prediction uniformly and cannot change rank order. Across the 13 participants with labels in both conditions, personalizing from the active pool gave a mean Brier of 0.228 against 0.240 from the fixed-timed pool. Because lower is better, this favors the active pool by 0.012, a small margin compared with the spread across participants, which ranged from -0.103 (favoring the fixed-timed pool) to $+0.201$. Per-participant label counts, AUC, and Brier values are reported in Appendix Table C.1 and Appendix Figure C.1.

That comparison evaluates the two label sources at their deployed budgets, which differ by design. It therefore cannot separate a benefit of uncertainty-based *selection* from a benefit of simply having fewer labels. It also carries a mismatch, since the on-device LR model selected the labels and the GLMM evaluated them. To address both problems, I re-simulated active learning offline on each participant’s pooled label stream, ignoring deployment condition and using the GLMM itself as the selector. Arriving EMAs were admitted for personalization by one of two gating rules: uncertainty gating, which accepts the labels the model is least confident about, or random gating at a matched prompt rate, where the prompt rate is the fraction of arriving EMAs spent on personalization. Scoring was prequential, so each EMA was predicted before it could be used to personalize the model, and I compared the two rules across the full range of prompt rates. Matching the prompt rate holds the number of labels fixed, and selecting with the GLMM makes the selector and the evaluated model the same, as they were not in the live deployment.

Uncertainty gating was not significantly better than random gating at any prompt rate (Fig-

ure 4.5). It was nominally better calibrated at the lowest prompt rates, but stayed well inside the interval spanned by random draws throughout. I also ran a base-rate control that personalized from the labels alone, using no sensor features, and it performed at least as well as either selection rule. What the labels recover, then, is each participant’s baseline tendency to report stress rather than any sensor-driven judgment about which moments are worth labeling.

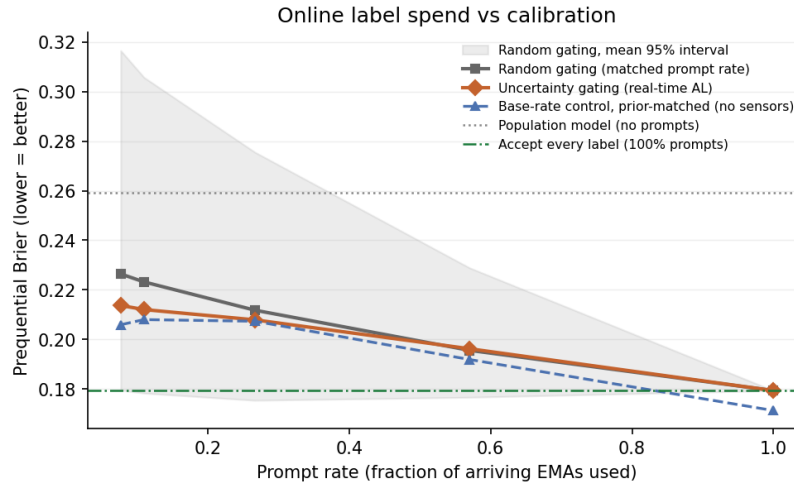


Figure 4.5: **Simulated active learning: prompt rate versus calibration.** Prequential Brier score (lower is better) against prompt rate, the fraction of arriving EMAs spent on personalization. Uncertainty gating, which accepts the labels the model is least confident about, is compared against random gating at a matched prompt rate; the shaded band is the 95% interval across random draws, and the reference lines mark the unpersonalized population model (no prompts) and accepting every arriving label. Uncertainty gating is nominally better calibrated at the lowest prompt rates. Still, it stays well inside the random band throughout, and a base-rate control personalized from the labels alone, with no sensor features, calibrates at least as well as either rule.

I also compared the discriminative performance of several model families on the same data, summarized in Table 4.4.

RQ3: Effect of the Timing Strategies on the Participant

RQ3 asks how each timing strategy affects the participant. For unlock-triggered EMA, the central finding is that moving prompts to unlock moments did not change participants’ reports. Unlock-triggered delivery samples moments non-randomly by design, since a prompt can only arrive when a participant picks up their phone, raising concerns that the labels it returns describe the moments participants choose to be on their phone rather than their day at large. For the stress label, I found no evidence of that. Partitioning answered EMAs by delivery timing, 23.0% were labeled stressed under unlock-triggered delivery against 21.0% under random-timed delivery, a difference of +2.0 percentage points, 95% CI $[-2.9, +6.9]$, an interval that includes zero (Figure 4.6). The stress label

Table 4.4: **Stress prediction by model family (Cohort 3).** Per-participant test AUC under the fixed-split leave-one-participant-out evaluation, over the $n = 14$ participants whose held-out test window contained both classes. Confidence intervals are reported on the mean AUC relative to chance (0.5). No family reached 0.60, and the two intervals that excluded chance did so by +0.003 and +0.006 across five families tested without multiplicity correction.

Model	Mean AUC	Median	Below chance	AUC $- 0.5$, 95% CI
Gaussian naive Bayes	0.586	0.577	5/14	[+0.006, +0.171]
Population GLMM	0.566	0.593	4/14	[+0.003, +0.126]
Logistic regression	0.516	0.487	7/14	[−0.072, +0.114]
2-layer NN	0.513	0.457	9/14	[−0.097, +0.138]
Random forest	0.487	0.488	7/14	[−0.124, +0.102]

was also the least divergent of the 22 measures compared between timing conditions. Combined with the compliance gain reported above, unlock-triggered delivery obtained substantially more labels without measurably shifting what those labels say. Appendix Figure D.1 shows how the other measures differed.

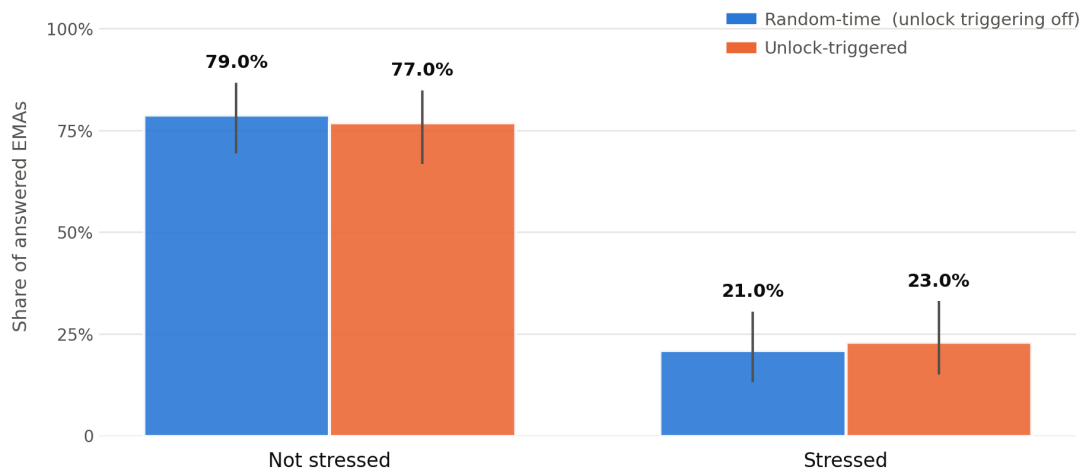


Figure 4.6: **Stress label distribution by delivery timing.** Share of answered Cohort 3 EMAs labeled not stressed (left) and stressed (right) under random-timed (blue) and unlock-triggered (orange) delivery; error bars are 95% confidence intervals. Unlock-triggered delivery returned a stressed share of 23.0% against 21.0% for random-timed delivery, a difference of +2.0 percentage points, 95% CI [−2.9, +6.9]. Delivering prompts at unlock moments did not measurably shift what participants reported.

Turning to active learning–timed EMA, the central finding is that suppression made participation measurably lighter without weakening engagement. Across all 19 Cohort 3 participants, response rates were 70.7% under active learning–on versus 73.5% under active learning–off, a non-

significant difference of 2.8 percentage points ($SD = 10.1$), $F(1, 18) = 1.51$, $p = .235$, $BF_{10} = 0.46$. The Bayes factor falls in the inconclusive range (between 0.33 and 3), so a small effect cannot be distinguished from none. Still, the data provide no evidence of a compliance penalty, and the observed difference is far smaller than the burden saving reported above. Those savings are pooled cohort totals: 64.7% fewer labels collected, from 61.1% fewer EMAs delivered.

The reduction in prompting also registered subjectively. Describing the contrast between condition days, P51 wrote: “There were some days where I was getting notified very frequently throughout the day, and other times when I only got the notifications a few times in a day. I definitely preferred when it was not asking me every hour how I was feeling.” P43 likewise found the prompting “too frequent,” adding that “being asked constantly adds stress to the experience itself,” a pointed observation in a study whose target construct is stress.

Suppression cut prompting without any compliance cost I could detect, and the active learning $\times$ timing interaction on response rates reported above was non-significant. The Bayes factor was inconclusive, though, so the null means I could not have detected a small compliance penalty, not that none exists.

4.6 CardioMate Analysis

4.6.1 Dataset and Simulation Procedure

To examine active learning–timed EMA beyond the live deployment, I ran a second analysis on another dataset I helped collect, called CardioMate. CardioMate comes from a longitudinal observational study of stress and blood pressure reactivity in adults with self-reported hypertension [149, 2]. Twenty adults were each monitored for about four weeks with a Fitbit for continuous heart rate and step counts, an Omron HeartGuide cuff for ambulatory blood pressure, and the CardioMate mobile application for EMA. Participants rated momentary stress on a 0–10 scale, which yielded 3,694 stress responses alongside 3,812 blood pressure readings. I treated a rating of six or greater as high stress, which accounted for 12.7% of responses.

I chose this dataset because personalized models fit to it predict far better than the model I deployed in this study. The original analysis reported a mean validation AUC of 0.849 for personalized high-stress prediction versus 0.641 for a single pooled population model [149], both above the near-chance discrimination I obtained here. Hence, a model trained on these data provides confidence estimates that uncertainty-based selection can act on. CardioMate also prompted EMAs throughout the day, at a median of six per participant per day, rather than only at phone unlocks, so the selection rule can rank a full day of candidate moments rather than moments when the participant unlocked their phone.

I then simulated active learning–timed EMA on these data, comparing a logistic regression, a GLMM with a random intercept only, and a GLMM with a random intercept and random slopes

across four targets. Two of the targets were clinical: high stress and a blood pressure spike taken from the cuff reading. The other two were methods-validation targets, both defined as an Omron cuff pulse reading above 80 beats per minute, one predicting it with the Fitbit heart-rate features withheld and one with them included. Fitbit heart rate and cuff pulse measure much the same physiology, so the version that keeps those features is partly leaked by construction, which is exactly what makes it useful. A null result on a target no model can predict says nothing, while a target the model predicts well shows that the analysis can detect a selection advantage when one exists. Within-participant discrimination on the held-out segment was an AUC of 0.658 for high stress, 0.605 for blood pressure spikes, 0.682 for cuff pulse above 80 without the Fitbit heart-rate features, and 0.846 with them included.

4.6.2 Results

Across all four targets and all three models, I did not find a case in which actively selected labels were significantly better than randomly selected labels at the same budget (Figure 4.7). Either there was not enough signal for the model to select better labels, or the model was already accurate enough that personalization made no difference.

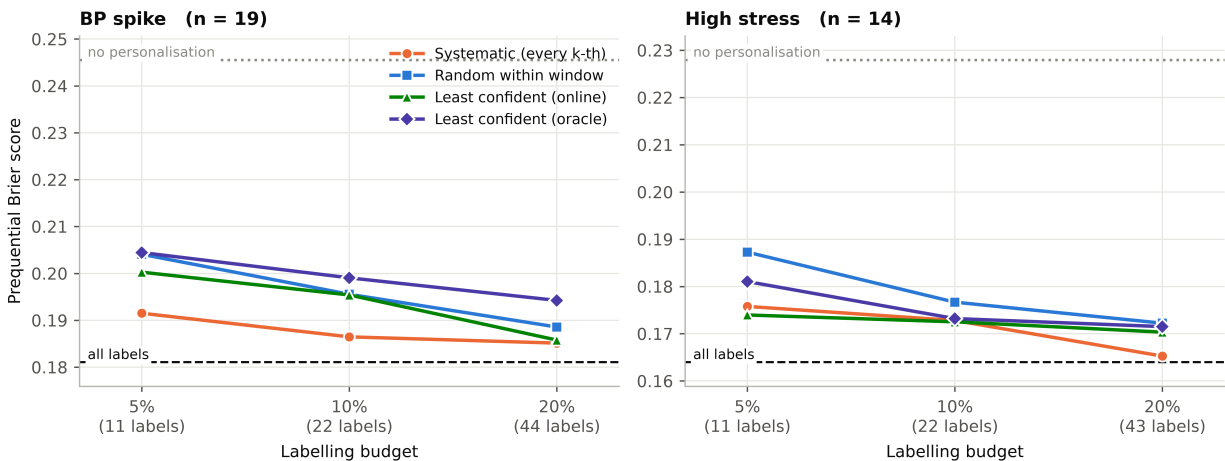


Figure 4.7: **CardioMate: label selection policies at matched budgets.** Prequential Brier score (lower is better) for the two clinical targets, blood pressure spikes (left, $n = 19$) and high stress (right, $n = 14$), as the labeling budget grows from 5% to 20% of each participant’s available labels. I compared four policies: systematic selection of every k -th moment, random selection within each window, online least-confident selection, and an offline least-confident oracle that sees the whole pool before choosing. Dotted lines mark the unpersonalized model (0.246 and 0.228) and dashed lines the all-labels ceiling (0.181 and 0.164). Personalization closes most of that gap at the smallest budget, but neither least-confident policy separates from the random or systematic baselines at any budget, and on blood pressure, the systematic policy matches both throughout.

4.7 Qualitative Analysis

To complement the quantitative findings, I analyzed the open-ended exit survey responses using the inductive thematic-analysis procedure described in the Methods. Thirty-six participants completed the five-question survey (Table 4.2). The preceding subsection reports themes concerning notification frequency; two further findings emerged from the remaining themes.

First, busyness was the most commonly cited barrier to responding, named by nearly every participant. As P10 explained, “It usually depends on how busy I am during the time it asked me. Juggling assignments and work played a huge part of my decision,” and P57 similarly noted that “if I didn’t respond it was usually because I was preoccupied with something or I was doing something.” Second, participants broadly valued the home screen widget as a passive engagement tool. Ten participants independently credited its visual presence with keeping the study salient; P10 observed, “I just needed to be reminded so I can answer the study. Since I love fidgeting my phone, I see the widget a lot of times so I am always reminded.” Several participants further described a gamification effect driven by a desire to maintain a complete response record: P47 said the widget “made me want to respond to keep the bars full,” and P49 reported that “it motivated me more to keep the graph perfect.” This finding aligns with the quantitative effect observed for unlock-triggered delivery and is consistent with the evidence from Chapter 3 that persistent reminders support sustained engagement. Taken together, these findings indicate that momentary availability, rather than motivation, governed whether any given prompt was answered, and that persistent visual cues such as the home screen widget played an important role in sustaining engagement across the deployment.

4.8 Additional Analysis

Analysis of PSS-10 and biographical data showed two relationships. First, the PSS-10 score participants gave at screening, before they had answered a single EMA, predicted how often they later reported feeling stressed in the app (Spearman $\rho = +.443$, $p = .006$, $n = 37$). The screening score is an independent measure collected before the study, and it agrees with app-recorded data. The second finding is that the more stressed a participant was, the fewer EMAs they answered (Spearman $\rho = -.547$, $p = .015$, $n = 19$ in Cohort 3; $\rho = -.566$, $p = .0007$, $n = 32$ across all cohorts with response data). Stressed participants were harder to reach, which matters for anyone designing an EMA deployment. That second relationship carries a caveat: a participant’s reported stress rate can be computed only from the EMAs they answered, so it and their response rate are drawn partly from the same pool. The screening score does not share that problem and points in the same direction in every cohort, but on its own it never reaches significance (PSS-10 vs. response rate: $\rho = -.414$, $p = .078$ in Cohort 3; $\rho = -.254$, $p = .161$ across all cohorts). I therefore treat “more stressed participants respond less” as a consistent pattern rather than a settled result. Both

relationships come from an exploratory battery of tests and are reported without correction for multiplicity.

No other factors were significant. Gender showed no significant difference in response rate, response time, engagement within 60 seconds, model AUC, Brier score, the benefit of personalization, on-device model accuracy, or PSS-10 screening score. Race showed no significant difference on any of those same eight outcomes, whether comparing Asian to Pacific Islander participants or running a four-way comparison that added the smaller race groups. Age showed no significant relationship with any of the eight outcomes. Pooling all three cohorts, no test of gender, race, or age was significant for these outcomes.

These groups were small enough that only a large difference would have been detectable, so the null results mean I could not have detected a modest difference, not that none exists.

4.9 Discussion

Unlock-triggered delivery was the strongest result in this chapter. Aligning EMA prompts with phone-unlock events raised the response rate by roughly 16 percentage points over random-timed delivery. Participants also responded faster to unlock-triggered prompts, but I read that difference as statistical rather than practical, for the reasons given with those results. I rest the practical case for the strategy on the response rate, where a 16-percentage-point gain translates directly into more labels collected. The engagement analysis points the same way: a model that predicted whether a prompt would be answered within 60 seconds matched unlock triggering (65.4% against 63.9%). It reached that rate only by suppressing five of every six opportunities, so it returned far fewer labels for the effort. The more complex machinery for choosing when to prompt performed no better than the simple unlock trigger.

I weigh that filtering against what it costs the label set, where the stressed labels are the binding constraint. Only about a fifth of answered EMAs were labeled stressed, so over the test window (about 21 days), the model-based suppression rule returned a mean of roughly 11 answered EMAs per participant, about 2 of them stressed, whereas prompting at every opportunity returned roughly 56 answered EMAs per participant and about 12 stressed (Figure 4.4 reports the same quantities as medians, which run higher for the unsuppressed strategy). Requiring an unlock as well narrows delivery further, to about 7 answered EMAs and fewer than 2 stressed labels per participant. Encouragingly, the prompts the combined rule kept did not differ in stress composition from those it discarded, so the rule appears to sample the two classes in similar proportion. The constraint is volume, not composition. The LOPO evaluation in this chapter gives a sense of where the floor sits. Personalization sets there had a mean of 18.4 active learning-timed labels per participant, and the one participant who contributed only three could not have an intercept re-estimated at all.

In contrast, the smallest pools that did support personalization had seven to nine labels (Appendix Table C.1). Re-estimating a random intercept is the cheapest form of personalization

available here, and participant-specific feature weights would still require more labels, with both classes represented. A rule tuned this conservatively is therefore best suited to a deployment that has already banked enough labels to personalize, or one long enough for the smaller per-day yield to accumulate. The unlock trigger sidesteps that trade-off, since it reaches nearly the same per-prompt response rate while still prompting at more than half of opportunities, which is what makes the size of its effect worth examining more closely.

The more useful question is why such a small change in timing worked so well. The mechanism appears to be *notification salience*. A random-timed prompt often arrives while the phone is stowed and the screen is off, so it competes for attention with whatever the participant is doing and is easily missed, deferred, or cleared unread. An unlock-triggered prompt arrives when the participant has just chosen to engage with the device. The screen is already on, attention is already directed at it, and the prompt is seen right away rather than after an unpredictable delay. The unlock event also marks a natural break in activity, so the participant is between tasks rather than interrupted mid-task, and the cost of a brief, single-item response is low. This salience comes without taking over the interaction. Unlock journaling secures attention by blocking the unlock itself [23], whereas surfacing the prompt just after a naturally occurring unlock keeps the same salience benefit without that friction, which likely helps compliance hold up rather than erode through deployment-related annoyance. That low cost to the participant is a precondition for the effect: I do not expect similar results from a prompt that took more than a few seconds to answer. What makes the finding notable is how little it cost to obtain. A timing change requiring no additional infrastructure, computation, or battery produced a larger compliance gain than the algorithmic intervention I evaluated alongside it. For applied EMA deployments, where compliance rather than modeling sophistication is often the limiting factor, tying delivery to existing device interactions is a cheap and broadly transferable lever.

Three properties of this deployment weakened it as a test of active learning–timed EMA. The selector was the on-device logistic regression while the evaluated model was the GLMM, iOS restricted the candidate pool to unlock moments rather than the full day, and the base model barely discriminated above chance, which left its confidence estimates uninformative enough that uncertainty-based selection degenerates toward random selection. An offline simulation removed the first of these by selecting with the same GLMM that was evaluated, and the null persisted at every budget. To explore active learning–timed EMA further, I also analyzed the CardioMate dataset, which had previously shown high accuracy for personalized models. There, a logistic regression or GLMM trained on population data and then personalized to a participant did not reproduce that accuracy on the blood pressure and stress targets, and again active learning showed no improvement over randomly selected labels. I then applied active learning to a target the models did predict well, pulse greater than 80, and under that target the gains from personalization were too small for any difference between actively selected and randomly selected labels to be detected.

These results align closely with previous work on when active learning is worth doing. Mussmann and Liang [150] tie the data efficiency of uncertainty sampling to the error rate of the model doing the selecting: uncertainty sampling pays off when the underlying model is already accurate, and its advantage over random sampling shrinks toward nothing as error grows. Two conditions must therefore hold together. The model must classify well enough for its confidence to carry information, and personalization must matter in the first place (large variance between participants). Neither my deployment nor the CardioMate dataset offered a target that met both: the stress models barely discriminated above chance, and the one target that was predicted well, pulse greater than 80, gained so little from personalization that no difference between selection rules could show. Before building a real-time active learning system, which is a substantial engineering effort, run that check on existing labeled data: how accurate is the base model, and how much does personalization buy?

What active learning did deliver was fewer prompts. Suppressing prompts when the model was confident cut the labels collected by 64.7%, and personalizing from that smaller pool was no worse on average than personalizing from the full fixed-timed pool, though with a near-chance base model that comparison cannot settle much either way. There could still be health targets for which the selection rule pays off, and the qualitative results suggest participants would welcome sparser prompting wherever possible. P51 explicitly preferred the sparser-notification days, and P43 reported that frequent prompting itself added stress.

4.9.1 Limitations

Several limitations warrant consideration when interpreting these findings. First, the analytic sample of 37 participants, 19 of them in the primary Cohort 3 analysis, came from a single institution and was predominantly male, so replication in more diverse and clinical populations is needed before generalizing the compliance and calibration effects reported here. Second, I operationalized momentary stress with a binary single-item microEMA, which captures whether stress is present but not its intensity, duration, or source. Richer multi-item or continuous outcome measures may reveal effects of active learning and delivery timing that the binary outcome obscured. Third, the deployment was platform-specific, using an iPhone and Apple Watch mediated by HealthKit’s background-delivery mechanism, and results may differ on other platforms. Because HealthKit delivered records only on unlock, candidate moments could be offered to the selector only at unlocks, and no analysis of these data can recover the moments it never saw.

The main limitation of evaluating active learning-timed EMA is that this deployment is a weak test, and the CardioMate analysis that supplements it is a secondary reanalysis of a dataset collected for other purposes, on different targets, and without a live prompting loop. Together, they show that active learning did not help *in these settings*, rather than that it cannot help in general.

4.10 Conclusion

I deployed a live, closed-loop mobile sensing system across three cohorts to test two adaptive EMA timing strategies: unlock-triggered delivery and real-time active learning-timed EMA. The final cohort used a within-subjects 2×2 factorial design that crossed the two. Unlock-triggered delivery produced the strongest result. Delivering the EMA shortly after a participant unlocked their phone raised the response rate from 64.2% to 80.0%, replicating a preliminary observation, and the gain held with or without active learning. Because these prompts only arrive when a participant picks up their phone, they do not sample the day at random, so I also checked whether they changed what participants reported. They did not. Participants labeled 23.0% of unlock-triggered EMAs as stressed compared to 21.0% of random-timed EMAs, a difference of 2.0 percentage points, 95% CI $[-2.9, +6.9]$, and the two label distributions were statistically indistinguishable (two-sample Kolmogorov-Smirnov $D = 0.014$, false-discovery-rate-corrected $q = 0.56$). Unlock-triggered delivery therefore collected more labels without changing what those labels said.

Active learning-timed EMA reduced prompting but did not choose better labels. Suppressing prompts when the model was confident cut the number of labels collected by 64.7%, and personalizing from that smaller set of labels was no worse on average than personalizing from the full fixed-timed set. However, the base model was too weak for that comparison to settle anything about active learning itself. That saving did not come from uncertainty sampling. In an offline simulation that held the label budget fixed and made the model doing the selecting also the model being scored, actively selected labels were no better than randomly selected ones. Personalization recovers each participant’s base rate, and any small sample of a participant’s labels estimates a base rate about as well as any other. This applies to the conditions I observed. The population model discriminated near chance ($AUC = 0.566$), so model confidence carried too little information for uncertainty-based selection, and I cannot say the same would hold with a stronger signal. Participants, for their part, lost nothing with suppressed prompts since response rates did not change significantly. In the final survey, respondents preferred being asked less often.

Taken together, these results suggest that small, simple HCI adjustments can have a large effect on active data collection. Unlock-triggered delivery required no additional infrastructure, computation, or battery, and it produced the largest and most reliable improvement reported in this chapter. The more complicated ML intervention, active learning, showed no significant benefit over collecting randomly selected labels. My conclusion is not that active learning-timed EMA should be set aside, since it may well pay off in the right circumstances, but that in this work the simple HCI intervention provided a clear and significant improvement in data collection.

CHAPTER 5

CONCLUSION

5.1 Summary of Contributions

Mobile and ubiquitous devices have changed how researchers collect health data in real-world settings, yet consumer platforms still pose challenges for both active and passive data collection. Active data collection, typically through ecological momentary assessment (EMA), must balance the need for frequent, high-quality labels against the burden imposed on participants. Passive data collection depends on appropriate user authorization and consistent background runtime that mobile operating systems do not guarantee. In this dissertation, I addressed these challenges through innovations at the intersection of Human-Computer Interaction (HCI) and Machine Learning (ML). I first conducted a systematic review of mobile health sensing studies to identify the most pressing gaps in active and passive data collection on consumer devices. I then evaluated HCI-driven solutions (persistent reminders and contextual authorization prompts) to sustain passive data streams and engage participants with EMA. Next, I deployed a live, closed-loop mobile sensing system to compare two adaptive EMA timing strategies: real-time active learning-timed EMA and unlock-triggered EMA delivery. Because the base model in that deployment discriminated too weakly for its confidence estimates to carry information, I re-examined label selection under simulation on a second, continuously sensed dataset whose published models had predicted far better. Together, these efforts support the central thesis of this dissertation: simple HCI modifications often outperform ML-based interventions that rely on accurate models in more complex systems. Interaction-aware design improved collection in every deployment that tested it, without requiring anything of the model. When the two were put head to head, a model trained to predict engagement did no better than the unlock trigger it was meant to improve. Whether selecting *which* moments to label pays off remains unresolved: it showed no measurable advantage over random selection in either dataset, even on a validation target the model predicted well enough for an advantage to emerge. The ML lever is therefore conditional. Its value depends on a predictive regime that neither dataset reached: a model confident enough that its uncertainty carries information, but not so accurate that personalization adds nothing. The work contributes both a demonstrated, low-cost compliance gain and a specification of the conditions under which the more expensive intervention is worth building.

Concretely, this dissertation makes the following contributions to mobile health sensing at the intersection of HCI and ML:

1. **A systematic review of mobile health sensing** that synthesized 77 studies and identified gaps and opportunities at the intersection of HCI and ML, together with design guidelines for improving active and passive data collection on consumer devices (Chapter 2).

2. **Empirical evidence that persistent reminders sustain passive data collection**, demonstrating a 266% increase in completed background sessions relative to traditional scheduled background tasks on consumer mobile devices (Chapter 3).
3. **A demonstration that contextual authorization prompts improve passive data collection**, showing a 55.5% advantage over setup prompts for securing background event authorization (Chapter 3).
4. **A live, closed-loop mobile sensing system** that integrates passive sensing, on-device logistic regression, real-time active learning–timed EMA, and unlock-triggered EMA delivery on iPhone with Apple Watch (Chapter 4).
5. **Empirical evidence that unlock-triggered EMA delivery substantially improves compliance**, raising response rates from 64.2% to 80.0% under controlled within-subjects factorial conditions (Chapter 4).
6. **Evidence that predicting engagement from sensor context does not outperform the unlock trigger**, where prompts delivered only at model-selected moments were answered within 60 seconds 65.4% of the time against 63.9% for unlock-triggered prompts, a non-significant difference the model reached only by suppressing five of every six delivery opportunities (Chapter 4).
7. **Evidence that the effectiveness of active learning–timed EMA remains inconclusive**, established across two independent analyses. In the live deployment, a matched-budget control found uncertainty-selected labels no better than randomly selected ones; in a matched-budget prequential simulation on the CardioMate dataset, where candidate moments were unrestricted and the selecting model was also the scored model, the absence of an advantage held on both clinical targets. Neither target was predicted much better than chance, so the results bound the conditions under which the strategy was tested rather than establishing that it does not work (Chapter 4).

5.2 Summary of Findings

Chapter 2 presented a systematic review of mobile health sensing studies that combine active and passive data collection to support machine learning for adaptive health interventions. The review traced trends across health domains, data collection devices, and methodological innovations, documenting a clear shift away from specialized wearables and environmental sensors toward consumer smartphones and smartwatches paired with on-device machine learning. Despite this maturation, challenges remained on both sides of the data collection pipeline: active data collection continued to suffer from participant burden and missed EMA opportunities, while passive data collection was

hindered by inconsistent authorization and unreliable background runtime on consumer devices. The review identified concrete opportunities at the intersection of HCI and ML to address these challenges, including persistent reminders to sustain background data collection and signal data continuity, contextual and event-driven prompts to simplify authorization, and uncertainty-based active learning to align EMA timing with both model needs and participant availability. These opportunities directly motivated the empirical work presented in Chapters 3 and 4.

Chapter 3 evaluated the HCI-driven opportunities identified by the review through a user study built on iOS and Android mobile sensing applications. The study examined three questions: whether contextual or setup prompting better supports notification-initiated EMA, which authorization paradigm more reliably secures background event access, and whether persistent reminders outperform scheduled background tasks for polling-based passive data collection. Contextual prompts secured background event authorization 55.5% more often than setup prompts, and persistent reminders produced 266% more completed background sessions than scheduled tasks. Prompting placement mattered less for notification-initiated EMA, though persistent reminders themselves showed promise for engaging participants with active data collection, and some participants resisted installing them. These techniques carried forward into Chapter 4, where they were combined with adaptive EMA timing.

Chapter 4 combined those techniques with adaptive EMA timing by deploying a live, closed-loop mobile sensing system across three cohorts, evaluating unlock-triggered EMA delivery alongside real-time active learning-timed EMA. The final cohort used a within-subjects 2×2 factorial design that crossed delivery timing (unlock-triggered vs. random-timed) with active learning (on vs. off). The system collected momentary stress labels via EMA and passively recorded vitals, activity, and sleep data from a connected Apple Watch to drive an on-device logistic regression model.

The clearest result was the simplest one. Delivering the EMA shortly after a participant unlocked their phone raised the response rate from 64.2% to 80.0%, replicating an earlier preliminary observation under controlled factorial conditions. The mechanism is notification salience: a randomly timed prompt frequently arrives while the phone is stowed and the screen is off, whereas an unlock-triggered prompt appears at a moment the participant has already chosen to engage with the device, and at a natural break between tasks rather than in the middle of one. Unlike unlock journaling, which secures that attention by blocking the unlock itself, surfacing the prompt just after a naturally occurring unlock preserves the salience benefit without the friction. What makes the finding notable is its cost: a timing change requiring no additional infrastructure, computation, or battery produced a larger compliance gain than the algorithmic intervention evaluated alongside it. Together with the persistent reminders and contextual prompts of Chapter 3, it completes a consistent pattern in which low-friction, interaction-aware design improved data collection in every deployment that tested it.

The algorithmic intervention was less conclusive. Suppressing prompts based on model confi-

dence reduced the number of collected labels by 64.7%, but three limitations prevented me from determining whether active learning-timed EMA was effective. First, the GLMM discriminated barely above chance with an AUC of 0.566. Second, the on-device model selecting the labels was not the model under evaluation. Third, all selection decisions were restricted to when the participant unlocked their phone. A reanalysis on a separate dataset was also inconclusive. On none of the four targets I examined did active learning-timed EMAs perform significantly better than randomly selected EMAs: blood pressure spikes (AUC = 0.605), high stress (AUC = 0.658), pulse above 80 beats per minute (AUC = 0.682), and pulse above 80 with heart rate signals included (AUC = 0.846). In each case, either accuracy was already high enough that personalization was unnecessary, or it was too low for the model to select better labels. Active learning may still be beneficial in other situations, but I did not encounter one. A parallel attempt to time prompts by predicted engagement rather than model uncertainty pointed to the same conclusion. A model of whether a delivered EMA would be answered within 60 seconds worked, in that prompting only when it was confident raised the 60-second response rate from 54.1% to 65.4%. Still, that rate was statistically indistinguishable from the 63.9% unlock triggering achieved on its own (+1.5 percentage points, $p = .574$), and the model reached it only by suppressing five of every six opportunities. Its value proved complementary rather than competitive since prompts that were both unlock-triggered and model-selected were answered 75.6% of the time, better than either strategy alone. But here too discrimination was modest (AUC = 0.595), and most of what the model identified was already captured by the unlock event itself.

Read together, these chapters suggest that small, simple HCI adjustments can greatly affect both passive and active data collection. The persistent reminder, the contextual prompt for background event authorization, and unlock-triggered EMA timing each significantly improved data collection. None of these depended on how well the model performed. The more complicated ML intervention, active learning, showed no significant benefit over collecting randomly selected labels in any study reported here.

5.3 Limitations

Although each chapter discusses its own limitations, several apply to the dissertation as a whole. The participants in the two deployed studies came primarily from a single institution and were largely college-aged; the CardioMate reanalysis, though drawn from an older, clinically defined sample, includes only 20 participants. Platform asymmetries constrained what I could test: contextual authorization prompts were available only on iOS, and the closed-loop deployment ran on iPhone with Apple Watch via HealthKit, so results may differ on Android or under more restrictive battery, notification, or privacy configurations. One-month deployments cannot capture novelty effects, habit formation, or compliance decay, and the deployed outcome was a binary single-item stress measure that richer outcomes might resolve differently. Most consequentially, no target ex-

amined fell in the regime that uncertainty-based selection needs: a model accurate enough for its confidence to be informative, but not so accurate that personalization adds nothing. The repeated null therefore characterizes the targets available to me rather than the strategy itself.

Much of this rests on the predictive ceiling of consumer wearable sensing, which bounds what any adaptive-timing strategy can achieve. Across two datasets and several health targets, a model light enough to personalize on-device never discriminated well on a clinically meaningful outcome. The one target it did predict well was a physiological indicator chosen to validate the method rather than a clinical outcome.

5.4 Future Work

The most consequential direction for future work follows directly from Chapter 4: active learning–timed EMA is worth its added complexity only after a population model has been shown to discriminate meaningfully on the intended target, so the priority is closing that gap rather than improving the selection rule. That work runs two ways. Richer sensing means denser or multi-modal streams than a consumer smartwatch provides. Richer modeling is the other, since the ensembles published on the CardioMate data reached AUCs of 0.849 and 0.803 whereas the intercept-only mixed model reached 0.658 and 0.605. The mixed model buys label efficiency and on-device tractability at a cost in discrimination. Reconciling the two would test whether selection and personalization might pay simultaneously.

5.5 Closing Remarks

The work presented across these chapters shows that small, simple HCI adjustments can have a large effect on both passive and active data collection. The persistent reminder, the contextual prompt for background event authorization, and unlock-triggered EMA timing each significantly improved data collection, and each did so without additional infrastructure or computation. The more complicated ML intervention, active learning, showed no significant benefit over collecting randomly selected labels in any of the studies reported here. It may prove beneficial in other situations, but I could not find one in which it significantly improved model performance. The conclusion I draw is not that complicated interventions like active learning–timed EMAs should be set aside, since they may well pay off in the right circumstances, but that in this work it was the simple HCI interventions that provided clear and significant improvements in data collection.

APPENDIX A

DATA EXTRACTION SHEET

Table A.1: **Data extraction sheet for 77 papers included in the systematic review.** The datasheet details the disease studied, study specifics, active and passive data collection methods, and the corresponding analysis and results. The data is categorized by the type of device used for data collection: phones, watches (any device worn on the wrist, regardless of time-display functionality), other wearable devices, and environmental sensors.¹

Paper	Disease	Study Details	Active Data	Improve Active Collection	Passive Data	Improve Passive Collection	Analysis & Results
Phone							
Aung 2016 [151] IEEE	Chronic Pain	3 Phase Study: Activity, Speech, Face Detection	EMA: Pain Level via Keppi	Portable device to report pain level (Keppi)	Phone: Physical Activity (GPS, Wi-Fi Location, accelerometer) Emotional State (Audio)	SAINT was developed to assist sensor data collection and signal processing inside phone. USED pub-sub.	SAINT (Various Models)
Huang 2016 [152] GS	Social Anxiety	18 Participants, undergraduate College psychology Students 10 Days	In-Person: Brief Psychological questionnaires.		Phone: GPS Used Foursquare to identify POI, home, work, food & leisure, and transportation.	Used Foursquare to identify POI, home, work, food & leisure, and transportation.	Correlation and LR showed significance.
Wang 2016 [61] GS	Schizophrenia	48 Participants recruited, 18 dropped out. 1 Year.	EMA: 3/Week on Mon., Wed., Fri. 10 Multiple Choice Questions	Participants are discarded if they have less than 50% EMA responses.	Phone: (Android, ChrossCheck App) Activity, Sleep Duration, Sociability, Accelerometer, Light Sensor, Location, App Usage	Days with fewer than 19 hours of sensing are discarded.	Bivariate Regression Analysis, GBRT. Accuracy: Mean Error rate of 7.6%
DeMasi 2017 [153] ACM	Well Being	87 Participants, College Students 8 Weeks	EMA: 4/Day, Mood and Energy Intro/Exit Survey	Had limited user participation	Phone: Location	Had incompatibility issues.	SVM, RF, LogR Accuracy: 0.75 better than baseline. Only 33 had enough data for analysis
Saha 2017 [74] GS/ACM	Mood Stability	51 Participants 5 Weeks	EMA: 4/Day (Quedget) Photo to Select Mood	Used Unlock Journaling to collect EMA	Social Media: (Facebook) Linguistic Content of Status updates		Naïve Bayes, LogR, RF, SVM. Accuracy: 91%

Paper	Disease	Study Details	Active Data	Improve Active Collection	Passive Data	Improve Passive Collection	Analysis & Results
Phone							
Spanakis 2017 [154] WofS	Eating	Study One: 100 Participants: 57 overweight, 43 healthy-weight 2 Weeks Study Two: 100 Overweight 8 Weeks	EMA: Every 2 hours Food Cravings, Emotions	Uses Ecological Momentary Interventions (Includes Assessments and Interventions)	Phone: Location, Activity		Clustering EMI can Improve Eating Habbits (Adaptive Intervention)
Bae 2018 [64] WofS	Alcohol Consumption	38 Participants Non-Treatment Seeking, Heavy Drinkers 28 Days	EMA: 1/Day	Initiated via Text	Phone: (AWARE) Time, Accelerometer, Phone Calls, Text Messages, Keyboard Usage		RF Accuracy: 90.9%
Cai 2018 [155] ACM	State Affect (Mood)	220 Participants 2 Weeks	EMA: 6/Day Once Every 2 Hours Mood		Phone: (Sensus App) Accelerometer, Phone Calls, Text Messages, Location		Statistical Analysis (ANOVA) Accuracy: Correlation is significant
Singh 2018 [78] GS	Risk-taking Propensity	50 Participants (33 male, 17 female) 10 Weeks	Surveys: Risk Propensity, Demographic, Personality		Phone: GPS, Message, Phone Log (Android smartphone app)	2 participants turned off location.	Activity, Reciprocity measure, feed into a OLS regression, Plus various ML algorithms. Accuracy: Outperforms demography-based models by 39%.
Wang 2018 [156] ACM	Schizophrenia	150 Participants 12 Months	In-Person: 7-Item Brief Psychiatric Rating Scale EMA: 3/Week		Phone: (CrossCheck App) Mobility, Sleep, Phone Usage, Phone Calls, Text Messages, Ambient Light, Ambient Sound		GBRT Accuracy: MAE +/- 1.45 with EMA +/- 1.59 w/o EMA
Zhou 2018 [11] WofS	Physical Activity	64 Participants Control: 10,00 Step Goal Intervention: Adaptive Step Goal 10 Weeks	Step Goal via Notification		Phone: (iOS) Steps	Data Collected via HealthKit	Behavior Analytics Algorithm determined goal. Results: 95% Less Step Reduction.

Paper	Disease	Study Details	Active Data	Improve Active Collection	Passive Data	Improve Passive Collection	Analysis & Results
Phone							
Zhang 2019 [77] GS/ACM	Mood Instability	86 Students Cohort 1: 42 Students 1 month. Cohort 2: 44 Participants 2 weeks. 68 participated 2 weeks, 18 dropped out.	EMA: 3/Day Ekman's Discrete Mood Survey		Phone: Accelerometer, Microphone, Audio, Light, Screen Status, WiFi, App Usage.	Background thread ran every 10 minutes. Some participants switched off or unauthorized, which were removed from data.	SVM, MLP, MVDS, CDMF. Accuracy: 78.33% F1: 77.78%
Etienne 2020 [157] IEEE	Stress	12 Participants College Students 2 Weeks	EMA: 3/Day Stress Level		Phone: Accelerometer, Light, Audio, Location		Transfer Learning, SVM Accuracy: 90%
Jacobson 2020 [65] WofS	Depression	31 Participants 7 Days	EMA: 1/Hour Mood and Heart Rate Assessment	Participants pre-reported awake hours.	Phone: Location, Heart Rate, Phone Calls, Weather		XGB, RF Accuracy: CI of 0.587
Rhim 2020 [79] GS/ACM	Subjective Well-being	78 college students 4 Months	EMA: 1/Daily Life Satisfaction, Emotional Experience		Phone: Activity, GPS, App Usage, and Screen Status.	Contacted participants when data was missing to resume collection.	Hierarchical Regression DT, RF Accuracy: Up to F1: 70%
Ameko 2020 [158] GS	Emotion Regulation	110 Participants 81 Female 29 Male 4 Excluded 5 Weeks	EMA: 6/Day Social, Interaction, Anxiety Level MetricWire Mobile App	Reminder notifications are sent if not completed in 30 minutes and removed if not completed in 45 minutes.	Phones: Activity and GPS		LogR and T-Tests Accuracy: Significant
Obuchi 2020 [159] ACM	Brain Functional Connectivity	105 Participants 1st Year Students 79 Days	In-Person: fMRI		Phone: (NeuroSense App) Location, Activity, Speech, Sleep	Used on-device pre-trained model to detect speech to preserve privacy.	SVM Accuracy: F1 0.793
Rashid 2020 [86] ACM	Social Anxiety	80 Participants with Social Anxiety. 5 Weeks	EMA: 1/Day at 10PM for two Hours Subject Measures of Anxiety		Phone: (MetricWire) GPS, Pedometer, Accelerometer, Activity, Phone Calls, Text Messages	Method to impute missing values: 1 MICE: regression, 2 Matrix Completion, 3KNN, 4 Last Observation Carried Forward	RF, MERF, CatBoost Accuracy: Imputation of missing values reduces error 22%

Paper	Disease	Study Details	Active Data	Improve Active Collection	Passive Data	Improve Passive Collection	Analysis & Results
Phone							
Buda 2021 [75] ACM/GS	Daily Happiness	303 Participants Included Cohort 1: 221 users from 5 different countries 2nd Cohort: 481 college students 2-3 Weeks	EMA: 5/Day Momentary Happiness Daily Happiness question (Likert)		Phone: Usage Pedometer, GPS, Ambient Light, Noise Level, Battery Level, Lock/Unlock, Phone Calls.	Users could opt out of certain types of data collection. iOS users were dropped from the results due to lower numbers and inconsistent data collection.	LogR, RF, Multilayer Perception and XGB Accuracy: 13% improvement in AUC and 27% in F-score.
Dong 2021 [10] ACM	Influenza Detection	448 Participants 24 States 1 year.	EMA: 1/Day Report Flu like symptoms		Phone: (ReadiSens) GPS, Wifi, Mobility, Social Interactions, App Usage, Activity	2700 Started, on 448 had 14 days of data.	GNN, LogR, SVM, RF, XGB, MLP Accuracy: F1 Score 0.8762 on GNN
Dong 2021 [8] IEEE	Mental Health	2700 Participants. Collection period over a year, 14 Days required.	EMA: Every 12 Days to Evaluate Mental Health		Phone: (ReadiSens) GPS, Wifi, Mobility, Social Interactions, App Usage	Developed GNN to deal with low number of labels.	Graph Instance Transformer Accuracy: Beat baseline by 8.8%
Meegahapola 2021 [160] GS	Eating	84 Participants College students in Mexico 60 Days	EMA: 3/Day Food intake self-reports: Eating Episodes, Food categories Concurrent Activities, Stress Level, etc.	Sent EMA after eating episodes to not interrupt and modify users.	Phone: App Usage, Accelerometer, Battery Events, Screen Events, Location	improve	Pearson and Point-Biserial correlation, RF, NB, GBM, NN, XGB, AdaBoost, SVM Accuracy: 83.49% - 87.81%
Srikanthan 2021 [76] IEEE	Traumatic Brain Injuries	375 Participants, 17 with TBI Recruited through Ads 12 Weeks	EMA: Daily Health Questions		Phone (iOS): Accelerometer, gyroscope, magnetometer, pressure, altitude, battery, pedometer, accessibility features Sensor data polled every second	Users had the option to turn off polling. Users with insufficient data were not included in results.	DEC, XGB Outlier Detection Accuracy: 88% and Sensitivity of 74%

Paper	Disease	Study Details	Active Data	Improve Active Collection	Passive Data	Improve Passive Collection	Analysis & Results
Phone							
Kim 2022 [22] ACM	Stress	36 Participants College Students 25 Day 10 Day Model 15 Day Prediction	EMA: 4/Day at 4 Hour Intervals Stress level	11 PM Was too late for EMA.	Phone: (MindScope) Accelerometer, Activity, GPS, App Usage, Noise Levels	Users Provide Common Locations. Explainable AI (XAI) Reports to User Reasons Why Stress is High (Location, Noise, etc). Provides Stress Intervention Ideas	Explainable AI, XGB Algorithm Used Statistical Analysis
Meegahapola 2021 [9] GS	Alcohol Consumption	241 Participants Young Adults in Switzerland 3 Months, Weekends only	EMA: Hourly from 8PM-4AM Log Drinks		Phone: (Android) Accelerometer, Battery, Bluetooth, Location, Wifi, App Usage, Screen Usage, Proximity	More details in additional papers. Data separated by continuous sensing and interaction sensing.	Pearson, and Point-biserial correlation. Accuracy: 75%-86%
Meegahapola 2022 [139] GS/ACM	Mood	678 Participants Eight Countries.	EMA: 20/Day (2 Hours to Complete) Start/End of Day		Phone: (Android) Steps, Location, Wifi App Usage, Screen Events, Phone Usage, Activity, Ambient Light, Notifications		Built both population and personalized (hybrid) models. Models for all participants, and each country. Accuracy: AUROC-score 0.78-0.98 and 0.76-0.94
Wang 2022 [84] GS/ACM	Mental Health of First-Gen College Students	80 Participants 27 First-Generation Students 1 Year	EMA: Weekly Mental Health: Depression and Anxiety (PHQ-4)		Phone: Activity, GPS, Phone Usage, Locks/Unlocks, Sleep Regularity Used iOS and Android API to Recognize Activity	Used App Center to monitor and update smartphone app. iOS used VoIP push alerts to run background tasks. Had dashboard to monitor collection activity	LogR, RF, GBM. LSTM, sub-NN. Accuracy: F1-0.71 (Best Deep learning)
Bae 2023 [161] WofS	Alcohol Consumption	75 Participants With Risky Behavior 14 Weeks	EMA: 2/Week Report Alcohol Consumption	SMS used to initiate EMA	Phone: (AWARE) Accelerometer, Location, Wifi, Gyroscope, App Usage, SMS, Ambient Light, Battery	Most informative features: Time of Day, GPS	XGB Accuracy: 95%

Paper	Disease	Study Details	Active Data	Improve Active Collection	Passive Data	Improve Passive Collection	Analysis & Results
Phone							
Bardram 2023 [88] GS	Type 2 Diabetes	12 Participants 6 Weeks Feasibility Study	EMA: Blood Glucose, Smoking, Weight, Alcohol, Lifestyle, Emotional distress, well-being, etc.		Phone: Battery, Screen, Noise, Location, Activity, Weather, Steps	Mentions event-based vs. polled-based collection. Health App (iOS) could have improved collection of Blood Glucose.	Analyzed the usability and feasibility of the app. No ML analysis
Lustrek 2023 [67] GS/ACM	Stress	Proposes design of JITAI, but does not include a study.	EMA: Diaries	Uses AI to automatically fill in parts of the EMA.	Phone: Detect context and stress from passively collected data.		AI helps users complete EMA and detects context and stress levels. Adaptively suggests exercises to reduce stress.
Ning 2023 [162] GS	Mood Disorders (Brain Health)	85 Participants 4-5 Weeks	In-Person: 2 Cognitive Assessments 14 Days Apart		Phone: (BiAffect Keyboard) Keyboard Usage	Used Keyboard Interactions as a passive sensing stream.	Statistical Analysis Diurnal Differences and Typing difference showed differences.
Vandelanotte 2023 [163] WofS	Physical Activity	Proposes design of JITAI but does not include a study.	Q&A: Physical Activity Related Question to proposed to generative	AI Uses generative AI as means of active data collection.	Phone: GPS, Location, Weather	Use location to determine weather and adapt “Nudges”	Proposal Only
Wang 2023 [164] GS/ACM	Auditory Verbal Hallucinations	384 Participants 30 Days	EMA: 4/Day Audio Diary		Phone: (Android) GPS, Lock/Unlock, Phone Usage, Activity, Ambient Light, Sleep		Deep Learning (RNN) Accuracy: F1: 0.78 (Voice Diary Only)
Nepal 2024 [83] GS/ACM	Mental Health	215 Participants Students at Dartmouth College 123 Completed to Graduation. 4 Years	EMA: Weekly Anxiety and Depression	75% Completeness Score	Phone: Activity, Locations, Phone Usage, Audio Play, Sleep	90% Completeness Score 20% Location Missing, 12% Audio, 22% Steps Updates Cause Challenges in Longitudinal Studies. Android introduces changes in background tasks.	Summary Analysis (No ML).

Paper	Disease	Study Details	Active Data	Improve Active Collection	Passive Data	Improve Passive Collection	Analysis & Results
Watch							
Gjoreski 2016 [165] ACM	Stress	5 Participants	EMA: Random Periods Stress Level		Watch: (Empatica) Activity, GSR HR, BVR		Accuracy: 92%
Biel 2018 [85] GS	Eating	122 participants (18-26 years old) 10 Weekdays	EMA: Food Intake with Photos, Where, With Whom EMA: End of Day Survey	Scheduled Notifications at mealtimes, end of day survey.	Watch (FitBit): Activity Data	End of Day survey to fill in missing FitBit Data.	RF Accuracy: 84%
Sefidgar 2019 [166] GS/ACM	Discrimination Events	209 Participants 1st Year College Students 6 Months (2-Terms)	EMA: 2/Week Discrimination Event 4/Week for 2 weeks.		Phone: (AWARE) Location, Screen Status, Call Logs, Activity Watch: (FitBit Flex 2) Steps, Sleep	Did not use FitBit and Phone all the Time	LR, Statistical Analysis Accuracy: N/A
Theilig 2019 [66] IEEE	Mental Health	6 Users 29 Days	Personal Surveys: Daily Activities	EMA to test receptivity. Time influenced by ML Model.	Watch: Activity Data	Read Health Data from HealthKit (iOS), or watch app. Includes JITAI feasibility.	ML model predicted the best time to send EMA about mental health from Activity Schedule (LSTM) Accuracy: 58% Activity Recognition
Hafiz 2020 [54] GS	Bipolar Disorder	15 Participants 9 healthy 6 with DB 7 Days.	EMA: (FitBit) 1/Day 3 Cognitive Tests (Choice Reaction Time, N-back, Stoop)	Used watch alarms as reminders. Monetary Incentive Paid per test taken.	Watch (FitBit): GPS, Activity, Sleep	Paid per night watch was worn.	KNN, SVM, RF, XGB. Accuracy: 74% (KNN)
Kunchay 2020 [73] ACM	Alcohol and Marijuana Consumption	1 Participant 1 Week	EMA: 5/Day Details of Substance Use	Used microEMA on Watch to reduce burden.	Phone: (AWARE) Accelerometer, Gyroscope, App Use, Noise, Batter, Phone Calls, Text Messages, Location, WiFi, Weather Watch: (HealthKit) Steps, Stand Hours, Activity, Heart Rate, Calories, Sleep		Feasibility Study With Only 1 Participant

Paper	Disease	Study Details	Active Data	Improve Active Collection	Passive Data	Improve Passive Collection	Analysis & Results
Watch							
Adler 2021 [60] GS/ACM	Stress	775 Participants 14 Months During Medical Internship	EMA: Daily Mood (Likert Scale)	Needed to filter participants with low data quality.	Watch: (FitBit Charge) Heart Rate, Steps, Sleep	Used EMA to help ensure FitBit Data synced to secure storage. Dropped hours without data.	Generalized Estimating Equations.
Bonaquist 2021 [167] IEEE	Mental Health	40 Participants Adolescents	EMA: (REDCap) 1/week PHQ-9		Phone: (AWARE) GPS, Accelerometer, Phone Calls, Wifi, Phone Usage Watch: (FitBit) No details included		XGB, XGBRF, RF, ExtraTrees, GBM, AdaBoost, Light GBM, Catboost Accuracy: RMSE 5.3 (XGBRF)
Lee 2022 [72] GS	Sleep	Preliminary: 136 Participants (University) and 54 (Major Airline) Initial Deployment: 11 Participants, 2 Weeks Deployment: 20 Participants, 8 Weeks	Initial Deployment: Online: Daily Sleep Survey Deployment: Online: 2/Day	Presents a Dashboard to User about progress and suggestions Overall compliance rate from 0.46 to 0.58	Watch: (FitBit) Fitness, Bedtime, Wake Time, Heart Rate, Steps Calendar: (Google) Updates	Implements JITAI for sleep. Adjusts for missed guidance	
Tsai 2022 [52] WofS	Panic Attacks	59 Participants with Panic Disorder 1 Year	In-Person: Panic Disorder Severity Screening EMA: Every 2 Weeks Anxiety Levels		Phone: Physiological Data Watch: (Garmin or FitBit) Steps, Heart Rate, Activity		RF, DT, Linear Discriminant Analysis, Adaptive Boosting, XGB, Regularized Greedy Forests. Accuracy: RF 67.4%-81.3%
Velmovitsky 2022 [87] ACM	Stress	1 Participant (Prosposal)	EMA: */Day Stress		Phone: (iOS HealthKit) Watch: (Apple) ECG (Empatica E4) HRV	Used iOS HealthKit to collect passive data.	RF, SVM Accuracy: 85% (RF) 70% (SVM)

Paper	Disease	Study Details	Active Data	Improve Active Collection	Passive Data	Improve Passive Collection	Analysis & Results
Watch							
Wang 2023 [168] GS/ACM	Social Anxiety	46 Participants College Students During Social Conversations Through Video Calls.	Online: Post Survey		Watch: (Empatica E4) Heart Rate, EDA, Accelerometer, Temperature		kNN, SVM, NB, Tree, AB, RF, MLP Accuracy: Best (MLP) F1: 0.69
Arakawa 2023 [69] GS	Hyperactivity in Children	61 Children	25 with hyperactivity Parents provided labels on hyperactivity.	ML models originally used parent labels, but later used contextualized data. 85% vs 82%.	Watch: (Apple Watch) Custom App: Heart Rate, Location, Bluetooth and Accelerometer	Apple's Library provided best data collection results for accelerometer.	SGD, DT, RF, GB Accuracy: 85.2%
Can 2023 [59] IEEE	Mood and Emotion	14 Participants 7 Days (Not Consecutive)	EMA: 1/3 Hour Session Perceived Emotion (7 Likert Questions). Initiated by emails.	Passive data was discarded when participants missed an EMA.	Watch: (Empatica E4) Accelerometer, Heart Rate, Temperature, EDA		RF, DT, kNN, MLP, LR, and SVM. LSTM and Bidirectional LSTM provided best results. Accuracy: 95%
Toshnazarov 2024 [169] IEEE	Stress	Lab: 26 Participants Real Life: 18 Participants 2 Weeks	EMA: 12/Day Stress Dynamics		Phone: Activity, Phone Calls, Screen State, Location Watch (Samsung) HR	Used an always running foreground service (Android)	Accuracy: F1 0.84 Lab, 0.71 Real Life

Paper	Disease	Study Details	Active Data	Improve Active Collection	Passive Data	Improve Passive Collection	Analysis & Results
Wearables							
Chatterjee 2016 [170] GS/ACM	Smoking Cessation	61 Participants 24 Hours Before Quit Date, 72 Hours After Quit Date	EMA: Cravings and Physical Activity	Daily in-person check-in to verify data.	Phone: GPS Watches: Accelerometer and Gyroscope. Wearable: (AutoSense): ECG, Accelerometer, Respiration.	improve	Developed mCrave model. Accuracy: 72%
Sarker 2016 [171] ACM	Stress	38 Participants	EMA: 5.8/day Stress, Drugs, and Smoking		Phone: Location, Accelerometer Wearable: Breathing Rate, ECG, HRV		Accuracy: F1 score 0.71

Paper	Disease	Study Details	Active Data	Improve Active Collection	Passive Data	Improve Passive Collection	Analysis & Results
Wearables							
King 2019 [68] ACM	Stress in Pregnancy	18 Women Not Pregnant 16 Lab Activities Real World: 18 Pregnant Women 2 Days	EMA: Hourly for 12 Hours Stress Test	Developed a ML Model to determine which EMA questions to use, prior to a study, to help reduce EMA burden.	Wearable: (Biostamp) ECG, (Polar H7) Heart Rate (Neurlog GSR), GSR/Skin Conductivity Only Biostamp in real world study.		SVM, DT, Ada Boost, NB, NN WorriedStress was the best EMA question to use.
Delmastro 2020 [172] IEEE	Stress detection with mild cognitive impairment.	9 Older adults, 5 women, 4 men.	In-person tests: SCWT	Data collected in lab	Wearables: (Bio Harness) Heart Rate, HRV, EDA, ECG (Shimmer3) EDA	Data collected in lab.	BN, SVM, k-NN, c4.5 DT, RF AB Accuracy: 80%-95.8%
Chatterjee 2020 [56] GS/ACM	Smoking “opportunity contexts”	126 Smokers 55 Female 71 Male. 4 Days Pre-Quit, 10 Days Post-Quit.	EMA: 12/Day Cigarette Availability and Smoking Allowance	67.2% compliance. 36 did not provide enough data to be included.	Wearable: (Chest and wristband) ECG, Respiration, Accelerometer During Waking Hours	Preprocessed data to account for noisy data.	RF, SVM-RBF, LogR, Adaboost classifiers Accuracy: F1: 74.3%
Gupta 2020 [142] GS/ACM	Couple Conflict	87 Couples 18-25 Years old 1 day	EMA: Hourly (9AM-12PM). Mood and Quality of Interactions		Phone: GPS 3-minute audio sample every 12 minutes. Watch: EDA (Q sensor) Wearable: ECG, (Actiwave)	Participants were provided a smartphone for the 1-day study.	Subpopulation clusters. K-means algorithm and FNN Accuracy: F1-76%
Mishra 2020 [91] ACM	Stress	27 Participants 3 Days	EMA: Every 30 minutes, for 8 hours per day. Stress level (1-5)	Collected on Watch Participants can self report	Watch: (Amulet) Accelerometer, HRV, Activity Wearable: (Polar H7)		SVM, RF Accuracy: F1-score: In field 0.72 (0.66 without activity)
Zhang 2020 [82] GS	Eating	20 Participants 10 Semi-free-living Study 10 Free-living Study 11 with 9 w/o Obesity 420 Total Hours Collected	Video	Used Video to provide input labels. Participants could delete video.	Wearable: (Necklace) IMU, Proximity, Ambient Light. Detects Eating Episodes.	Need to ensure devices time is correct.	GBM Classifier Accuracy: F1 81.6% and 77.1%

Paper	Disease	Study Details	Active Data	Improve Active Collection	Passive Data	Improve Passive Collection	Analysis & Results
Wearables							
Elvitigala 2021 [173] ACM	Stress	10 Participants Office Workers 4 Week	In-Person: Weekly, Rate Agreement with Detected Stress	Allowed Users to Select Interventions	Phone: Calendar Integration Wearable: (Inertial Measuring Unit Sensor Tag), Mount on Shoe		Linear-Discriminant Analysis ML Model
Kumar 2022 [71] GS/ACM	Ambulatory Arrhythmia	24 Participants 2 weeks	EMA: Daily Stress Level, Sleep, Dietary Details Any Time: Self-Report of Symptoms	UI included visual display of both active and passive data collection rates. Helped participants fix collection errors.	Phone: Activity, Location, Sleep, Weather, Noise Level. Wearable: (EcgMove4): ECG HR, HRV, and MET		Only analyzed the feasibility of the system.
Ullah 2022 [174] GS/ACM	Smoking	92 Participants	EMA: 12/Day (3 per 4 Hour Block)		Phone: GPS Chest band: ECG, Respiration Watch: (both Arms) Accelerometer/gyroscope	Used Event-based encoding to reduce noise.	Deep learning model (LSTM). Includes historical influence and recent Accuracy: 85% of lapses
Daryabeygi-Khotbehsara 2024 [175] WofS	Activity Recognition	15 Participants	In-Person: Cameras to Record Activity		Wearable: (SORD) Accelerometer, Gyroscope (ActivPal) Detects Standing, Walking, Lying, and Sitting	Compared Devices	Deep Learning Accuracy: Above 95%
King 2024 [63] WofS	Emotional State	45 Participants 10 Days	EMA: 10/Day Access Mood, 13 Questions, 9 negative, 4 positive	Initiated Via Text Message, Paid per response.	Watch: EDA, Temperature, Accelerometer Wearable: (Chest Patch) ECG, Accelerometer		k-means Clustering, RF and NN Accuracy: Decrease negative effect by 3 points
Presseller 2024 [53] WofS	Binge-Spectrum Eating Disorders	22 Participants 2 Weeks	EMA: Self-Initiated Reports	Used high incentives (\$500) for compliance. Participants were monitored and reminded	Wearable: (Dexcom G6 CGM) Blood Glucose		RF Accuracy: 82% for Meals, 78% Non-Meals

Paper	Disease	Study Details	Active Data	Improve Active Collection	Passive Data	Improve Passive Collection	Analysis & Results
Environment Sensors							

Paper	Disease	Study Details	Active Data	Improve Active Collection	Passive Data	Improve Passive Collection	Analysis & Results
Environment Sensors							
Diethe 2018 [176] GS	General Health	30 Homes	EMA: Questionnaires		Home Sensors: Temperature, Humidity, Noise Level, Luminosity Video Wearable: Accelerometer	Actors act out a script to create labels. Unsupervised learning. Post-Hoc analysis of data.	Discussed Methods LR, Unsupervised approaches as well. Accuracy: Not reported.
Chen 2019 [55] GS	Moderate Cognitive Impairment and Dementia	119 Participants 82 Healthy 32 MCI or Dementia 12 Weeks	EMA: Daily, 2-Questions, Bi-Weekly Assessment Test (iPhone app)	Participants contacted directly to use devices when not collecting data.	iPhone: Phone Usage, Messages, Unlocks, App History. Apple Watch: Heart Rate, Breathing Beddit: Sleep Interpolation for missing Heart Rate. Non-compliance used as a feature.		EXGB. Accuracy: AUROC = 0.8
Tong 2019 [57] GS/ACM	Multiple Sclerosis (Fatigue and Health State)	198 Patients 6 Months	EMA: Weekly Surveys, Daily Questions	24 patients did not complete surveys. Only 73 attempted daily questions.	Watch: Sleep and Activities Scale: Weight Sleep Tracker: Sleep		Regression Model Accuracy: MAE ranges from 0.58, 1.0, 0.093, 0.097 to predict scores.
Baglione 2020 [21] IEEE	Medication Adherence	Study Protocol, No Participants	EMA: 1/Day Stress Ratings.		Phone: Acceleration, Location, Sound, Proximity, Communications, App Usage, Light Level Wearables: Heart Rate GSR, ECG, Activity Beacons: Proximity, Temperature.	Use ML and context to turn off sensors to improve energy efficient sensing.	Develop personalized models to adapt to various patients.
Goldstein 2020 [177] WofS	Weight Control	121 Participants Overweight 10 Weeks	EMA: 6/Day Eating Relapses and Causes	EMA determined cause.	Scale: Weight		DT Accuracy: 79.8% Lapse Prediction

Paper	Disease	Study Details	Active Data	Improve Active Collection	Passive Data	Improve Passive Collection	Analysis & Results
Environment Sensors							
Kao 2020 [70] GS	General Health	212 Health Workers 3 Cohorts 10 Weeks	EMA: 1st Daily job, health, personality 2nd Daily - Psychological Flexibility or Capital	Participants could monitor compliance rate in app	Fitbit: Heart Rate, Steps, Sleep, Activity OMSignal Garment: ECG, breathing, motion. Unihertz Jelly Pro: Accoustic Data Phone: Phone Usage, Battery Life, Screen time Hospital Sensors: Location in Hospital, Light, Temperature, Motion	Participants could monitor their data quality in the app.	Radial Basis function. Personalized models used as well as a baseline model. Accuracy: Better than baseline w/o historical data.
Cook 2021 [81] IEEE/GS	Clinical Scores related to Brain Function, Mobility, and Verbal Ability.	21 Participants, < 45 Years Old 1 Month	EMA: Functioning, Socialization, Activity, Fatigue and Mood In-Person: Clinical Assessments		Watch: Location, Accelerometer, Gyroscope Home: Infrared Motion/Light, Temperature, Door	Used two watches to collect data continuously. Still missing time periods that was filled in with RF regression.	GBM Accuracy: Best 0.881
Saha 2021 [178] GS/ACM	Psychological Constructs	754 Participants 1 year	During Enrollment: Cognitive Ability, Personality Traits, Anxiety, Well-Being Tests		Phone: Physical Activity, Mobility, Phone usage Watch: (Garmin) Step Count, Sleep Bluetooth Beacons: Desk Use Social Media: Facebook	Includes several different devices and social media.	Compares general models, clustering, and personal models. Generalized: Better for cognitive ability Contextualized: Better for personality traits, anxiety, and sleep quality

Paper	Disease	Study Details	Active Data	Improve Active Collection	Passive Data	Improve Passive Collection	Analysis & Results
Environment Sensors							
Shin 2021 [179] ACM	Stress	50 Participants In-Lab Study	In-Person: Self Report		Watch: (Empatica E4, MetamotionR) HR, HRV Wearable: (Zephyr Bioharness) Motion Environment: (Radar) Reflected Radar Signals		Accuracy: 76.22% (11.57% better than wearables)
Booth 2022 [58] IEEE	Stress	606 Participants Full-Time, Information Workers 4 cohorts 1 Year	EMA: Daily for first 56 Days. (Had 4 Hours to Complete) Semi-Random Schedule (8AM, Noon, 4PM)	Sent Notifications via Text Message Required 7 EMA for inclusion.	Phone: Phone Usage, Locks/Unlocks, GPS Watch: (Garmin Vivomart 3) Heart Rate, Steps, Activity Beacons: (with key fob) Home, Work, Weather, Other Participants	Includes Literature Search about Missing Data, Reliability for Stress Studies. 9.5% of the data was missing.	EN, RF, FFMLP, GRU, and LSTM. Accuracy: 0.19 Spearman correlation = 0.39 Cohen's d
Teh 2022 [80] WofS	Mild Cognitive Impairment	49 Participants 21 with MCI, 28 Healthy 2 Months	In-Person: Cognitive Assessment		Watch: Heart Rate, Steps Home: Motion Sensors, Door Sensors, Bed Sensor Beacons: Key and Wallet	Missing data, caused by not wearing watch, beacons. Fuzzy ARAM model deals with missing values.	Fuzzy ARAM Accuracy: 58.3% with missing data, 63.6% w/o missing data
Zakaria 2022 [180] GS	Sleep	46 Participants College Students 4 Weeks Additional Smaller Study	In-Person: Sleep Health Assessment. Additional Study: EMA: Sleep Lop		Wifi: Device usage. Wearable: (Oura Ring) Sleep	Wifi: Included all user devices, instead of just smartphone.	RF with Infinitesimal Kackknife Variance Estimation Accuracy: 5% uncertainty rate.
Liang 2023 [51] GS	Stress	10 Participants 4 weeks.	EMA: Every 30 minutes Stress-level		Watch: (Garmin) Stress Level (Ground Truth) Home: Millimeter Wave Radar to Detect Stress		NN. Accuracy: 80%

Paper	Disease	Study Details	Active Data	Improve Active Collection	Passive Data	Improve Passive Collection	Analysis & Results
Environment Sensors							
Wyant 2023 [62] WofS	Alcohol Use Disorder	154 Participants 3 Months	EMA: 4/Day (Survey) 4/Day Alcohol Use, Current State/Mood, Exposure to Risks, Sleep 1/Day (Audio) About Recovery	Initiated Via Text Message	Phone: Location, Call Logs, Text Messages Watch: (Empatica E4) Heart Rate, EDA, Temperature Home: (Beddit) Sleep Monitor	Quit using watch due to complications and syncing issues.	Measured Acceptability Results: Passive methods were more acceptable.

¹Machine-Learning Abbreviations: LR: Linear Regression, LogR: Logistic Regression, DT: Decision Tree, RF: Random Forest, GBM: Gradient Boosting Machines, NB: Naive Bayes, SVM: Support Vector Machine, RF Random Forest, GBRT: Gradient Boosted Regression Trees, XGBRF: XGBoost Random Forest, EN: Elastic Net, MLP: Multi-Layer Perceptron, FFMLP: Feed-Forward Multi-Layer Perceptron, GRU: Gated Recurrent Unit, DEC: Deep Embedded Clustering, MVMDs: Multi-View Multi-Dimensional Scaling, CDMF: Collective Deep Matrix Factorization

APPENDIX B

OVERVIEW OF THE DIFFERENCE BETWEEN AUTHORIZATION PROCEDURES ON ANDROID AND IOS DEVICES

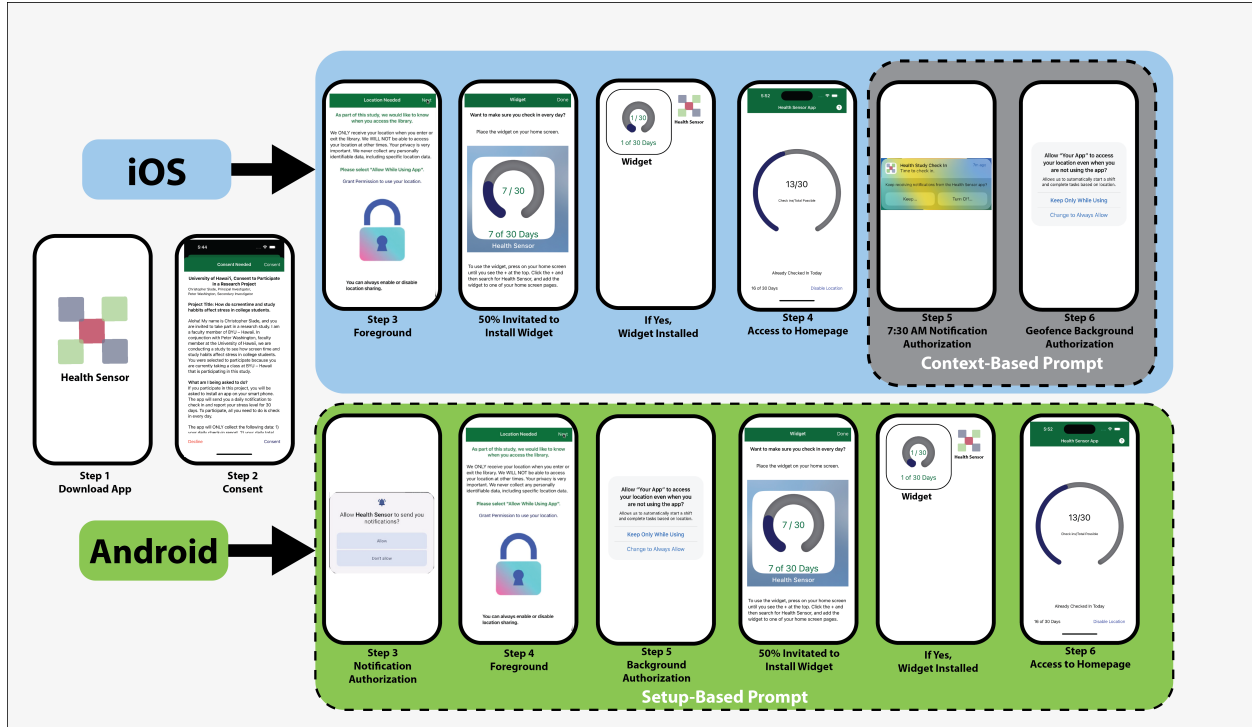


Figure B.1: Overview of the difference between authorization procedures on Android and iOS devices. On iOS devices, both notification and background location authorization prompts are received in context. The notification authorization prompt is presented with the first EMA reminder notification, and the background location authorization is presented when the first location event occurs. For Android devices, those prompts are presented during the initial launch of the application.

APPENDIX C

PER-PARTICIPANT STRESS PREDICTION RESULTS (COHORT 3)

Table C.1: **Per-participant stress prediction results (Cohort 3).** All columns come from a single leave-one-participant-out evaluation: for each held-out Cohort 3 participant, a Bayesian random-intercept GLMM is fit on the pooled labeled EMAs of all remaining Cohort 2 and Cohort 3 participants. Population AUC and Brier use that model’s fixed effects only (no personalization); base and active Brier add a MAP random intercept estimated from the held-out participant’s base (fixed-timed) versus active (active learning–timed) labels. Lower Brier is better. Sample sizes n_{base} and n_{active} count the number of labeled EMAs available from the held-out participant in each personalization condition, and n_{test} is the size of the chronologically held-out 30% test set. Δ denotes the base-personalized minus active-personalized Brier; positive values favor active learning, matching the convention used in Chapter 4. Mean population AUC and Brier are computed over all 14 evaluated participants; mean base, active, and Δ are computed over the 13 participants with both personalization conditions. P53’s active-personalized Brier and Δ are omitted because too few active learning–timed labels were available (four in total; $n_{\text{active}} = 3$ in the personalization split). Participants P46, P47, P49, P50, and P55 were excluded entirely because their held-out test sets contained only “Not Stressed” labels, leaving no basis for evaluating discrimination or calibration.

PID	n_{base}	n_{active}	n_{test}	Pop. AUC	Pop. Brier	Base Brier	Active Brier	Δ (base–act)
P40	27	14	18	0.633	0.583	0.237	0.335	−0.097
P41	55	26	35	0.557	0.145	0.143	0.158	−0.015
P42	26	11	16	0.521	0.552	0.648	0.575	+0.073
P43	63	21	37	0.294	0.129	0.160	0.263	−0.103
P44	9	9	9	0.650	0.293	0.378	0.258	+0.121
P45	74	7	36	0.657	0.067	0.027	0.033	−0.006
P48	51	30	36	0.771	0.057	0.033	0.039	−0.006
P51	61	31	40	0.455	0.385	0.285	0.282	+0.004
P52	80	41	53	0.459	0.087	0.078	0.151	−0.073
P53	16	3	9	0.571	0.537	0.298	—	—
P54	47	16	28	0.615	0.275	0.240	0.241	−0.001
P56	26	12	17	0.443	0.407	0.301	0.280	+0.021
P57	57	24	36	0.646	0.084	0.113	0.076	+0.037
P58	45	13	25	0.652	0.107	0.476	0.275	+0.201
Mean	45.5	18.4	28.2	0.566	0.265	0.240	0.228	+0.012

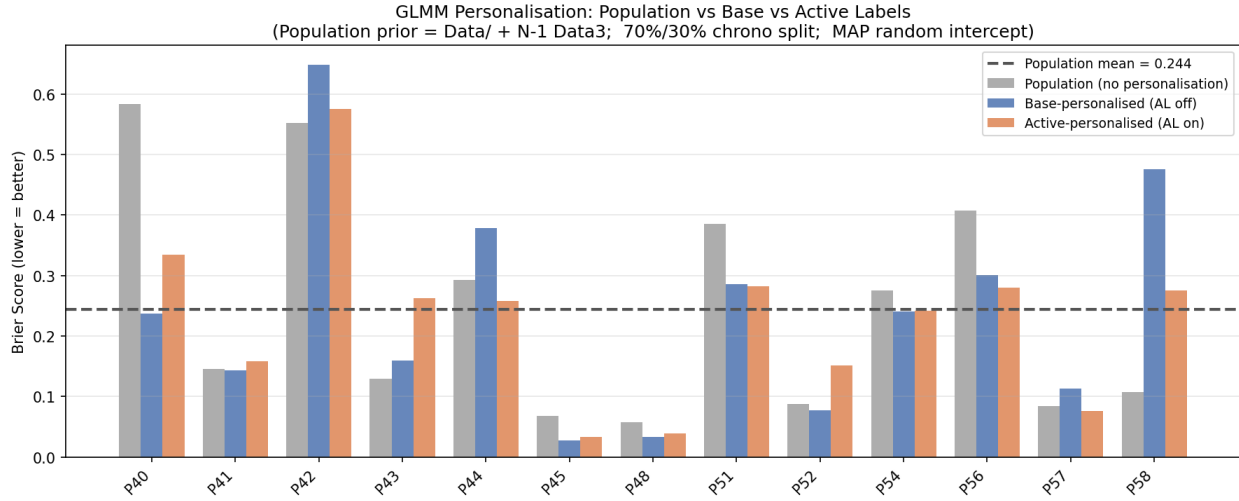


Figure C.1: **Per-participant model calibration: population vs. base- vs. active-personalized.** Brier scores (lower is better) for the 13 Cohort 3 participants with sufficient labels in both personalization conditions. The unpersonalized population GLMM (gray) is compared against base-personalized (blue, active learning off) and active-personalized (orange, active learning on) variants in which only the participant random intercept was re-estimated on the 70% chronological personalization split. Both personalization strategies improved calibration on average over the unpersonalized population model, and active learning-timed labels matched base labels in calibration despite being roughly 60% fewer in number for these participants (258 vs. 637 in Table C.1), against the 64.7% reduction across the full cohort. However, active learning-timed labels show no improvement over randomly selected labels.

APPENDIX D
SENSOR FEATURES THAT DIFFERED BY DELIVERY
TIMING (COHORT 3)

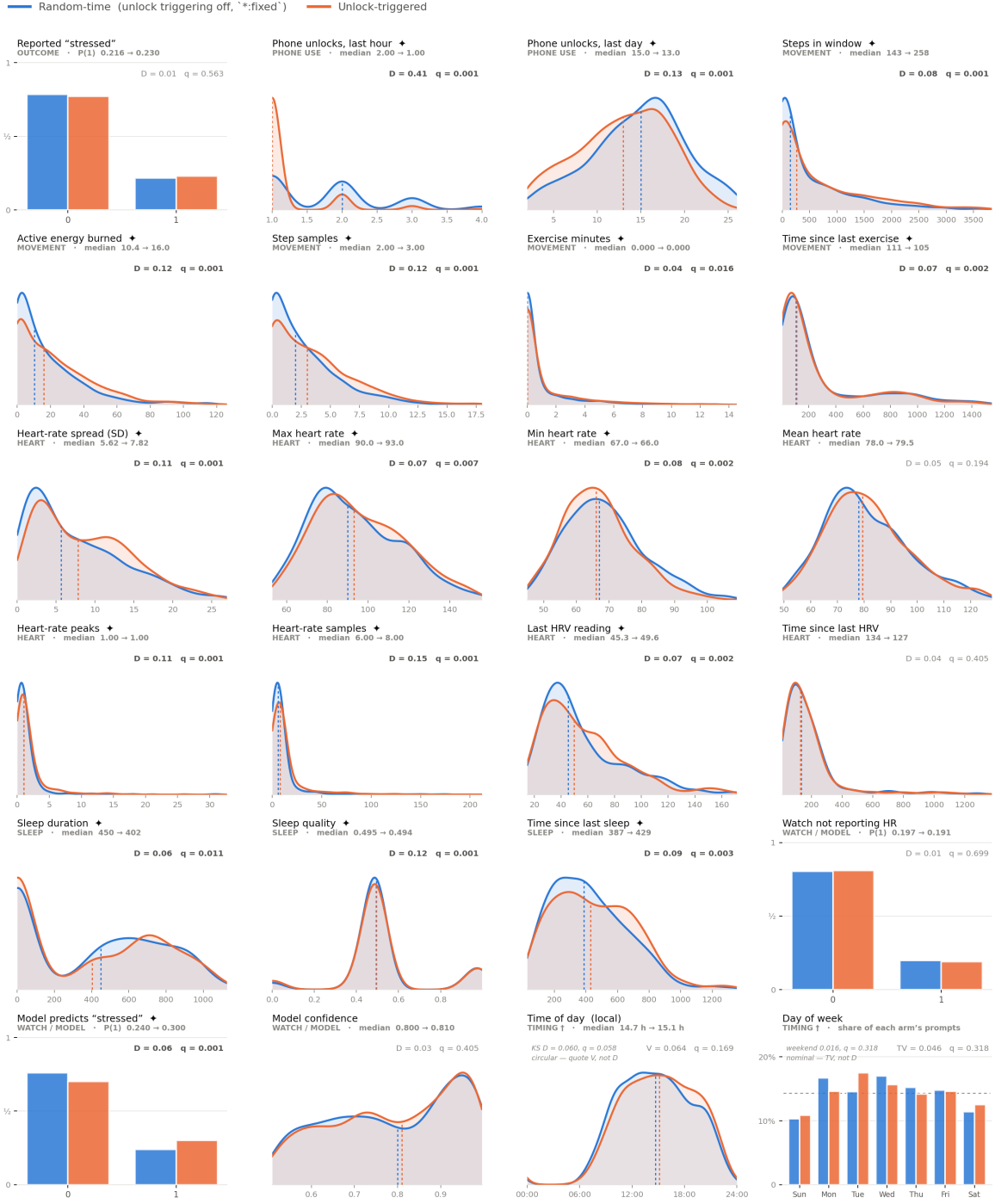


Figure D.1: **Sensor features by delivery timing.** Distributions of every Cohort 3 measure under random-timed (blue) and unlock-triggered (orange) delivery. Dashed lines mark condition medians; D is the two-sample Kolmogorov–Smirnov statistic and q its Benjamini–Hochberg false-discovery-rate-corrected p -value; starred panels mark significant differences. Apart from the recent-unlock counts, which unlock triggering fixes by construction, the divergences are small.

APPENDIX E

PERCEIVED STRESS SCALE (PSS-10) SCREENING INSTRUMENT

Table E.1: **Perceived Stress Scale (PSS-10) items used for eligibility screening.** Participants responded to each item using a 5-point Likert scale ranging from “Never” to “Very Often.” Positively worded items were reverse-coded during scoring following standard PSS guidelines.

Item	Question Text
1	In the last month, how often have you been upset because of something that happened unexpectedly?
2	In the last month, how often have you felt that you were unable to control the important things in your life?
3	In the last month, how often have you felt nervous and “stressed”?
4	In the last month, how often have you felt confident about your ability to handle your personal problems?
5	In the last month, how often have you felt that things were going your way?
6	In the last month, how often have you found that you could not cope with all the things that you had to do?
7	In the last month, how often have you been able to control irritations in your life?
8	In the last month, how often have you felt that you were on top of things?
9	In the last month, how often have you been angered because of things that were outside of your control?
10	In the last month, how often have you felt difficulties were piling up so high that you could not overcome them?

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